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# Advanced liquid crystal devices for augmented reality and virtual reality displays: principles and applications

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## Abstract

Liquid crystal displays (LCDs) and photonic devices play a pivotal role to augmented reality (AR) and virtual reality (VR). The recently emerging high-dynamic-range (HDR) mini-LED backlit LCDs significantly boost the image quality and brightness and reduce the power consumption for VR displays. Such a light engine is particularly attractive for compensating the optical loss of pancake structure to achieve compact and lightweight VR headsets. On the other hand, high-resolution-density, and high-brightness liquid-crystal-on-silicon (LCoS) is a promising image source for the see-through AR displays, especially under high ambient lighting conditions. Meanwhile, the high-speed LCoS spatial light modulators open a new door for holographic displays and focal surface displays. Finally, the ultrathin planar diffractive LC optical elements, such as geometric phase LC grating and lens, have found useful applications in AR and VR for enhancing resolution, widening field-of-view, suppressing chromatic aberrations, creating multiplanes to overcome the vergence-accommodation conflict, and dynamic pupil steering to achieve gaze-matched Maxwellian displays, just to name a few. The operation principles, potential applications, and future challenges of these advanced LC devices will be discussed.

## Introduction

Display devices have become ubiquitous in our daily lives; their applications range from smartwatches, smartphones, pads, computer monitors, TVs, data projectors, to augmented reality (AR) and virtual reality (VR) headsets, just to name a few. After several decades of extensive research and development in materials and devices, and heavy investment in manufacturing technologies, liquid crystal displays (LCDs)<sup>1–3</sup> and organic light-emitting diode (OLED)<sup>4–6</sup> displays have become the dominant technologies. Recently, micro-LEDs<sup>7</sup> and mini-LEDs<sup>8</sup> are emerging; the former can offer an extremely high luminance while the latter can serve either as a locally dimmable backlight for high-dynamic-range (HDR)<sup>9</sup> LCDs or as an emissive display itself. Micro-LEDs offer an

ultrahigh luminance, long lifetime, and high pixel density, and are particularly attractive for the see-through AR displays<sup>10</sup>, especially when the ambient light is strong. Nevertheless, the biggest challenge for micro-LED to overcome is the mass transfer yield and defect repair<sup>11</sup>, which in turn affects the cost.

AR and VR have potential to revolutionize the ways we perceive and interact with digital information as promising next-generation displays. Since the 1990s, AR and VR experienced their first boom ushering in the dawn of evolution in display and information platforms<sup>12</sup>. Over the past decades, emerging technologies such as advanced liquid crystal (LC) materials and functional devices<sup>13–17</sup> have greatly reshaped the AR/VR display systems, enabling this advanced information technology to be truly integrated into people's lives. These new display platforms are mutually reinforcing with LC optics, resulting in many impressive, AR/VR-ready devices.

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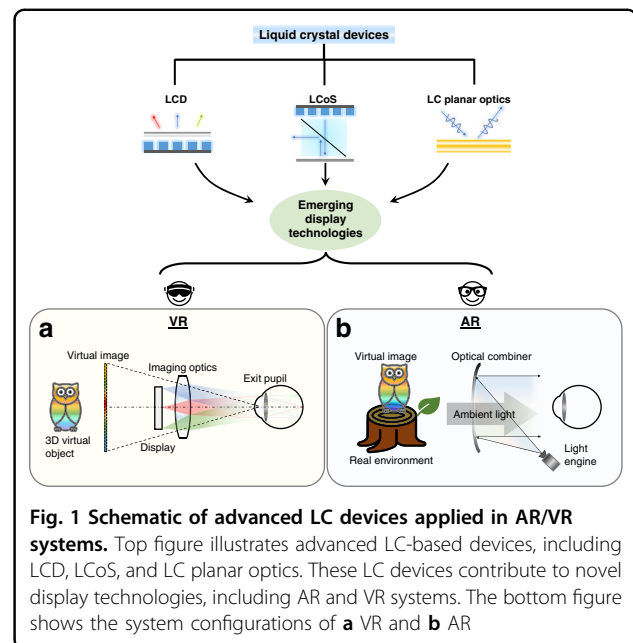
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In this paper, we focus on advanced LC-based light engines and optical components for AR/VR applications. First, we review the basic structures of AR/VR displays and the key parameters with associated criteria. Then, we dive into three main categories of LC functional devices: HDR microdisplay for VR, high-resolution-density, high-brightness LCoS for AR/VR light engines, and nearly 100% light efficiency planar LC optics. On the microdisplay part, we discuss how an efficient, high brightness light engine improves the VR imaging system. For the LCoS panels, we firstly review the working principles and present the status of both amplitude and phase modulators, and then introduce various unique applications adopted for near-eye displays. For LC planar optics, the working principles, and promising solutions to the pressing challenges of AR/VR displays are discussed. These three LC functional devices cover the entire AR/VR systems from light engines, imaging optics, to various functional elements.

### AR and VR display systems

More recently, the next-generation display technologies under dedicated development are no longer limited to flat panels that are just placed in front of the audiences but aimed at revolutionizing the way of interactions between the users and their surrounding environment. At one end of the spectrum is VR display, which effectively extends the field of view (FoV), blocks the entire ambient, and offers a completely immersive virtual environment. At the other end of the spectrum is AR display, which pursues a high-quality see-through performance to enrich the real world by overlaying digital contents<sup>12</sup>. With the refreshing visual experiences, AR/VR displays have demonstrated the promising potential for a wide range of attractive applications, including but not limited to healthcare, education, engineering, manufacturing, and entertainment<sup>18–20</sup>.

For VR systems, the emitting light from the display module, which is usually 2–3 inches in diagonal, generates a virtual image with an adequate FoV via the well-designed magnification lens (normally 2 inches aperture size with 35–45 mm focal length). Figure 1a depicts the schematic diagram of a VR optical system. The eyebox defines the region within which the whole image FoV can be viewed without vignetting, and its size is usually correlated with FoV due to the basic Etendue conservation. Since VR provides an immersive experience with virtual images, the 3D virtual object with depth cue is an important feature. Displaying two different images to the left eye and right eye can form the vergence cue. But the fixed image plane often mismatches with the actual depth of the intended 3D image, leading to vergence-accommodation conflict (VAC)<sup>21,22</sup>, which will be discussed in detail later. Perceived image resolution for VR



displays can be assessed from the angular resolution and calculated by dividing the total resolution of the display panel with the FoV. To achieve human visual acuity of 1 arcminute angular resolution<sup>23</sup>, 60 pixels per degree (ppd) is considered a common goal. Regarding AR systems, the emitting light from the microdisplay module (usually < 1 inch) generates virtual content, which overlaps with the real world. Figure 1b depicts the generic sketch of an optical see-through AR system with an optical combiner. Throughout various optical architectures from free-space combiners<sup>20,24</sup>, total internal reflection (TIR) freeform combiners<sup>25,26</sup>, to waveguides with diffractive or reflective combiners<sup>12</sup>, the definition of the above parameters remains the same. But the VAC issue is more essential in AR than VR, as the image in an AR display is directly superimposed onto the real world with correct depth cues. Compared to VR with a 2–3-inch display panel, the light engine in AR is much smaller, which leads to the conflict between FoV and eyebox size, especially for a glasses-like AR system. To meet above requirements, advanced LC devices with unique optical properties and optoelectronic responses have been widely used in AR/VR displays, covering all key components from light engines to optical components.

### LCDs for VR

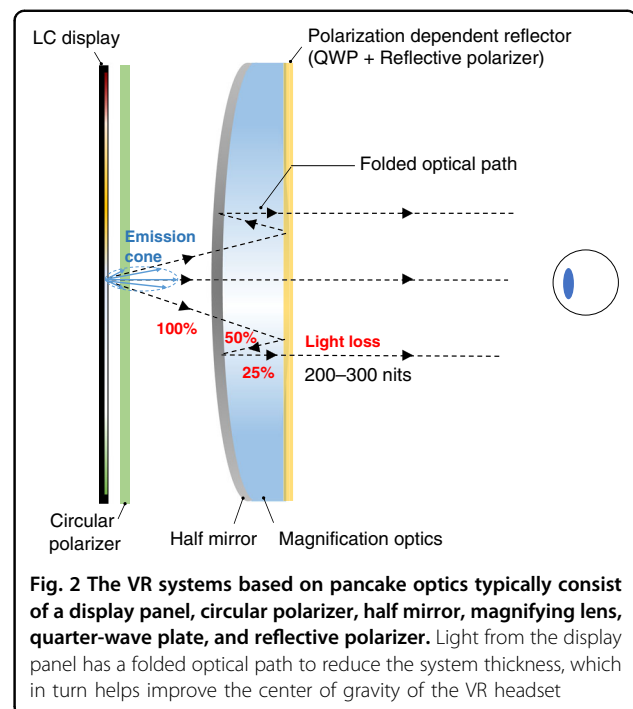
An LCD consists of two main parts: backlight unit and LC panel. The backlight unit provides a uniform illumination on the LC panel, and the transmittance of each pixel is controlled by a thin-film-transistor (TFT). In the past five decades, both backlight unit and LC panel have witnessed tremendous growth and significant improvement<sup>2,27–33</sup>.

Recently, mini-LED backlight units with sufficient local dimming zones have successfully suppressed the halo effect and achieved a contrast ratio of  $>10^5:1$ <sup>9,34,35</sup>. Today, TFT LCDs remain the dominant flat panel display technology, especially in TVs, tablets, and monitors, despite the strong competitions from OLED displays. The working principles and applications of LCDs as direct-view displays have been discussed in several review papers<sup>3,8</sup> and will not be repeated here.

However, in a VR system, due to the emergence of many new requirements, there are some problems to directly use a traditional LCD as the light engine. To clarify these issues, in the following we will highlight the key differences between the display devices for direct-view systems and VR systems and present the challenges and potential solutions. Before diving into the details, we will briefly introduce active-matrix OLED display, which is also a strong contender for VR light engines. Compared to LCDs, the self-emissive OLED displays show an unprecedented contrast ratio, high efficiency, and fast-response time. However, the resolution density of OLED displays is not sufficient to eliminate the screen-door effect. Such a resolution limitation results from two parts: For the front plane, the traditional fine metal mesh (FMM) fabrication method can typically support organic material patterning of about 600–800 pixels per inch (PPI)<sup>36</sup>. For the back-plane, it is necessary to squeeze the complex compensation circuit (typically, eight thin-film transistors (TFTs) and 1 capacitor) into each small subpixel ( $<8\ \mu\text{m}$ : 3000 PPI). To overcome the above-mentioned PPI issue, the OLED-on-silicon has attracted lots of attention<sup>37–39</sup>. The CMOS (complementary metal-oxide semiconductor) with high electron mobility significantly reduces the circuit size. Moreover, new technologies such as laser patterned FMM<sup>40</sup>, silicon nitride mask<sup>41</sup>, photolithography method<sup>42</sup>, and electro-hydrodynamic printing<sup>43</sup> are also emerging to help achieve high-resolution density. The other method for OLED-on-silicon to fulfill high-resolution density is to use white-OLED (WOLED) with color filters. However, color crosstalk, high driving voltage, and low optical efficiency are the key drawbacks. To address these problems, WOLEDs with microlens arrays<sup>44</sup> and tandem structures<sup>45</sup> have been proposed. Moreover, the high cost of silicon wafer normally restricts the display panel size to be about one inch. According to the Etendue conservation in the VR system, such a small size display panel further limits the FoV to be below  $100^\circ$ . Therefore, the competition between OLEDs and LCDs has extended from traditional direct-view displays to AR/VR headsets. In this review, we will focus on LCDs.

In a VR headset, an enlarged virtual image is formed in front of the user through a magnifying lens, resulting in an immersive experience. Such a projection process demands the display panel to have a much higher resolution density

than that of a direct-view display. The visual acuity of the human vision system is 1 arcmin<sup>46</sup>, then a 6 K resolution display is typically required to support a  $100^\circ$  FoV VR headset<sup>47</sup>. To achieve a compact form factor, such a  $6\text{ K} \times 6\text{ K}$  resolution needs to be assembled on a 2–3-inch panel, resulting in a 2000 PPI of the microdisplay panel. Moreover, the motion blur becomes more severe and obvious for a near-eye system with fast eye moving<sup>18</sup>, especially during a video game<sup>48</sup>. Another major difference between traditional direct-view display and VR headset is the viewing condition. For a VR headset, the housing blocks the surrounding illumination, which significantly reduces the ambient light effect. Based on the dark adaptation of human visual system, a display with about 200–300 nits is bright enough<sup>49</sup>. In contrast, a TV usually requires 1000 nits. Additionally, the head-mounted devices require a compact form factor and lightweight for comfortable wearing. Recently, the folded optical path based on polarization selectivity has been widely applied in VR systems to reduce the volume size<sup>50,51</sup>. This thin and light design by using folded optical path is also known as pancake optics. In such a pancake system (Fig. 2), due to the light loss caused by the multiple partial reflections, the optical efficiency is reduced about four times<sup>50</sup>. To achieve the same brightness (200–300 nits) as a normal VR display, the required brightness from the display module should be boosted by four times<sup>20</sup>, which is around 800–1200 nits. Fortunately, the fixed viewing position in a VR system provides an opportunity to



**Fig. 2** The VR systems based on pancake optics typically consist of a display panel, circular polarizer, half mirror, magnifying lens, quarter-wave plate, and reflective polarizer. Light from the display panel has a folded optical path to reduce the system thickness, which in turn helps improve the center of gravity of the VR headset



**Table 1** Total light transmission (TLT) calculation of an LC panel for VR headsets

| Factors             | Backlight emission                                     | Aperture ratio   | Two polarizers | Color filters | Dead zone coverage | Total |
|---------------------|--------------------------------------------------------|------------------|----------------|---------------|--------------------|-------|
| values              | 80–95% <sup>a</sup>                                    | 38% <sup>b</sup> | 35%            | 33%           | 50%                |       |
| Calculation formula | $95\% \times 38\% \times 35\% \times 33\% \times 50\%$ |                  |                |               |                    | 2%    |

<sup>a</sup>Depending on the LC operation mode and cell gap design employed<sup>b</sup>1500-PPI condition

achieve higher efficiency by concentrating the emitted light into the eyebox area<sup>52</sup>.

Based on the above-mentioned requirements, there are several challenges when using LCDs as light engines in VR headsets. The first challenge is the reduced aperture ratio resulting from high-resolution density<sup>53</sup>. LCD panels with 100 PPI (TVs), 300 PPI (laptops), 600 PPI (smartphones), and 1500 PPI (VR light engine) have aperture ratios of 95%, 86%, 73–38%, respectively. The smaller aperture ratio reduces the optical efficiency, which in turns increases the power consumption and thermal effect of the VR headset. To mitigate the motion artifacts in VR headsets, display with high frame rate ( $\geq 120$  Hz) and low duty ratio ( $\leq 20\%$ ) are preferred<sup>54,55</sup>. Therefore, the LC response time should be fast enough to support the high frame rate and low duty ratio<sup>56–62</sup>. To achieve a response time faster than 3 ms, the standing layer approach (also known as dead zones) has been demonstrated. The existence of dead zones helps shorten the response time, but the tradeoff is decreased transmittance<sup>61</sup>.

Overall, to achieve high-resolution density and fast-response time, LCDs for VR headsets suffer from low transmittance caused by small aperture ratios and disclination lines. If we further consider the absorption losses from color filters and polarizers, the total light transmittance of the LC panel for VR headsets is merely about 2%. Details are listed in Table 1. Therefore, how to enhance the optical efficiency of VR display panels is critically important.

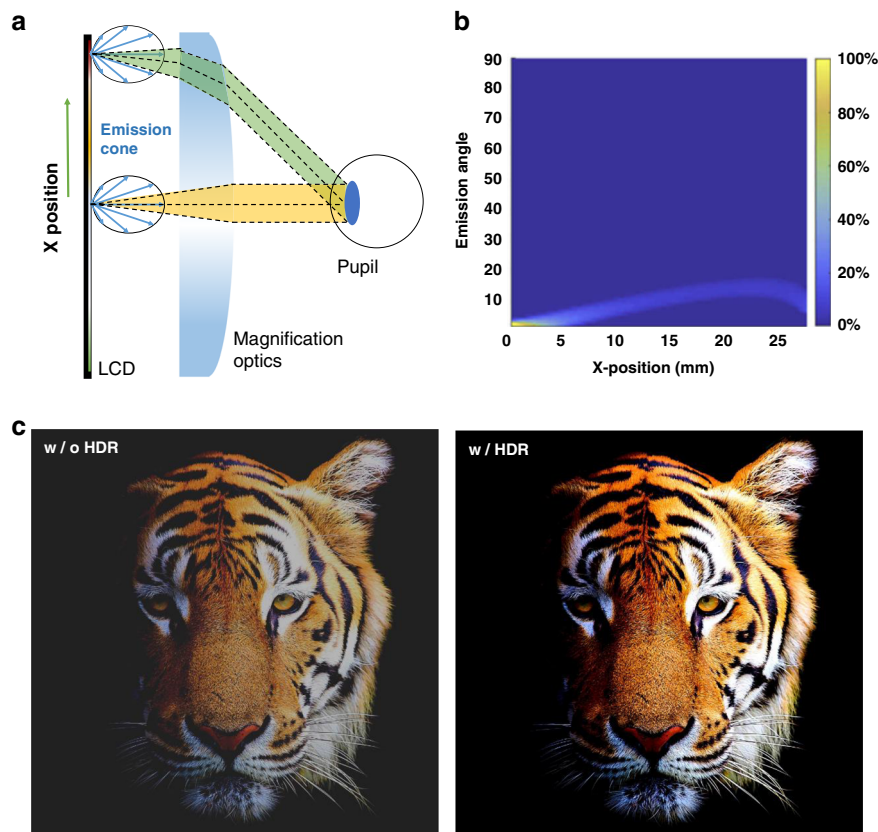
### Efficiency enhancement in VR

In this section, we will discuss how to improve LCD's optical efficiency for VR headsets. The total optical efficiency of a VR system consists of two main components: one is the panel efficiency, and another is the imaging optics. The panel efficiency has been clearly discussed above (Table 1), here we will focus on improving the optical system efficiency using a directional LCD backlight. For a typical Lambertian display, only a small portion of the light can be received by the viewing eyebox<sup>63</sup> (Fig. 3a, green and yellow cones), and the rest is wasted<sup>64</sup>. The system consists of an LCD, a Fresnel lens with focal length  $\sim 35$  mm, and a circular receiver with a diameter of 4 mm (pupil size). The eye relief is about 15 mm, and the panel is placed on the back focal point of the Fresnel lens. The relationship between the emission angle ( $y$ -axis) and the light efficiency

is plotted in Fig. 3b, which indicates that only a small portion of the emitted light can be received by the viewer's eye. Therefore, a directional backlight is preferred to boost the optical efficiency of VR systems. Due to the limited volume size in a VR system, many thin and compact designs for directional backlight have been achieved by applying nanometer-sized grating, microlens arrays, micro-pyramid films, or reflective microstructures<sup>65–70</sup>.

The implementation of directional backlighting can significantly improve the overall optical efficiency but will also cause severe vignetting effects. As illustrated in the green cone of Fig. 3a, the chief ray of the emission cones is mismatched with the receiving angle of the eyebox. Figure 3b shows the optical efficiency at different pixel positions ( $x$ -axis). Only when the chief ray of the emission angle matches the receiving angle, maximum efficiency can be reached<sup>63</sup>. Therefore, to spatially control the direction of the emission angle is crucial to avoid vignetting while maintaining a high efficiency. To address this issue, functional optical films can be added to spatially match the radiation pattern of the display panel to the receiving cone of the VR optical system. Furthermore, Peng et al.<sup>71</sup> found that the spatial radiation-pattern modulator can mitigate the ghost image in an VR system. Using a directional backlight, when the pixel position is far away from the center, the ratio of ghost to signal energy increases. The spatial radiation-pattern modulator can modulate the local radiation of each pixel and avoid receiving the ghost image. As a result, the ratio of ghost to signal energy decreases.

Aside from optical efficiency improvement, adding functional films with intensity modulation can help to enhance the VR image quality. Generally, an LCD's contrast ratio ranges from 2500:1 (FFS mode) to 5000:1 (MVA mode). Such a grayish dark state severely degrades the user's immersive experience. Recently, spatial intensity modulation is realized by the zoned mini-LED backlight. Depending on the displayed image, such a locally dimmable backlight can boost the LEDs in bright image areas and dim them in dark areas<sup>72</sup>. The mini-LED backlight technology is applied in the VR light engine to display HDR images and provide a better immersive experience. Figure 3c shows the image quality comparison of images with and without local dimming functions (2.16 inch LCD module with the 1411 PPI and 2304 dimming zones)<sup>53</sup>. Wu et al.<sup>73</sup>



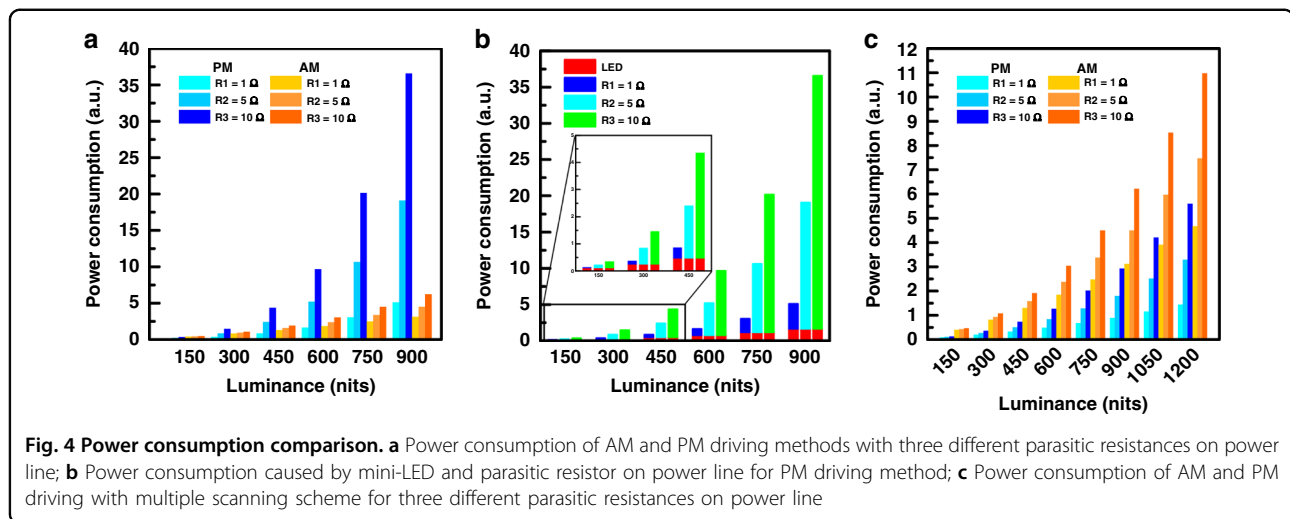
**Fig. 3 Directional backlight displays.** **a** Schematic of Fresnel lens in a normal VR system. **b** Simulated light collection efficiency of different pixel positions. **c** Image comparison with and without local dimming. **b** ref. <sup>63</sup>, with permission from The Optical Society, and **c** ref. <sup>53</sup> with permission from John Wiley and Sons

packed thousands of mini-LEDs into a 2.6-inch backlight, and the brightness of each LED is modulated according to the image content. However, a notorious visual artifact, called halo effect, in the mini-LED backlit LCDs could occur, depending on the zone number, native LC contrast ratio, image content, ambient light, and viewing angle<sup>9,35,74</sup>. From the perspective of backlight unit, to mitigate the halo effect, it is required that the light distribution of each local dimming zone is uniform, and the light crosstalk between adjacent zones is minimized. Huang et al. described the desired light profile by a super-Gaussian function and discovered that good light confinement (power of super-Gaussian function about 4.5) helps to improve image quality. To achieve such a desired light profile, two approaches are often considered. One is to optimize the LED chip geometry, mount a scattering lightguide on top of the LED array, and fabricate reflective sidewalls and microstructures on the outer surface of the LED. Another approach is to laminate optical films in the backlight unit to shape the light profile before it enters the LC panel. Based on these two methods, novel mini-LED backlight systems supporting spatial light intensity modulation are

proposed to prevent the light crosstalk and to maintain the luminance and color uniformity of the zones<sup>34,75–79</sup>. On the software side, Seetzen et al.<sup>80</sup> proposed a local dimming algorithm (called maximum method algorithm<sup>80</sup>) to systematically optimize the LCD transmittance and spatial profile of mini-LED backlight. The brightness of the mini-LEDs in each zone is determined by the brightest pixel in the zone. Although a sufficient brightness can be maintained, the drastic brightness change per frame may cause severe flickering and image artifacts. The averaging algorithm, on the other hand, suppresses the peak brightness of each area to avoid such image artifacts, but the highlighted images may become dimmer and less vivid. In order to solve this dilemma, extensive works<sup>81–83</sup> on novel local dimming algorithms to improve the image quality of mini-LED backlit LCDs have been proposed.

### Driving methods and power consumption of mini-LED backlit LCD

Previous sections have fully discussed the optical efficiency, in this section, we will briefly analyze the electrical efficiency to provide a comprehensive and systematic



discussion. Due to the limited battery storage, an effective driving system is essential to lower the power consumption and the associated thermal effect of a compact VR headset. To drive the mini-LED backlight, both active matrix (AM) and passive matrix (PM) driving methods have been explored<sup>53,73,84,85</sup>. The AM driving method integrates the mini-LEDs with the TFT backplane to control the luminance of each zone and reduce the required number of IC chips<sup>84</sup>. For an AM mini-LED backlight, because the desired driving voltage is stored on the capacitor, the driving current can be kept constant for a long period. Therefore, the driving current is low while achieving the required luminance<sup>86,87</sup>. In addition, the TFT also consumes power as the driving current passing through. On the contrary, the PM driving method, which directly integrates the LCD panel with the mini-LEDs through printed circuit board (PCB), is a simpler way to drive the mini-LEDs<sup>88</sup>. The PM method drives the backlight row-by-row, and due to the lack of storage capability, the mini-LEDs in each row are only driven for a short time<sup>89</sup>. Therefore, to attain the same luminance as the AM driving, the PM method requires a higher current for the mini-LEDs.

To compare the power consumption between the AM and the PM driving methods, a mini-LED backlight with  $48 \times 48$  mini-LEDs for VR displays is studied. Herein, the duty ratio of the AM and PM driving methods is set at 8.33% and 2.08%, respectively. Generally, the resistance of the parasitic resistor on the power line for AM driving is several ohms, so the resistance of 1  $\Omega$ , 5  $\Omega$ , and 10  $\Omega$  are selected for the evaluation. Figure 4a shows the simulated power consumption of AM and PM driving methods with different resistance. Overall, the PM driving consumes more power than the AM method except at low resistance (1  $\Omega$ ) and luminance  $\leq 600$  nits. The higher power consumption originates from the larger driving current

because of the shorter emission period in the PM driving method. As shown in Fig. 4b, for the PM driving method, most of the power is consumed when the large driving current passes through parasitic resistor ( $P = I \times R^2$ ). Recently, to overcome the impact from short emission period, the multiple scanning scheme, which drives multiple rows at one time, is used in the PM driving method to increase the duty ratio<sup>90</sup>. In this work, the total rows are divided into 12 groups and the mini-LEDs of each group are driven at the same time, extending the duty ratio from 2.08% to 8.33% to reduce the required driving currents. The power consumption difference between the AM and the PM driving methods with the multiple scanning scheme are plotted in Fig. 4c. Notably, to compensate for the optical loss in the pancake configuration where the maximum transmittance drops by about four times, the display luminance should be boosted from 300 nits to 1200 nits. Within the maximum luminance of 1200 nits, the PM driving method with the multiple scanning scheme offers a lower power consumption than the AM driving method. This confirms that extending the duty ratio to reduce the driving current of mini-LEDs by the multiple scanning scheme significantly lowers the power consumption. However, more ICs are required to individually control the mini-LED in each zone, which in turn increases the cost.

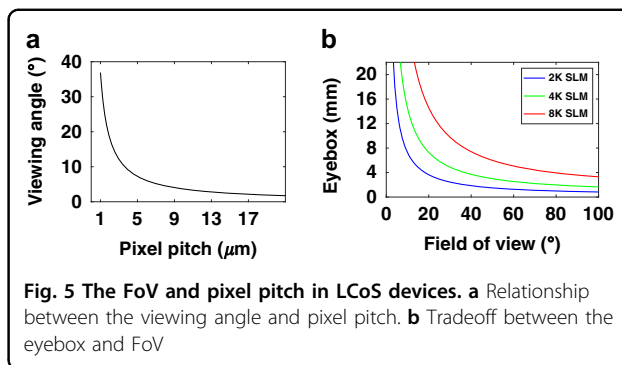
### LCoS devices for AR/VR

In contrast to conventional transmissive TFT-LCDs, the reflective LCoS panel combines the electro-optic effect of LCs and high-performance silicon CMOS electronics to offer a high fill factor ( $>90\%$ ), high resolution, compact form factor, and high frame rate<sup>14</sup>. Due to the excellent light modulation ability, both amplitude and phase-only LCoS devices are important light engines for AR/VR applications<sup>91,92</sup>. Especially, compared to other phase

modulators, such as microelectromechanical system (MEMS)<sup>93,94</sup>, the phase-only LCoS devices (also called spatial light modulators (SLMs)) stand out in multilevel phase modulation, low driving voltage, and comparably low cost<sup>95</sup>. As a result, most holographic displays for AR/VR applications are implemented with SLMs. In this section, we will first briefly describe the working principles of LCoS devices, then stress some remaining technical challenges with recent progress, and finally present their various applications in AR/VR systems.

A typical reflective LCoS device consists of a CMOS silicon backplane, pixelated aluminum reflectors, LC layer, and an indium tin oxide (ITO)-coated cover glass. The aluminum electrodes deposited onto the silicon backplane act as reflective pixel arrays. The LC layer is sandwiched between the ITO-glass substrate and the CMOS silicon backplane. When an incident light traverses the LC layer, the voltage-dependent phase retardation (for an amplitude modulator) or phase change (for a phase modulator) can be obtained<sup>96–98</sup>. For an amplitude modulating LCoS device in a projection system, the LED source emits an unpolarized light. Let us assume the s-polarized light (*s*-wave) is reflected, while the p-polarized light (*p*-wave) transmits through the polarizing beam splitter (PBS). The modulated *p*-wave from LCoS is reflected by the PBS onto the projection optics of the near-eye system. When a phase-only LCoS is used as an SLM with a coherent light source<sup>99</sup>, by spatially controlling the wavefront, the SLM can be applied in near-eye displays with holographic imaging, which will be discussed in detail later. It should be noted that the beam splitter can be replaced by other holographic optical elements (HOEs) in an off-axis configuration<sup>92</sup> or waveguide geometry<sup>100</sup>, leading to a more compact form factor.

For the LCoS based near-eye systems, the two main requirements are wide FoV and high resolution. These two factors are interrelated. In an unmagnified near-eye system, the FoV is equal to twice the maximum diffraction angle, and the eyebox is equal to the size of the employed SLM. At normal incidence, the maximum diffraction angle in a binary grating is  $\sin^{-1}[\lambda/(2p_o)]$ , where  $p_o$  is the pixel pitch of the LCoS SLM, and  $\lambda$  is the incident wavelength. Figure 5a shows the relationship between FoV and pixel pitch at  $\lambda = 633$  nm of the SLM. If the pixel pitch is reduced to 1  $\mu\text{m}$ , the FoV can be widened to  $\sim 37^\circ$ . The tradeoff between FoV and one-dimensional eyebox is governed by  $\sin(\text{FoV}/2) \cdot \text{eyebox} = \lambda N/2$ , where  $N$  is the 1D number of pixels<sup>101</sup>. As shown in Fig. 5b, FoV can be increased for a given number of pixels at the expense of reduced eyebox size. By employing an SLM with a smaller pixel pitch and more pixels, a wider FoV can be achieved while maintaining the same eyebox size. Based on the above analysis and the requirements of AR/VR, an ultrahigh resolution of 25,400 PPI corresponding to a pixel pitch of 1  $\mu\text{m}$  is critically needed.



To achieve an ultrahigh-resolution LCoS panel, the main challenges are the CMOS backplane and the fringing field effect (FFE). New LCoS foundries supporting a smaller feature size need to be developed to accommodate enough transistors in an ultrasmall pixel. Because of the limited voltage swing the CMOS backplane can support, the LC materials with a larger dielectric anisotropy are desired to reduce the required driving voltage. The layout optimization also helps achieve a smaller pixel size<sup>102</sup>. For the second challenge, the fringing field effect is caused by the unequal voltages on adjacent pixels. As the pixel pitch gets comparable or even smaller than the cell gap, the FFE will become much stronger, which deteriorates the LCoS performance. Currently, the state-of-the-art pixel pitch is  $\sim 3$   $\mu\text{m}$  for both amplitude and phase LCoS devices<sup>103</sup>. For an amplitude LCoS, a straightforward method to mitigate the FFE is to reduce the cell gap by employing a high  $\Delta n$  LC material<sup>104,105</sup>. Besides, several other strategies have been proposed to suppress the FFE, such as optimizing the slope of pixel electrodes<sup>106</sup>, adopting circularly polarized light<sup>107</sup>, fabricating double-side electrode structures<sup>108</sup>, and introducing patterned pretilt angles<sup>109</sup>. For an SLM panel, the phase disturbance caused by the FFE could lower the diffraction efficiency and exacerbate the reconstruction error. Compared to amplitude LCoS devices, it is more challenging to mitigate the FFE of phase-only LCoS devices because of their thicker cell gap and more stringent requirements such as linearly polarized light and sharp phase edge. To achieve 1- $\mu\text{m}$ -pitch phase modulators, Isomae et al.<sup>110,111</sup> proposed a dielectric shield wall structure, but the device fabrication is challenging. Besides the material-based and device-based methods, model-based<sup>112</sup> or algorithm-based methods<sup>113</sup> have also been introduced to compensate the pixel crosstalk. Nevertheless, rigorous simulations on the LC dynamics are necessary when the pixel pitch becomes comparable or even smaller than the cell gap. Persson et al.<sup>114</sup> proposed a modified algorithm to include the FFE in the hologram optimization, but the real-time operation is a main concern. Despite decades of extensive efforts,



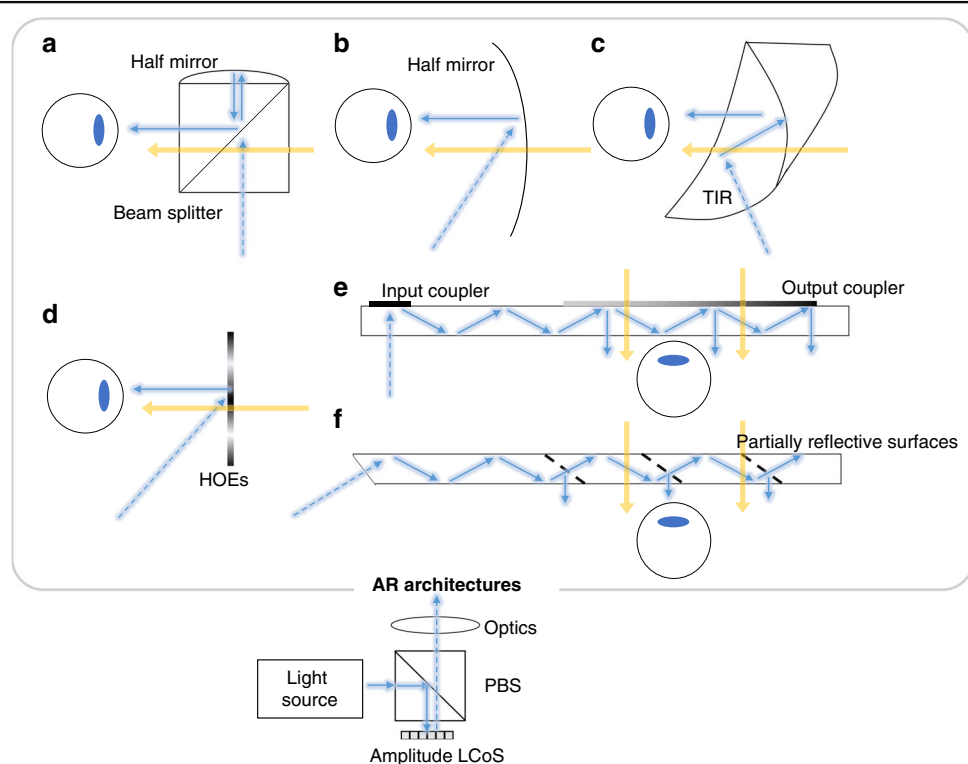
the ultrahigh-resolution phase modulators for wide FoV holographic displays are still awaiting to be developed.

### Amplitude LCoS in AR

For AR displays, several types of light engines have been developed as the image source to provide the virtual images, such as LCoS, digital light processor (DLP), OLED microdisplay ( $\mu$ OLED), micro-LED, and laser beam scanner (LBS)<sup>12,115</sup>. LCoS can provide high brightness ( $>50$  Knits) and has come to fabrication maturity after more than 20 years of development. Companies like Sony, Himax Displays, Compound Photonics, etc., are devoted to developing LCoS based light engines with high resolution for AR applications and the total volume is only about  $1\text{ cm}^3$  including the illumination optics<sup>15,116</sup>. As shown in Fig. 6, an amplitude LCoS has been integrated into several types of AR systems. For example, Google Glass adopts a traditional PBS as optical combiner and utilizes a Field Sequential Color (FSC) LCoS with resolution of  $640 \times 360$  as its light engine. The combiner designs used in Google Glass are often referred to as birdbath optics (Fig. 6a). Another simpler design using a single partially reflective off-axis freeform mirror as the combiner can be seen in later products such as Meta 2<sup>117</sup>, achieving a large FoV (up to  $90^\circ$ ) with the cost of form factor (Fig. 6b). Another way to fold the optical path is to use a TIR-prism (Fig. 6c). In this design, all the surfaces

are freeform, which offers an excellent image quality<sup>25</sup>. To cancel the optical power for the transmitted environmental light, a compensator is added to the TIR prism. The whole system has a well-balanced performance between FoV, eye-box, and size. To further reduce the form factor, the prism/half mirror can be replaced by thinner and angular-sensitive HOEs (Fig. 6d) and achieve a glasses-like form factor<sup>118,119</sup>. However, these HOEs show a strong wavelength dependence thereby suffering severe dispersion issues. In the early days, amplitude LCoS are widely used in data projectors, where real images are projected to a scattering screen for audiences to see with a large viewing angle. With the aid of see-through in AR displays, Lee et al.<sup>120</sup> proposed a projection system based on diffusive HOEs, where most ambient lights pass through the diffusers due to the narrow angular selectivity of the HOEs. One common configuration for generating such images is to modulate a collimated beam with an amplitude LCoS. Another configuration for using LCoS as a display panel is to add pinholes to a  $4-f$  system, where a collimated light source is no longer required<sup>121</sup>.

Currently, most AR products adopt waveguide design with diffractive (Fig. 6e) or reflective (Fig. 6f) combiners due to the attractive thin form factor. Among them, HoloLens 1 and Magic Leap 1 and 2 adopt LCoS as light engines<sup>122,123</sup>. A newly developed AR prototype, Lumus Maximus<sup>124</sup> combines a  $2\text{ K} \times 2\text{ K}$  LCoS with its partial



**Fig. 6** Amplitude LCoS for different AR architectures. **a** Birdbath optics with beam splitter. **b** Single freeform surface as combiner. **c** Freeform TIR prism as the combiner. **d** HOEs-based combiner with compact form factor. Waveguide AR with **e** diffractive couplers and **f** partially refractive couplers

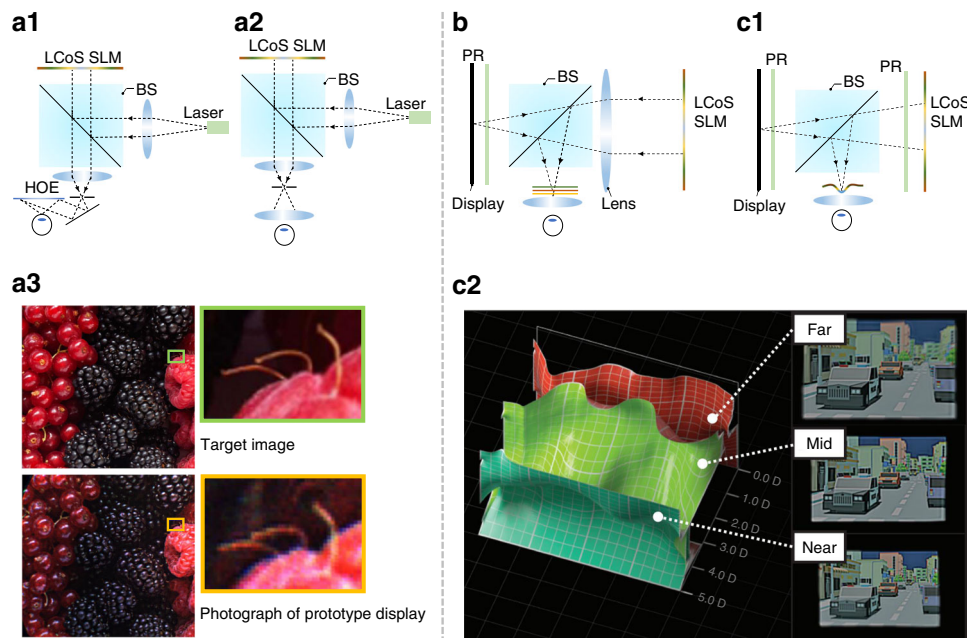
reflective surfaces waveguide, achieving a  $50^\circ$  FoV and showing an impressive image quality. It is worth mentioning that LCoS itself is a reflective display operating with a polarized light, while most light sources are unpolarized, including traditional lamps and energy-efficient LEDs. To improve the light efficiency, a polarization converter can be integrated into the amplitude LCoS thereby increasing the light efficiency<sup>125–129</sup>.

### SLMs for AR/VR

A high-resolution LCoS SLM can directly control the properties of the illumination wavefront, allowing to support holographic view<sup>130</sup>, which is also referred to as computer-generated holography (CGH)<sup>101</sup>. The generated holographic image exhibits the natural focus-and-blur effect like a real 3D object. Such an SLM is particularly useful for AR/VR displays to provide the 3D scenes with correct depth cues. Maimone et al.<sup>92</sup> demonstrated a high-quality imagery with a compact holographic view for VR and AR displays using a SLM. Based on Fresnel holographic principles, the system can integrate most of the imaging complexity into a SLM, enabling precise and spatially variant control of image focus. This control is the core for generating holograms. Figure 7a1 and a2 show the system configurations for AR and VR, respectively. Here the pitch size is  $8\ \mu\text{m}$ , and the active area is  $15.36 \times 8.64\ \text{mm}^2$ . Figure 7a3 shows an image quality comparison

between the target and the photograph of the prototype VR display. It is encouraging to achieve holograms with SLM for near-eye displays. However, the hologram rendering requires complicated computation and is time-consuming. Another concern for further using an SLM in near-eye displays is the limited Etendue. Presently, the smallest pixel size of an SLM is  $\sim 3\ \mu\text{m}$  (usually  $3\text{--}12\ \mu\text{m}$ ), therefore the diffraction angle is generally  $<5^\circ$  under plane-wave illumination. A typical SLM panel size is only 0.5–1 inch, as a result, the tradeoff between eyebox size and FoV needs to be balanced.

To use a LCoS SLM for AR/VR displays with an expanded Etendue, one strategy is to increase its diffraction angle by relay optics or applying spherical-wave illumination<sup>131–133</sup>. Chen and Chu<sup>131</sup> demonstrated a 4- $f$  relay system to demagnify the SLM. The imaged pixel size becomes smaller via the 4- $f$  system, leading to a larger diffraction angle at the cost of a smaller display panel. Then Chen et al.<sup>132</sup> used a spherical wavefront to illuminate the SLM, increasing the diffraction angle and thereby the FoV. Meanwhile, Shi et al.<sup>133</sup> also demonstrated a near-eye holographic system with spherical waves for a wider FoV. Aside from above solutions, Kuo et al.<sup>134</sup> developed a scattering-based Etendue expansion method, in which the outgoing coherent beam from the SLM is scattered by a static mask with randomized scattering properties. The wavefront coming from the SLM



**Fig. 7** Phase-only LCoS for AR/VR systems. **a1** Schematic of phase LCoS based AR system. **a2** Schematic of phase LCoS based VR system. **a3** Image quality comparison between the target and the photograph of the prototype. **b** 4  $f$  system with a LCoS SLM. **c1** Focal surface system configuration with a LCoS SLM. **c2** The optimized focal surfaces and color images displayed across three frames. Adapted from **a3** ref. <sup>92</sup>, and **c2** ref. <sup>137</sup>, under the Creative Commons Attribution 4.0 License



can be scattered by the mask to a larger range of angles, thus increasing the Etendue. As a result, the FoV can be extended without decreasing the eyebox size. By applying this approach, the FoV was experimentally expanded by four times in both horizontal and vertical directions. It is worth noting that the scattering masks are randomly patterned, leading to the inevitable noise of these randomly scattered lights, which will cause low fidelity problems when extending the system Etendue.

Recently, based on the scattering mask concept, Baek et al.<sup>135</sup> proposed a neural Etendue expander with wide-angle and high fidelity for near-eye systems. Here, the generation of neural Etendue expander and the state of SLM were jointly trained via neural network across a natural image dataset. As a result, the trained Etendue expander possesses the benefits of scattering elements for enlarging FoV, while improving the signal-to-noise ratio (SNR). In the simulation, this approach can increase the Etendue by sixty-four times, which means a wide FoV and a large eyebox can be achieved simultaneously.

Due to the relatively high cost and limited panel size with today's hardware, using an SLM as the light engine for AR/VR displays is not yet practical. However, SLMs can be used as an extra booster to combine with traditional display panel, such as LCD, for near-eye displays to overcome the VAC issue. Due to its phase modulation ability, an SLM can be placed in the light path as a spatially controllable lens. Cui and Gao<sup>136</sup> demonstrated a multifocal system by dividing a display panel into four subpanels and optically mapping them to different depths with an SLM located at the Fourier plane of the 4-*f* system (Fig. 7b). Here the SLM presents a static phase profile, including the quadratic phases that image the subpanels to different depths. Meanwhile, Matsuda et al.<sup>137</sup> also designed and demonstrated a full-color focal surface display to address the accommodation issue for near-eye displays. In this design, the SLM works as a programmable lens with spatially varying focal length, which can optically map the pixels at different spatial locations to different depths. As shown in Fig. 7c1, the focal surface display has a more compact form factor than the design reported by Cui and Gao due to the absence of the 4-*f* system. Moreover, the focal surface display can support arbitrary depth maps compared to conventional multifocal displays thereby displaying more accurate depth and blur (Fig. 7c2). Here, the burden to generate correct depth is placed on the computation, which makes real-time global optimization of phase patterns on SLM very challenging. The good news is that with the advancement of algorithms and convolutional neural networks (CNNs)<sup>138,139</sup>, solving this problem is just around the corner. Recently, Shi et al.<sup>139</sup> demonstrated a real-time CGH synthesis pipeline with a high image quality, achieving a 60-Hz frame rate through a desktop computing unit.

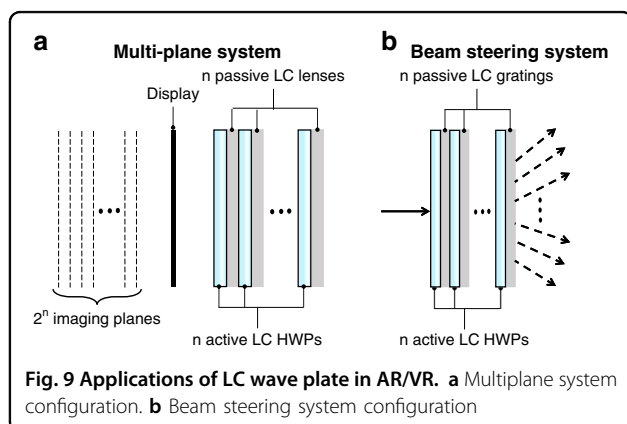
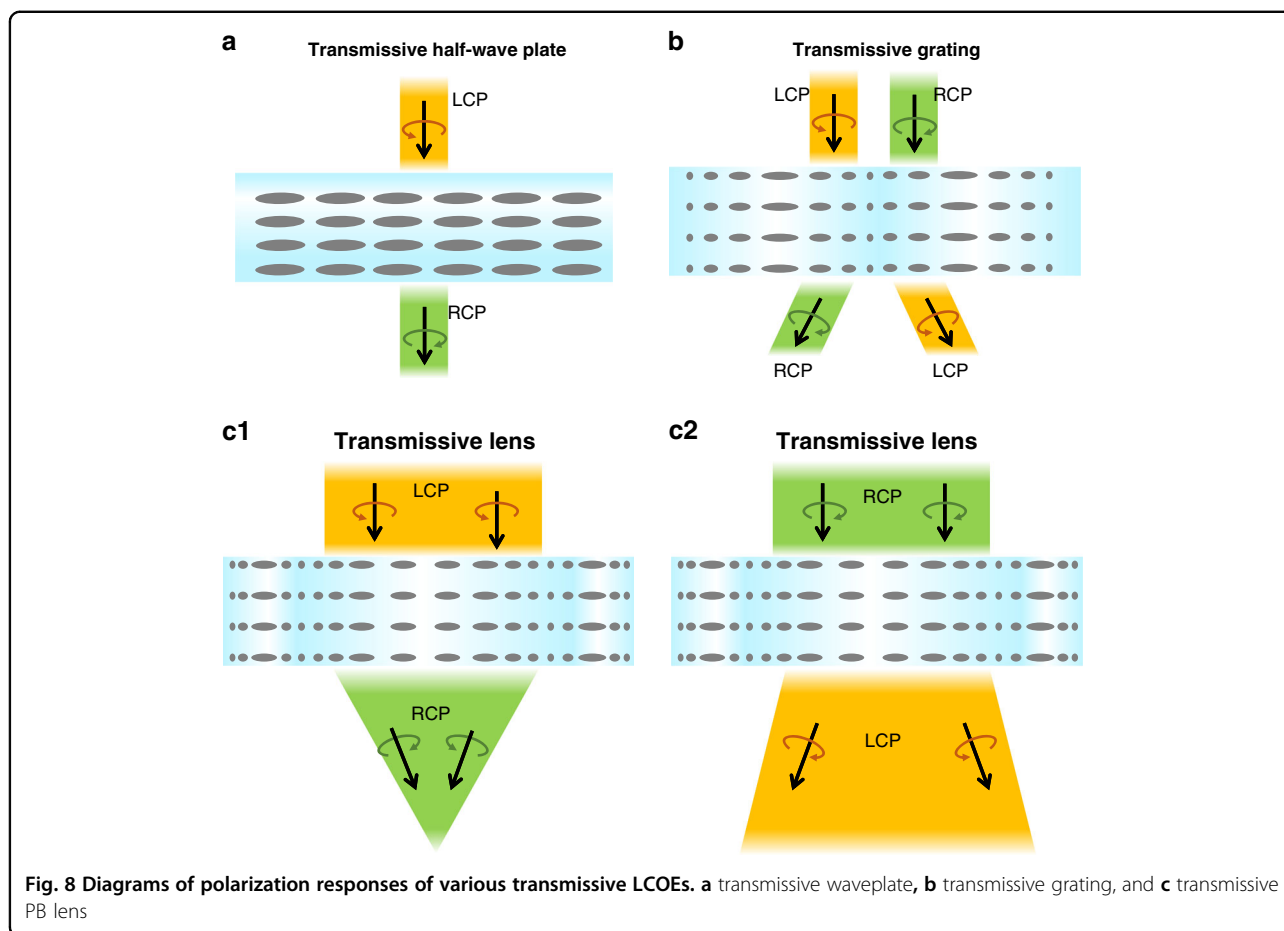
## LC planar optics

In addition to modulating the image as a display or SLM, LCs exhibit some other attractive features, such as polymerization and photo-patternable characteristics<sup>140,141</sup>. These properties can be used to create new photonic devices, called LC planar optics or LC optical elements (LCOEs)<sup>142–144</sup>, which exhibit a ultrathin form factor, nearly 100% efficiency, strong polarization selectivity, and switching ability. These LCOEs show promising applications in AR/VR systems. different from light engine. In this section, we will discuss the LCOEs from transmissive and reflective aspects, respectively.

### Transmissive LC planar optics

In the transmissive LCOEs, we will focus on the geometric phase optical elements, whose phase retardation depends on the LC alignment. This is the main reason that the transmissive type optical element can achieve sharp phase modulation within a thin layer. Being a diffractive optical element, this type of device is working in the Raman-Nath regime, which is regarded as the “thin” optical element (e.g., surface relief grating), and its diffraction efficiency can be maximized at an optimal thickness. Such a transmissive LCOE will show strong polarization selectivity and can work as a waveplate, grating, and lens, respectively, depending on the LC alignments. The polarization responses of each element are summarized in Fig. 8. In the following, we will discuss each device in detail including its working principles and promising applications in AR/VR displays.

Being an anisotropic optical material, LC can introduce phase retardation for a polarized light<sup>145</sup>, thereby it is a promising candidate for waveplates (or phase retarders). Both passive (LC polymer) and active (LC cell) waveplates can be fabricated<sup>146,147</sup>. The passive LC waveplate is a polymer film after UV stabilized process, which is ultrathin and light. The active LC waveplate is switchable between on- and off- states by applying a voltage. The schematic of a simple LC half-waveplate (HWP) is depicted in Fig. 8a. Lu et al.<sup>148</sup> proposed a method to achieve multiplane near-eye display by using active LC waveplates, as shown in Fig. 9a. In the system, the LC-based active HWP works together with a passive LC lens. Since the LC lens is polarization dependent, switching the active HWP will rotate the polarization state, which in turn changes the optical power. If the passive LC lens is replaced by a polarization selective grating, then the focal plane modulation will turn to output beam angle shifting for beam steering, as Fig. 9b shows<sup>149</sup>. This compact beam steering has potential for pupil steering<sup>150</sup> and foveated imaging<sup>151</sup> in AR/VR systems. To have a broadband active waveplate, the simplest device is a TN cell. Moreover, we can fabricate a broadband active waveplate by using a negative-dispersion LC material to compensate for the



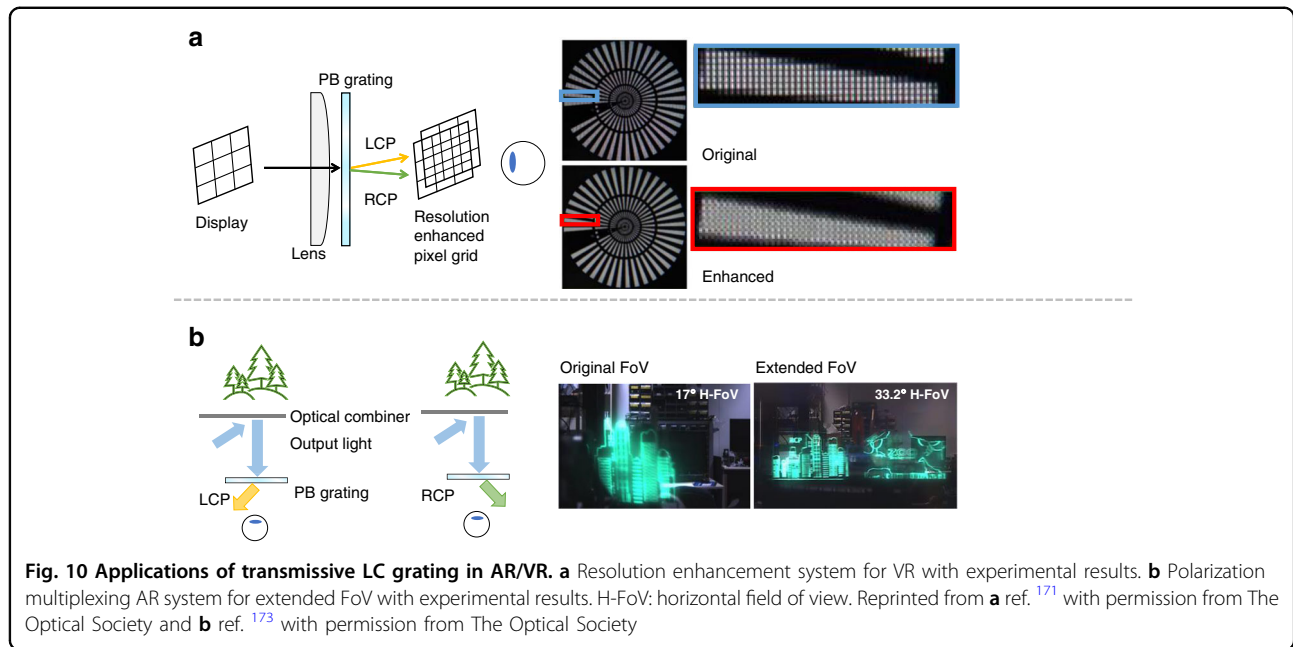
wavelength dependent phase retardation in the visible range. Merck has developed such a negative-dispersion LC monomer (e.g., RMM 1705) for passive LC waveplates. However, for multiplexing and steering applications in the AR/VR systems with an active LC waveplate, the response time needs to be fast ( $\sim 1$  ms). At present, no negative-dispersion LC with such a fast-response time has been reported yet. Another approach is to build multi-twist

structure<sup>152</sup>. According to the requirements of active driving, each twisted layer needs two ITO-glass substrates, which increases the thickness and weight of the system substantially.

An LC grating can be obtained by introducing a linearly varying phase profile to the LC waveplate. Several methods have been developed for generating LC grating patterns: one is to create a periodic electric field by making periodic electrodes<sup>153–155</sup> or periodically modulating the electric field distribution<sup>156</sup>, and another is to provide volume/bulk/surface alignment to the LC molecules<sup>157–164</sup>. Currently, the most widely employed method is surface alignment with interference exposure, which can achieve a smaller grating period and smooth phase change. This method is also suitable for low-cost mass production. In the interference exposure, the grating pattern can be described as:

$$\begin{bmatrix} 1 \\ i \end{bmatrix} e^{-ik_0 x \sin \theta} + \begin{bmatrix} 1 \\ -i \end{bmatrix} e^{ik_0 x \sin \theta} = 2 \begin{bmatrix} \cos(k_0 x \sin \theta) \\ \sin(k_0 x \sin \theta) \end{bmatrix} \quad (1)$$

from Eq. (1), the grating pattern is sinusoidal linear polarization whose diffraction efficiency at normal



incidence is:

$$\eta_0 = \cos^2\left(\frac{\pi\Delta nd}{\lambda}\right), \eta_{\pm 1} = \frac{1 \mp S'_3}{2} \sin^2\left(\frac{\pi\Delta nd}{\lambda}\right) \quad (2)$$

where  $S'_3 = S_3/S_0$  is the normalized Stokes parameter of the incident light. Due to linear phase change of the grating pattern, only the 0<sup>th</sup> and  $\pm 1^{\text{st}}$  orders have nonzero solution, when the grating transfer matrix has the Fourier transform for the far field diffraction<sup>165</sup>. In other words, according to Eq. (2), when the grating thickness satisfies the half-wave condition, all the diffracted light will contribute to the  $\pm 1^{\text{st}}$  orders. Furthermore, if the incident light is circularly polarized, the diffraction efficiency can reach 100% in theory. Under this condition, the light propagation process can be expressed by Jones matrix:

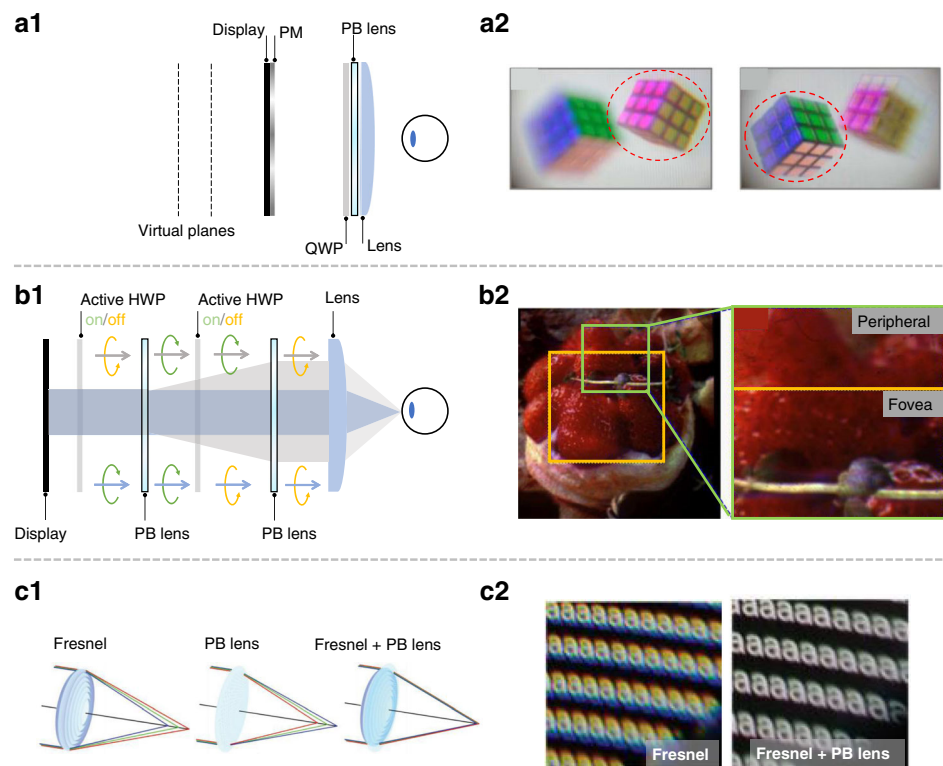
$$J_{HWP} \begin{bmatrix} 1 \\ \pm i \end{bmatrix} = \begin{bmatrix} \cos(2k_0 x \sin \theta) & \sin(2k_0 x \sin \theta) \\ \sin(2k_0 x \sin \theta) & -\cos(2k_0 x \sin \theta) \end{bmatrix} \begin{bmatrix} 1 \\ \pm i \end{bmatrix} \\ = \begin{bmatrix} 1 \\ \mp i \end{bmatrix} e^{\pm i 2k_0 x \sin \theta} \quad (3)$$

as we plotted in Fig. 8b, the incident circularly polarized light will be converted to the opposite handedness with an extra phase, and this phase is also known as geometric phase or Pancharatnam-Berry (PB) phase<sup>166</sup>. PB phase grating has attracted lots of attention because of its nearly 100% diffraction efficiency<sup>149</sup>. It can be achieved with an electro-optically controlled active grating<sup>167,168</sup> or a polymer-based passive grating. Moreover, such a passive PB grating can adopt dual-twist<sup>142,169</sup> and multi-twist<sup>170</sup>

structures to achieve broadband and wide-view performance.

For VR applications, Lee et al.<sup>171</sup> used an active PB grating to enhance resolution by modulating the imaging position with time multiplexing. As a result, the original pixel grid and the shifted pixel grid appear alternatively by frames, thereby the apparent pixel density is doubled (Fig. 10a). Zhan et al.<sup>172</sup> also applied a passive PB grating to realize similar pixel shifting by polarization multiplexing rather than time multiplexing. To precisely control the polarization state, a pixelated polarization rotator is added to align with the original display panel. For AR applications, Yoo et al.<sup>173</sup> proposed a waveguide AR system with an extended FoV by using such a PB grating. As depicted in Fig. 10b, the output light with different circular polarization states is deflected into opposite directions. By combining images from two orthogonal polarizations (LCP and RCP), the FoV is doubled in horizontal direction. Besides, PB phase grating can also be used to enlarge the eyepiece of Maxwellian system. For example, Lin et al.<sup>174</sup> demonstrated a 2D pupil duplication in Maxwellian view by stacking two 1D PB gratings and one QWP together. Such a stacked 1D device can also be replaced by a 2D PB phase grating. Shi et al.<sup>175</sup> proposed a multi-beam interference method to generate the 2D grating pattern, and He et al.<sup>176</sup> also fabricated a PB Damman grating using projection lithography.

An LC lens can be obtained by introducing a parabolically varying phase to the LC waveplate. The lens pattern can also be generated by electric field modulation and pattern alignment, as described in the LC grating section<sup>177–180</sup>. The surface alignment method enables the



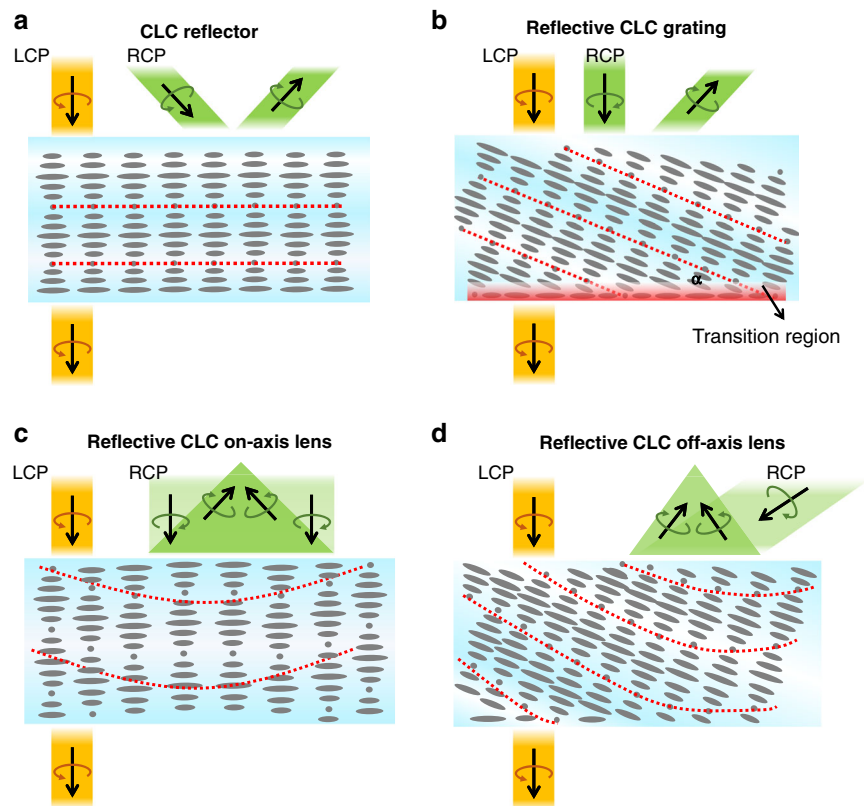
**Fig. 11 Applications of transmissive LC lens in AR/VR.** **a1** Polarization multiplexing based multiplane VR system with **a2** experimental results. **b1** Foveated VR system with **b2** experimental results. **c1** Chromatic aberration correction for VR system with **c2** experimental results. Reprinted from **a2** ref. <sup>185</sup> with permission from The Optical Society, **b2** ref. <sup>186</sup> with permission from The Optical Society, **c2** ref. <sup>188</sup> with permission from John Wiley and Sons

PB phase to generate an LC lens whose optical power is independent of the LC birefringence<sup>181</sup>. Similar to a PB phase grating, the PB phase based LC lens (PB lens)<sup>161,182,183</sup> is also a polarization selective optical element. Its focal length depends on the LCP or RCP input (Fig. 8c). Both passive and active PB lenses can be fabricated using a polymer film or an LC cell, respectively<sup>184</sup>.

Based on this unique polarization dependent focal length, Tan et al.<sup>185</sup> realized a polarization multiplexed multiplane VR display by using a passive PB lens. The working principle of polarization multiplexing is depicted in Fig. 11a1. The system requires a pixelated polarization modulator (PM) to control the ratio between RCP and LCP, allowing independent generation of image content on two virtual planes (Fig. 11a2). Polarization multiplexing is an elegant solution for creating multiplane displays without sacrificing the frame rate, while keeping the headset lightweight for long-term wearing. However, its limitation is that there are only two orthogonal polarization states, and theoretically only two imaging planes can be generated. Yoo et al.<sup>186</sup> presented a foveated display system by using two PB lenses. In this system (Fig. 11b1), the opposite focal lengths of LCP and RCP lead to two modes with high- and low-angular resolutions.

By switching between LCP and RCP, both high and low-angular resolutions can be perceived with different spatial positions, thereby a foveated imaging can be generated (Fig. 11b2). The reported system has an optical efficiency of 12.5%, and both active and passive PB lenses can be used for this system. However, this approach requires a space between lenses for the beams to expand to different diameters. How to accomplish a small form factor could be challenging.

For current near-eye displays, a high optical power lens with compact form factor is required, which usually has significant chromatic aberrations<sup>187</sup> near the edge field. To solve this issue, a hybrid lens (combining a passive diffractive PB lens with a refractive lens, Fig. 11c1) has been proposed. Due to the opposite dispersion, their chromatic aberrations can be compensated each other (Fig. 11c2)<sup>188</sup>. Besides, active PB lens is a good adaptive optics candidate for the AR/VR displays to generate light field display by switching between each polarization responses<sup>189</sup>. Based on recent advance in fast-response LC materials<sup>190</sup>, the switching time of 1–2 ms has been demonstrated in active PB lens or LC HWP. Besides, LC microlens array (MLA)<sup>191,192</sup> and multi-domain LC lens also play important roles in AR/VR systems. One of the



**Fig. 12** Diagrams of polarization responses of various reflective LCOEs. **a** CLC reflector, **b** reflective CLC grating, **c** reflective on-axis CLC lens, and **d** reflective of-axis CLC lens

important applications of MLA is the integral imaging light field display<sup>193</sup>. The system can generate multiple views to reconstruct the real object light field, thereby the user can observe 3D images with both amplitude and angular information. Currently, the reported integral imaging systems use traditional glass MLA because of its good imaging quality and acceptable form factor. Potential applications of LC MLA with tunable focal length and polarization selectivity in integral imaging display systems are foreseeable<sup>183</sup>.

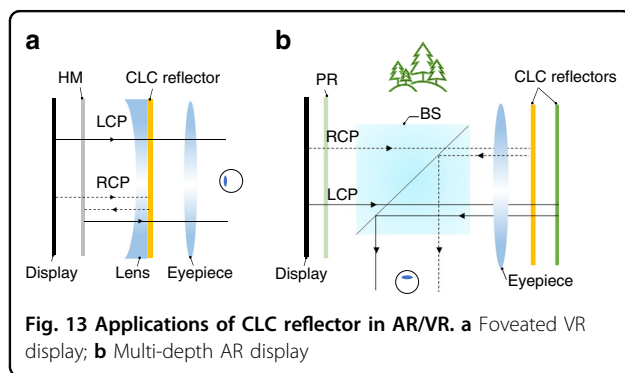
### Reflective LC planar optics

In a reflective LCOE, the reflection mechanism is based on the Bragg reflection of the employed cholesteric liquid crystal (CLC)<sup>194–196</sup>. Therefore, this type of diffractive optical element is called as “thick” optical element (e.g., Bragg volume grating). The efficiency of such a reflective LCOE is closely related to the device thickness. To establish Bragg reflection, the minimal LC layer thickness is about ten pitches. Besides, the helical CLC structure can be achieved by doping some chiral dopants to a nematic LC host<sup>197</sup>. To accommodate a wide viewing angle and broadband, we can apply more complex structures, such

as multilayers and gradient pitch<sup>198–200</sup>. As plotted in Fig. 12, the red dashed lines connect the short axes of the LC directors and represent the Bragg surface. Depending on different Bragg surfaces generated by the LC alignments, such a reflective LCOE exhibits strong polarization selectivity and can work as a reflector, grating, and on-/off-axis lens, respectively. The polarization responses of each element are summarized in Fig. 12.

The CLC reflector (Fig. 12a) has a simple structure but can contribute to various applications for AR/VR. In the VR pancake structure, a CLC reflector can replace the QWP and reflective polarizer to obtain a simpler structure. Furthermore, a foveated VR display can be achieved via encoding two images into orthogonal circular polarizations by using CLC reflector<sup>201</sup>. As shown in Fig. 13a, the images undergoing different paths experience different magnifications, and therefore the spatial resolution of the foveal image could be greatly enhanced. Leveraging the polarization selectivity, CLC reflectors can also generate two optical paths in an AR system. Chen et al.<sup>202</sup> demonstrated a dual-depth AR system by using two CLC reflectors which are separately placed in sequence to produce two different diopters and optical paths, where





each path corresponds to one image depth (Fig. 13b). By employing multiple CLC reflectors, more image depths can be produced.

A reflective LC grating can be obtained by introducing a linearly varying phase to the CLC reflector, called polarization volume grating (PVG). This linearly changed phase can be generated by surface alignment<sup>203–206</sup>. Compared to a CLC reflector, the PVG has a more complex structure, where the azo compounds on the photoalignment layer exhibit a sinusoidal pattern along the horizontal axis. Figure 12b depicts the LC distribution of the PVG structure. The Bragg surface is still the line connecting the short axes of the LC directors but with a slanted angle ( $\alpha$ ). The bottom LC directors tend to follow the sinusoidal pattern, but the helical axis of the bulk CLC is tilted ( $\sim 25^\circ$ ) to minimize the free energy, therefore producing a transition region from the bottom to top. When the incident beam satisfies the Bragg condition, the diffraction angle is twice the slanted angle, and the diffraction efficiency is the highest<sup>207</sup>.

In recent years, the reflective LC diffractive couplers (e.g., PVGs) have also been widely studied and demonstrated in lab-level waveguide-based AR prototypes<sup>208</sup>. Before diving into the PVG-based waveguide structure, we briefly introduce the basic operation principles of the diffractive waveguide AR. As shown in Fig. 14a, the waveguide architecture has diffractive input and out-couplers to deliver the image from light engine to human eye<sup>209</sup>. A large Etendue benefits from the exit pupil expansion (EPE) process. In practicality, the FoV is still a bottleneck, and the EPE is expected to be 2D in consideration of eyebox expansion in both horizontal and vertical directions. The practical FoV is jointly determined by the refractive index ( $n_g$ ) of the waveguides and the angular bandwidth provided by the couplers. A large  $n_g$  waveguide with wide angular response of the couplers is conducive to expand the FoV.

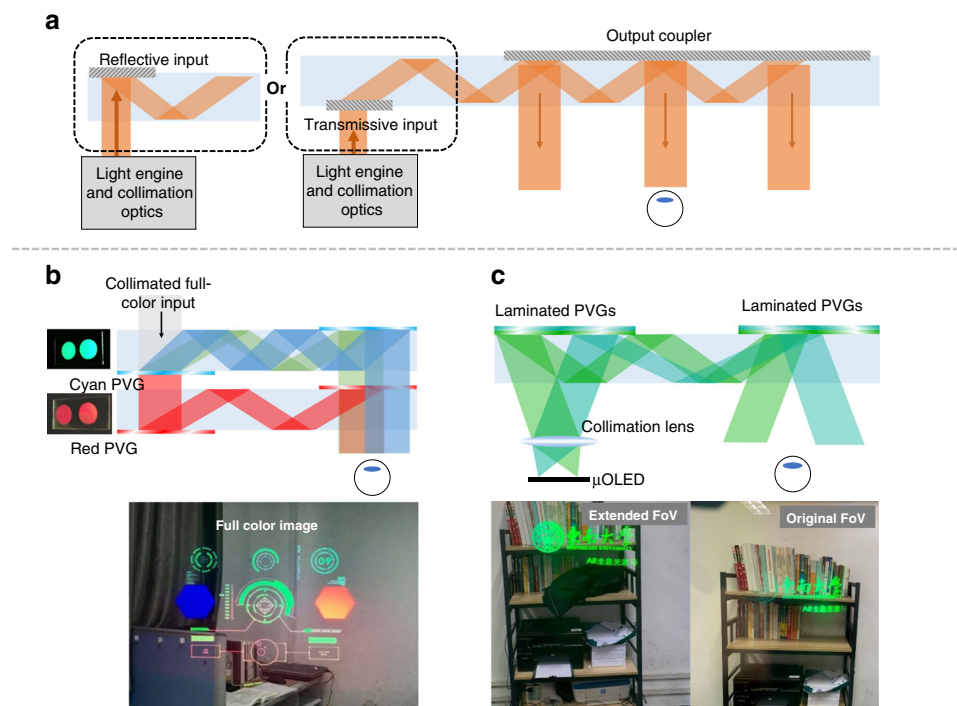
Currently, surface relief gratings (SRGs) are dominant couplers in waveguide AR products. SRGs have a large index modulation to produce wide angular response and

can achieve a uniform EPE, although the design is somewhat sophisticated<sup>210</sup>. As the input coupler, both transmissive and reflective SRGs can be considered (Fig. 14a). For the output coupler, a reflective type with suppressed transmission orders is preferred<sup>211</sup>. Compared to SRGs, PVGs exhibit a smaller index difference, but spatially variant LC orientations, strong circularly polarized responses, thin-film form factor, and easy fabrication process. To obtain a uniform light output, a polarization management layer consisting of an LC layer with spatially varying orientations can be utilized<sup>212</sup>. It also offers an additional degree of freedom to control the polarization state of the TIR light.

In the following, we will review some methods employing PVGs as the couplers in the waveguides. To display full-color images, it is feasible to stack R/G/B PVGs as the in- and out-couplers on one waveguide. Weng et al.<sup>213</sup> proposed a dual-waveguide structure for a full-color AR. The blue and green PVGs are laminated together on one waveguide, and the red PVG is put solely on the other waveguide (Fig. 14b). Aside from full-color images, for practical applications, it is necessary to take the PVG's angular response into account to analyze the FoV, especially focusing on the angular bandwidth with  $0^\circ$  as the symmetric center for the in- and out-couplers. The analysis with simulation parameters is listed in Supplementary S1. The angular response of a PVG can be adjusted by two methods: changing the LC birefringence and stacking several layers with different Bragg pitches<sup>208</sup>. Recently, Gu et al.<sup>214</sup> experimentally demonstrated this approach based on stacking two PVGs with opposite polarization responses to extend the angular response at the target wavelength, thereby enlarged the observed FoV (Fig. 14c).

In addition to enlarging the angular response, in a waveguide AR, a uniform eyebox is also critical. Generally, there are two methods to create a uniform eyebox. The first solution is to manipulate the local efficiency at different out-coupling regions on the PVG<sup>20</sup>. Nevertheless, the gradient efficiency of PVG (out-coupler) will cause uneven ambient light transmittance to the user's eye. Another solution is to use a PVG with uniform efficiency but attach an LC-based polarization modulation layer (PML) to the bottom of the waveguide<sup>212</sup>. By modulating the azimuthal angles of the LC directors in the PML, the phase control and thus the polarization control can be achieved. As a result, the output light intensity could be uniform because the reflection efficiency of the PVG is dependent on the light polarization.

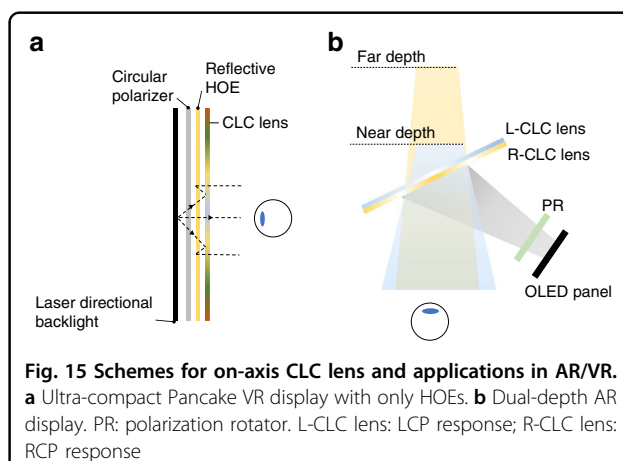
A reflective LC lens can be obtained by introducing a parabolically varying phase to the CLC reflector. The lens phase profile can be superposed onto either planar-CLC or slanted-PVG pattern to form an on-axis or off-axis CLC lens (Fig. 12c, d). The CLC lens still follows the



**Fig. 14 Schemes of diffractive waveguide combiners and applications of PVG in waveguide AR.** **a** Working principles of waveguide AR (transmissive input coupler and reflective input coupler). **b** System configuration of full-color waveguide AR display with experimental results. **c** Enlarged-FoV waveguide AR display with experimental results. Reproduced from **b** ref. <sup>213</sup> with permission from The Optical Society. Reprinted from **c** ref. <sup>214</sup> under the Creative Commons Attribution 4.0 License

polarization selectivity rule as the CLC reflector but exhibits opposite phase profiles when the incident light impinges from different sides<sup>215,216</sup>. In other words, the incident light with the target polarization can be converged or diverged, depending on which side the beam enters. Leveraging the polarization selectivity of CLC, multiple CLC films can be stacked together to perform different functionalities for a wide range of applications.

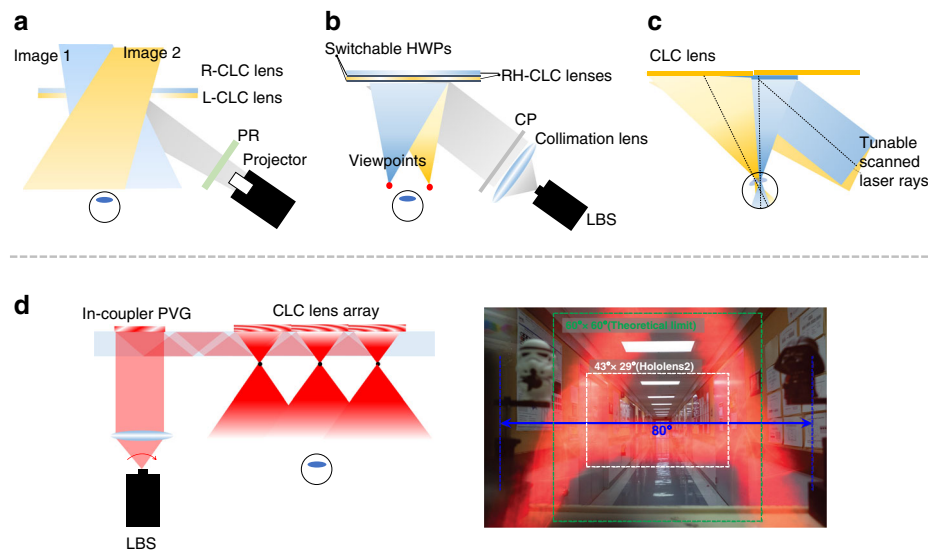
For a VR system, the CLC reflector in pancake design can be further replaced by an on-axis CLC lens. Such a CLC lens provides more degrees of freedom for optimizing the optical aberrations<sup>217</sup>. The combination of refractive and diffractive elements suppresses the chromatic aberrations due to their opposite dispersion behaviors, like the result shown in Fig. 11c. The pancake structure can be further evolved to a more compact form factor, if all the refractive optical elements are replaced by the HOEs<sup>51</sup> (Fig. 15a). To accommodate the angular selectivity and wavelength selectivity of the HOEs, a monochromatic directional backlight (e.g., 532 nm laser source) is preferred as the light engine. Besides, the on-axis CLC lenses can be implemented in projection AR systems (Fig. 1b) to serve as optical combiners. Li et al.<sup>218</sup> proposed a dual-depth AR system<sup>219,220</sup> by tilting the CLC lens at 22.5° relative to the eye pupil (Fig. 15d). Since the



**Fig. 15 Schemes for on-axis CLC lens and applications in AR/VR.** **a** Ultra-compact Pancake VR display with only HOEs. **b** Dual-depth AR display. PR: polarization rotator. L-CLC lens: LCP response; R-CLC lens: RCP response

polarization dependent CLC lenses could distinguish two circular polarizations and apply different diopters, two image depths can be obtained.

For off-axis CLC lenses, Yin et al.<sup>221</sup> proposed a two off-axis CLC lenses based multiplexing system for projection AR systems (Fig. 1b) to expand the FoV. As shown in Fig. 16a, each lens responds to a different circular polarization and works for a specific FoV. By combining the two views, an expanded tiled FoV can be presented in



**Fig. 16 Schemes for off-axis CLC lens and applications in AR system.** **a** Doubled FoV AR display; **b** Pupil steering Maxwellian AR display; **c** Gaze-matched pupil steering Maxwellian display. **d** Scanning waveguide AR display. LBS: laser beam scanning. Reproduced from **d** ref. <sup>229</sup> with permission from The Optical Society

front of the observer<sup>222</sup>. It should be mentioned that the image aberration could be caused by the dispersion of diffractive elements and large off-axis angle<sup>223</sup>, especially with a wideband light source. To mitigate these aberrations, digital compensation or a light source with narrow bandwidth can be adopted. The off-axis CLC lenses can also serve as the diffractive couplers in Maxwellian displays<sup>224–226</sup>. Maxwellian view has a tiny eyebox, which is usually assumed to be the pupil size (3–4 mm). The implementation of CLC lenses could perform pupil duplication or pupil steering to expand the eyebox. Pupil duplication refers to the scenario in that multiple viewpoints are formed simultaneously to cover a larger eyebox. The light intensity in each viewpoint is determined by the total light intensity divided by the viewpoint numbers. For pupil steering, the viewpoint position is timely shifted in response to the pupil location detected by the eye-tracking technique. This approach offers an excellent optical efficiency. Xiong et al.<sup>224,227</sup> proposed a pupil steering structure by using Maxwellian system with off-axis CLC lenses (Fig. 16b). Each CLC lens is coupled with a switchable HWP. In the voltage-on state, the HWP converts the LCP light into RCP light, which is reflected by the first CLC lens and steered into the first viewpoint. As the pupil moves to a second viewpoint, the first HWP is turned off and the second HWP is switched on. As a result, the second CLC lens reflects the image toward the pupil. More viewpoints can be accommodated by stacking more CLC lenses and using ultrathin ( $\sim 100\ \mu\text{m}$ ) glass substrates to keep the package compact and lightweight.

Apart from generating or switching between several viewpoints, a gaze-matched function can also be realized to accommodate the eye rotation in a Maxwellian system. In the scenario of gaze-matching, the exit pupil should be aligned with the ocular pupil. In other words, the chief ray should match with the eye gazing direction. To meet this requirement, the incident light angle on the imaging couplers should be modulated in real time. Kim et al.<sup>228</sup> proposed a mechanical shifter to shift the horizontal position of the imaging coupler. This approach can directly achieve gaze-matching; however, the mechanical shifter increases the complexity and weight of the system. The aberration will also arise when the same holographic lens coupler is shifted, since the probing wave on the holographic lens coupler fails to match the wavefront of the reference wave used in the exposure process. Zou et al.<sup>150</sup> developed a gaze-matched Maxwellian system by employing multiple off-axis CLC lenses with different patterns to accommodate different incident light directions. As shown in Fig. 16c, three lenses are designed for three exit pupils to align with the ocular pupil positions when the eye rotates. Since each lens can be specifically designed for one incident angle, the aberration will be negligible. However, this approach could become bulkier if more viewpoints are required to cover a large range of eye rotation. To make it reasonably practical, some compromise can be made between the viewpoint numbers and the degree of gaze-matching. Although the Maxwellian display is limited to a tiny eyebox, recalling that the waveguide framework can support a large eyebox. By

combining these two advantages together, Xiong et al.<sup>229</sup> proposed a scanning waveguide display to achieve a wide horizontal FoV ( $\sim 80^\circ$ ) by deploying an off-axis CLC lens array as the out-coupler (Fig. 16d). The collimating light is outcoupled by the off-axis CLC lens array and then forms multiple viewpoints. In this framework, the FoV is no longer constrained by the refractive index of the waveguide but solely decided by the  $f$ -number of each lens element. Because of the robust CLC alignment, the off-axis CLC lens can reach a small  $f$ -number ( $\sim 0.6$ ).

## Conclusion and outlook

AR/VR technologies combined with advanced LC devices have emerged as a significant new direction for novel displays and information platforms, with potential for a wide range of applications. To enable a compact form factor and high image quality in wearable AR/VR headsets, advanced LC devices play important roles, ranging from light engines to various functional optics. In this review, we divide these advanced LC devices into three categories: high-dynamic-range LCDs, LCoS including amplitude and phase modulators, and planar optics. Although LCDs for VR headsets suffer from low transmittance issues caused by small aperture ratios and disclination lines blocking light, achieving a high-performing VR that is compatible with directional backlight and spatial patterned modulator techniques represents a promising solution. At the same time, LCoS SLMs offer unrivaled phase modulation to realize holographic views beyond conventional displays. Further advancements in device engineering and fabrication processes are expected to improve the performance of LCDs and LCoS SLMs in AR/VR applications. Emerging LC planar optics provide excellent optical properties with an ultrathin form factor and high efficiency. These advanced LC-based devices play pivotal roles for systematically improving the image quality and form factor of the AR/VR displays.

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## Author contributions

K.Y. conceived the idea and initiated the project. K.Y. mainly wrote the manuscript and conducted the experiment. E.-L.H., J.Z., Y.L. Z.Y., and Q.Y. helped with the experiment. P.-C.L. conducted the power consumption simulation. C.-L.L., and S.-T.W. supervised the project and edited the manuscript.

## Data availability

All data needed to evaluate the conclusions in the paper are present in the paper. Additional data related to this paper may be requested from the authors.

## Conflict of interest

The authors declare no competing interests.

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# Resize Me! Exploring the user experience of embodied realistic modulatable avatars for body image intervention in virtual reality

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Obesity is a serious disease that can affect both physical and psychological well-being. Due to weight stigmatization, many affected individuals suffer from body image disturbances whereby they perceive their body in a distorted way, evaluate it negatively, or neglect it. Beyond established interventions such as mirror exposure, recent advancements aim to complement body image treatments by the embodiment of visually altered virtual bodies in virtual reality (VR). We present a high-fidelity prototype of an advanced VR system that allows users to embody a rapidly generated personalized, photorealistic avatar and to realistically modulate its body weight in real-time within a carefully designed virtual environment. In a formative multi-method approach, a total of 12 participants rated the general user experience (UX) of our system during body scan and VR experience using semi-structured qualitative interviews and multiple quantitative UX measures. Using body weight modification tasks, we further compared three different interaction methods for real-time body weight modification and measured our system's impact on the body image relevant measures body awareness and body weight perception. From the feedback received, demonstrating an already solid UX of our overall system and providing constructive input for further improvement, we derived a set of design guidelines to guide future development and evaluation processes of systems supporting body image interventions.

## KEYWORDS

virtual reality, avatar embodiment, user experience, body awareness, body weight perception, body weight modification, body image disturbance, eating and body weight disorders



# 1 Introduction

Obesity is a complex chronic disease characterized by severe overweight and an above-average percentage of body fat (World Health Organization, 2019). Its prevalence has more than doubled within recent decades and is expected to rise (Venegas and Mehrzad, 2020; World Health Organization, 2021). Besides the physical burdens (e.g., an increased risk of several secondary diseases (Stefan et al., 2021), affected individuals deal with an external or internalized stigmatization that can lead to body image disturbances (Thompson and Tantleff-Dunn, 1998; Rosen, 2001; Meadows and Calogero, 2018). Body image disturbances are composed of a misperception of body dimensions (body image distortion) and the inability to like, accept, or value one's own body (body image dissatisfaction) and are also associated with a reduced body awareness (Todd et al., 2019b; Turbyne et al., 2021). Various interventions (e.g., cognitive-behavioral therapy supported by mirror exposition or fitness training) have been designed to target persisting disturbances but often only achieve small improvements in the body image (Alleva et al., 2015). In recent years, novel virtual reality (VR)-based methods complementing the therapy of body image disturbances have successfully been explored in research with promising results (Ferrer-Garcia et al., 2013; Wiederhold et al., 2016; Riva et al., 2019). The further improvement of these approaches in the context of obesity forms the frame of our current work.

VR-based approaches for supporting body image interventions often use 3D models of human beings (Horne et al., 2020; Turbyne et al., 2021), so-called avatars (Bailenson and Blascovich, 2004). VR in general, and the confrontation with embodied avatars in particular, have great potential to influence human perception and behavior (Yee and Bailenson, 2007; Ratan et al., 2020; Wienrich et al., 2021). In the context of body image, avatars have been utilized to expose users of a VR system to generic virtual bodies or body parts varying in size or shape to investigate the principles of body weight perception (Thaler, 2019; Wolf et al., 2020, 2021, 2022a) or to influence the perception or attitude towards the user's own body (Turbyne et al., 2021). Recent developments in computer graphics allow for the generation of photorealistic avatars that match a person's real-life appearance within a short duration at a low-cost (Achenbach et al., 2017; Wenninger et al., 2020; Bartl et al., 2021) and for a realistic modulation of body dimensions in pictures and videos (Zhou et al., 2010; Zhao et al., 2018; Xiao et al., 2020; Tang et al., 2021) or in VR (Piryankova et al., 2014a; Hudson et al., 2020; Maalin et al., 2020). However, no work has yet been presented where users embody their photorealistically personalized avatar in VR while also having the ability to manipulate that avatar's body shape in real-time actively, nor has the impact of such a system on the users and their experiences been evaluated.

To address this gap, we present the development of a VR system allowing users to embody a photorealistic, personalized

avatar within a virtual environment and to actively modify its body weight in real-time using different interaction methods. In a further step, we evaluated the system with regard to a later usage in clinically relevant settings within our research project Virtual Reality Therapy by Stimulation of Modulated Body Perception (ViTraS) (Döllinger et al., 2019). In particular, we performed a formative user evaluation of the avatar generation process and interactive VR exposure with a small sample of healthy participants. Following Wienrich and Gramlich (2020) and considering the future potential user group, we assessed relevant factors such as security, physical comfort, accessibility, usability, and user experience, which we also already considered during the development process. Based on our evaluation's results, we derive a set of design guidelines for the future design and development of similar avatar-based body image therapy support tools.

# 2 Related work

Body image disturbance is characterized by an "excessively negative, distorted, or inaccurate perception of one's own body or parts of it" (World Health Organization, 2019). It may manifest in body image distortion, the misperception of one's body weight and dimensions that have repeatedly been reported based on underestimations (Valtolina, 1998; Maximova et al., 2008) or overestimations (Docteur et al., 2010; Thaler et al., 2018a), or body image dissatisfaction, a negative attitude towards the body that is associated with body image avoidance (Walker et al., 2018) and reduced body awareness (awareness for bodily signals) (Peat and Muehlenkamp, 2011; Todd et al., 2019a,b; Zanetti et al., 2013). While often caused by internalized weight stigma and a fear of being stigmatized by others (Meadows and Calogero, 2018), body image disturbance interferes with efforts to stabilize body weight in the long term (Rosen, 2001). Treatments for body image disturbance mainly rely on cognitive-behavioral therapy, typically combining psychoeducation and self-monitoring tasks, mirror exposure, or video feedback (Farrell et al., 2006; Griffen et al., 2018; Ziser et al., 2018). Based on the fundamentals of these established methods, an increasing number of researchers have started to explore VR applications as additional support for attitude and behavior change in general (Wienrich et al., 2021) and therapy of body image disturbance (Riva, 1997; Ferrer-Garcia et al., 2009, 2013; Riva et al., 2019; Turbyne et al., 2021) and obesity in particular (Döllinger et al., 2019; Horne et al., 2020).

## 2.1 The unique potential of modulatable avatars in VR

VR offers the opportunity to immerse oneself in an alternative reality and experience scenarios that are otherwise

only achievable via imagination. Endowed with this unique power, mainly the use of avatars has attracted attention in treating body image disturbance (Horne et al., 2020; Turbyne et al., 2021). Image processing methods for simulating body changes are well established. Using parametric models, it is possible to retouch images to simulate different face or body shapes (Zhou et al., 2010; Zhao et al., 2018) and even manipulate them in real-time during video playback (Xiao et al., 2020; Tang et al., 2021). Avatars in VR allow simulating rapid changes in body shape or weight in an immersive environment using life-sized avatars going beyond the presentation of pictures and videos. They enable further general investigation of body weight perception (Thaler, 2019; Wolf et al., 2020, 2021). While some researchers are using multiple generic avatars differing in body weight (Normand et al., 2011; Piryankova et al., 2014b; Keizer et al., 2016; Ferrer-Garcia et al., 2018; Preston and Ehrsson, 2018), others have developed methods for dynamic body weight modification in VR (Alcañiz et al., 2000; Johnstone et al., 2008; Piryankova et al., 2014a; Nimcharoen et al., 2018; Hudson et al., 2020; Maalin et al., 2020; Neyret et al., 2020). A huge advantage when using advanced body weight modification methods is that the avatar's body weight can be realistically changed to a desired numeric reference value. For this purpose, mainly the body mass index calculated as  $BMI = \text{Body Weight in kg} / (\text{Body Height in m})^2$  (World Health Organization, 2000) is used. One example is the work of Thaler et al. (2018a), who trained a statistical model to apply realistic BMI-based body weight modification to their generated personalized, photorealistic avatars. But also other factors like muscle mass could be included in such models (Maalin et al., 2020). However, while picture and video-retouching methods tend to focus on facial features, the statistical models of weight gain/loss of avatars in VR are usually trained on the whole body (Piryankova et al., 2014a) or neglect the head region completely (Maalin et al., 2020). For our system, we also learned a statistical model of weight gain/loss for the head region but kept small parts of the face region fixed to preserve the identity of the users when applying the body weight modification.

Besides the shape of the used avatar, application or system-related properties also might alter how we perceive the avatar, and particularly its body weight, in VR. Wolf et al. (2020) presented an overview of potentially influencing factors, noting that while the used display or the observation perspective might unintentionally alter body weight perception (Wolf et al., 2022b), especially the personalization and embodiment of avatars hold potential for application in body image interventions. For example, Thaler et al. (2018a) found that the estimator's BMI influences body weight estimations of a realistic and modulatable avatar, but only when the avatar's shape and texture matched the estimator's appearance. This comes along with a recent review by Horne et al. (2020), who identified the personalization of avatars as an important factor when using avatars. For

embodiment, Wolf et al. (2021) recently found, for example, that females' own BMI influences body weight estimations of a generic avatar only when embodying it.

## 2.2 User experience of a VR-based body image intervention

In the design process of a VR application, it is of utter importance to test the system's user experience (UX). UX refers to the sum of all perceptions and reactions of a user to the interaction with an interface before, during, and after its use (International Organization for Standardization, 2019). It combines a variety of hedonic qualities, such as the joy of the user during an experience, and pragmatic qualities, such as the efficiency of interactions. Concerning the UX evaluation of VR systems, it is suggested to include the assessment of further VR-specific variables (Tcha-Tokey et al., 2016; Wienrich and Gramlich, 2020), namely simulation sickness (Kennedy et al., 1993), the feeling of presence (Slater, 2009), and the feeling of embodiment (Kilteni et al., 2012). Concerning avatar-based body image interventions, particularly the user's feeling of embodiment towards their avatar is of interest (Turbyne et al., 2021). It can be evoked by visuomotor congruence, for example, when the user sees the avatar moving like their real body (Slater et al., 2009, 2010) and is divided into the feeling of being inside (self-location), of owning (virtual body ownership), and of controlling (agency) an avatar (Kilteni et al., 2012).

In addition to a system's classical UX evaluation, it is important to embed the development into an iterative design process. This typically involves understanding and establishing the context of use, defining the requirements for use, developing prototypes, and an iterative evaluation. Wienrich and Gramlich (2020) recently presented the *appRaiseVR* framework for UX evaluation in VR, which adapts the general process of UX design to the context of VR. In their VR-adapted design cycle, they include four steps: 1) defining the setting of the experience, including the details of the system, the planned usage context, and the target user group; 2) defining the level of evaluation, including either an evaluation of the system itself, the task, the narrative, the effect on the user, or the relation between different users; 3) rating the plausibility of the experience, namely its realism, its virtual and physical components; and 4) selecting the time of measure, whether evaluating the expectancy towards a system, the immediate reaction within the experience a post-experience evaluation or follow-up assessments.

Considering this framework, our research evaluates a highly immersive VR system, including a realistic environment and photorealistic, modulatable avatars (1, 3). The design aims at a realistic, clinical setting with a target group dealing with obesity and a disturbed body image (1). Based on our target group, our

evaluation focuses on security, physical comfort, and accessibility of our system, next to VR-specific UX and usability of different interaction tasks and the plausibility of avatar modifications (2), which we test during and after the experience (4).

In the context of our application, we further define the effects on body weight perception and physical body awareness during the use as essential parts of the users' experience. For example, Riva et al. (2019) stated that embodiment could potentially help update the misperception of body dimensions by experiencing the ownership over a differently shaped or sized avatar. This goes along with a recent review by Turbyne et al. (2021), summarizing that participants' body image conformed to a modified virtual body size when participants felt embodied in it. VR further interferes with the user's physical body awareness. Filippetti and Tsakiris (2017) showed an increase in body awareness when embodying an avatar for individuals with initially low body awareness. Döllinger et al. (2022) revealed that especially the feeling of body ownership towards a personalized avatar is positively related to body awareness. However, there is no research on body awareness in a VR-based body image treatment task.

## 2.3 User interaction for body weight modification

Most VR applications for body image interventions aim for enhanced mirror confrontation. They surpass real mirror confrontation by modifying the mirror image or the shown avatars into different body shapes. In our system, we want to go one step further and allow users to adjust the shape of their avatar interactively. Our idea is to give users the opportunity to actively engage in analyzing their body image and develop a novel feeling for their own body. Object manipulation in VR has been widely researched and can serve as a reference in the development of body weight modification interaction methods. For example, LaViola et al. (2017) presented a set of design guidelines for different types of object manipulation, including object scaling by virtual buttons or other control elements, the inclusion of physical interfaces as provided on most VR controllers, or the design of gesture-based object manipulation. Furthermore, Williams et al. (2020) and Wu et al. (2019) investigated the preference of users towards different gestures in object manipulation, and both proposed using two-handed gestures (e.g., moving the hands apart or bringing them together) for size modification of large objects.

## 2.4 Summary and outline

VR in general, and the embodiment of modulatable avatars in specific, hold great potential to innovate interventions for body image

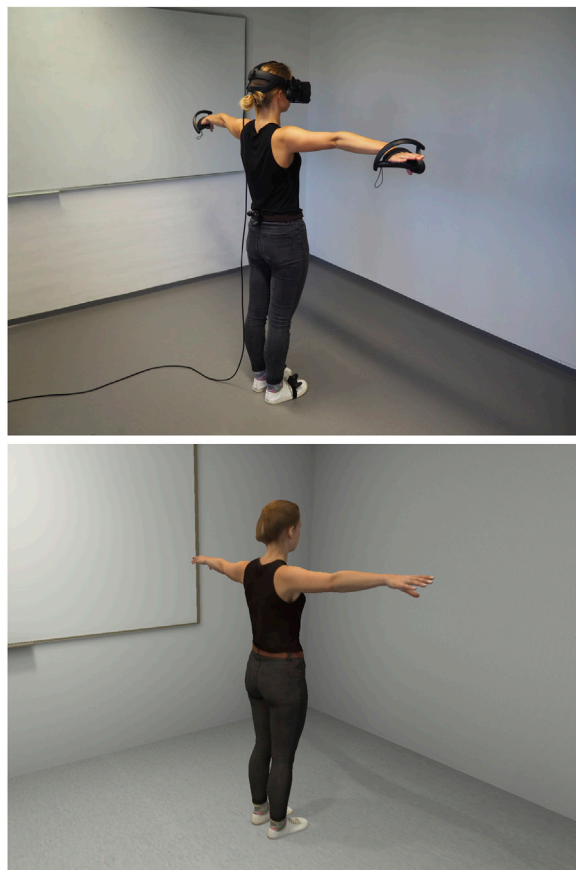
disturbances. The introduced research shows that there exist promising developments toward avatar-based interventions for body image disturbances. However, no work to date undertakes a comprehensive VR-specific UX investigation of such an intervention system.

Our current work within the interdisciplinary research project ViTraS (Döllinger et al., 2019) addresses this research gap and aims toward a novel approach for supporting body image therapy. We present a high-fidelity prototype of a body image therapy support system that allows users to embody their rapidly generated personalized, photorealistic avatar within a carefully designed virtual environment. Our system allows users to dynamically alter their body weight while being embodied in VR using three different interaction methods (joystick, gestures, and virtual objects). We focus on a user experience evaluation with normal-weight participants performed within our first design cycle. In a comprehensive mixed-methods evaluation, we assessed 1) the body scan experience during the avatar generation process, 2) the general, VR-specific UX of the exposure and different modification methods, and 3) their impact on body image-related UX, including body awareness and body weight perception. To sum up our results, we derive a set of guidelines for the design and implementation of future VR systems supporting body image interventions.

## 3 System description

The technical implementation of our system is realized using the game engine Unity version 2019.4.15f1 LTS (Unity Technologies, 2019). As VR HMD, we use a Valve Index (Valve Corporation, 2020a), providing the user a resolution of  $1440 \times 1600$  pixels per eye with a total field of view of  $120^\circ$  running at a refresh rate of 90 Hz. For motion tracking, we use the two handheld Valve Index controllers, one HTC Vive Tracker 3.0 positioned on a belt at the lower spine, and two HTC Vive Tracker 3.0 on each foot fixed by a velcro strap. The tracking area was set up using four SteamVR Base Stations 2.0. All VR hardware is integrated using SteamVR in version 1.16.10 (Valve Corporation, 2020b) and its corresponding Unity plugin in version 2.7.3.<sup>1</sup> In our evaluation, the system was driven by a high-end PC composed of an Intel Core i7-9700K, an Nvidia RTX2080 Super, and 32 GB RAM running Windows 10. To ensure that users always received a fluent VR experience and to preclude a possible cause of simulator sickness, we measured the motion-to-photon latency of our system by frame-counting (He et al., 2000; Stauffert et al., 2021). For this purpose, the video output was split into two signals using an Aten VanCryst VS192 display

<sup>1</sup> <https://assetstore.unity.com/packages/tools/integration/steamvr-plugin-32647>.



**FIGURE 1**

The figure depicts a comparison between the real environment where the experiment took place (left) and the replicated virtual environment used for preparation (right). Both environments contain a user, respectively the avatar, performing the embodiment calibration.

port splitter. One signal still led to the HMD, the other to an ASUS ROG SWIFT PG43UQ low-latency gaming monitor. A high-speed camera of an iPhone 8 recorded the user's motions and the corresponding reactions on the monitor screen at 240 fps. The motion-to-photon latency for the HMD matched the refresh rate of the used Valve Index closely, as it averaged 14.4 ms ( $SD = 2.8$  ms). The motion-to-photon latency for the body movements was considered low enough to provide a high feeling of agency towards the avatar (Waltemate et al., 2016), as it averaged 40.9 ms ( $SD = 5.4$  ms).

### 3.1 Virtual environments

We realized two virtual environments. The first environment replicates the real environment, in which the user was located physically during our evaluation, and which

is automatically calibrated accurately to overlay the physical environment spatially (see Figure 1). Here, all preparatory steps required for exposure are performed and tested (e.g., ground calibration, vision test, equipment adjustments, embodiment calibration). For spatial calibration, we use a customized implementation of the Kabsch algorithm<sup>2</sup> (Müller et al., 2016), based on the positions of the SteamVR base stations in real and virtual environments. Additionally, the virtual ground height is calibrated by briefly placing the controller onto the physical ground.

The second environment is originally based on an asset taken from the Unity Asset Store<sup>3</sup> that was modified to match our requirements. This exposition environment is inspired by a typical office of a psychotherapist with a desk and chairs and an exposure area in which the mirror exposure takes place (see Figure 2). The exposure area includes a virtual mirror allowing for an allocentric observation of the embodied avatar. We aimed for a realistic and coherent virtual environment to enhance the overall plausibility of the exposure (Slater, 2009; Latoschik and Wienrich, 2022).

### 3.2 Generation and animation of personalized avatars

In our system, the user embodies a personalized avatar from an egocentric perspective while the avatar is animated according to the user's body movements in real-time. The following sections describe the generation of the avatars as well as the animation system.

#### 3.2.1 Generation process

The generation of the avatars closely follows the method of Achenbach et al. (2017). First, the subject is scanned with a custom-made photogrammetry rig consisting of 94 DSLR cameras, where four studio lights equipped with diffuser balls ensure uniform illumination (Bartl et al., 2021). Instead of employing a separate face scanner like Achenbach et al. (2017) did, ten of the 94 DSLR cameras are zoomed in on the subject's face to capture more detail in this region. The images taken by the cameras are then automatically processed with the commercial photogrammetry software Agisoft Metashape (Agisoft, 2021), resulting in a dense point cloud of the subject. We manually select 23 landmarks on the point cloud in order to guide the subsequent template fitting process. The counterparts of these landmarks are pre-selected on the

<sup>2</sup> <https://github.com/zalo/MathUtilities/#kabsch>.

<sup>3</sup> <https://assetstore.unity.com/packages/3d/props/interior/manager-office-interior-107709>.





**FIGURE 2**

The images show a participant's personalized avatar standing in front of a mirror within the virtual exposition environment of our concept prototype with a reduced (left), normal (center), or increased (right) body weight.

template model, which comes from the Autodesk Character Generator (Autodesk, 2014) and consists of  $N \approx 21k$  vertices, an animation skeleton with skinning weights, facial blendshapes, as well as auxiliary meshes for eyes and teeth. Achenbach et al. (2017) enhance the template with a statistical model of human shape variation. This statistical, animatable human template model is then fitted to the point cloud by optimizing for alignment, pose, and shape by employing non-rigid ICP (Bouaziz et al., 2014). This optimization of the model parameters defines the initial registration of the template, which is then further refined by allowing fine-scale deformation of the vertices to match the scanner data more closely. For more details, we refer to Achenbach et al. (2017).

### 3.2.2 Animation process

For avatar animation, the participants' movements are continuously captured using the SteamVR motion tracking devices. The tracking solution provides for our work a sufficiently solid and rapid infrared-based tracking solution for the crucial body parts required for animation without aligning different tracking spaces (Niehorster et al., 2017). To calibrate the tracking devices to the user's associated body parts and capture the user's body height, arm length, and current limb orientations, we use a custom-written calibration script that requires the user to stand in T-pose for a short moment (see Figure 1). The calibrated tracking targets of the head, left hand, right hand, pelvis, left foot, and right foot were then used to drive an Inverse Kinematics (IK) (Aristidou et al., 2018) approach realized by the Unity plugin FinalIK version 2.0.<sup>4</sup> FinalIK's integrated VRIK solver continuously calculates the user's body pose according to the provided tracking targets. The tracking

pose is automatically adjusted to the determined body height and arm length in order to match the user's body. In the next step, the tracking pose is continuously retargeted to the imported personalized avatar. Potentially occurring inaccuracy in the alignment of the pose or the end-effectors can be compensated by a post-retargeting IK-supported pose optimization step. This leads to high positional conformity between the participant's body and the embodied avatar and avoids sliding feet due to the retargeting process.

## 3.3 Body weight modification of avatars

Our system allows the user to modify their body weight during runtime dynamically. The statistical model of weight gain/loss and the implemented user interaction methods are described in the following.

### 3.3.1 Data-driven body weight modification

To build a statistical model of body weight modification, we follow the approach of Piryanova et al. (2014a), who first create a statistical model of body shape using Principal Component Analysis (PCA) and then estimate a linear function from anthropometric measurements to PCA coefficients. For computing the statistical model of human body shape, we use the template fitting process described above to fit our template model to the European subset of the CAESAR scan database (Robinette et al., 2002). It consists of  $M = 1700$  3D scans, each annotated with anthropometric measurements such as weight, height, arm span, inseam, waist width, etc. After bringing the scans into dense correspondence via template fitting, we are left with  $M$  pose-normalized meshes consisting of  $N$  vertices each. Our approach for data-driven weight gain/loss simulation differs from the method of Piryanova et al. (2014a) in the following ways: 1) Instead of encoding body shape as a  $3 \times 3$  deformation matrix per mesh face (Angelov et al., 2005), we encode body

<sup>4</sup> <https://assetstore.unity.com/packages/tools/animation/final-ik-14290>.





**FIGURE 3**

The figure illustrates our approach of facial weight gain simulation. When modifying the weight of an avatar (left), part of the face region (highlighted in red) is fixed (center left). The modified vertices are stitched to the face region in a seamless manner using differential coordinates (Sorkine, 2005) (center right). Not keeping these vertices fixed would require recalculating the position of all auxiliary meshes such as eyes and teeth due to the undesired change in facial proportions for nose, mouth and eyes stemming from changing the parameters of the underlying face model (right). For the right image, eyes are copied from the unmodified avatar in order to better highlight the change in shape and position.

shape directly via vertex positions. 2) Modelling weight gain/loss as a change in parameters of a statistical parametric shape model (Piryankova et al., 2014a; Xiao et al., 2020) changes face identity during weight modification due to the fact that the learnt direction of change is not subject-specific. This leads to effects such as changing the shape of the eye socket, the pupillary distance or other unrealistic changes in face proportions. To mitigate these effects, we keep vertices in the face region fixed while deforming the rest of the mesh in order to preserve the identity of the participants.

To this end, we define a set  $\mathcal{S}$  with cardinality  $V$  containing all vertices outside of the face region (see Figure 3) as well as a selector matrix  $\mathbf{S} \in \mathbb{R}^{3V \times 3N}$  which extracts all coordinates belonging to vertices in  $\mathcal{S}$ . Let  $\mathbf{x}_j = (\mathbf{p}_1^T, \dots, \mathbf{p}_N^T)^T \in \mathbb{R}^{3N}$  be the vector containing the stacked vertex positions of the  $j$ th training mesh and  $\bar{\mathbf{x}} = \frac{1}{M} \sum_j \mathbf{x}_j \in \mathbb{R}^{3N}$  be the corresponding mean. Performing PCA on the data matrix  $\mathbf{X} = (\mathbf{S}\mathbf{x}_1 - \bar{\mathbf{S}}\bar{\mathbf{x}}, \dots, \mathbf{S}\mathbf{x}_M - \bar{\mathbf{S}}\bar{\mathbf{x}}) \in \mathbb{R}^{3V \times M}$  and taking the first  $k = 30$  components then yields the PCA matrix  $\mathbf{P} \in \mathbb{R}^{3V \times k}$ . Let  $\mathbf{W} = (\mathbf{w}_1, \dots, \mathbf{w}_M) \in \mathbb{R}^{k \times M}$  contain the PCA coefficients  $w_j$  of the  $M$  training meshes, computed by  $\mathbf{w}_j = \mathbf{P}^T (\mathbf{S}\mathbf{x}_j - \bar{\mathbf{S}}\bar{\mathbf{x}})$ . If we denote by  $\mathbf{D} \in \mathbb{R}^{M \times 4}$  the matrix containing the anthropometric measurements weight, height, armspan and inseam of the  $j$ th subject in its  $j$ th row, we can then compute a linear mapping from anthropometric measurements  $\mathbf{D}$  to PCA coefficients  $\mathbf{W}$  by solving the linear system of equations  $(\mathbf{D} \mid \mathbf{1})\mathbf{C} = \mathbf{W}^T$  in a least-squares sense via normal equations.

New vertex positions for a subject with initial vertex positions  $\mathbf{x}$  and a desired change in anthropometric measurements  $\Delta \mathbf{d} \in \mathbb{R}^5$  can then be calculated via  $\tilde{\mathbf{x}} = \mathbf{S}\mathbf{x} + \mathbf{P}(\mathbf{C}^T \Delta \mathbf{d})$ , i.e., by first projecting the desired change in measurements into PCA space via the learned linear function and then into vertex position space via the PCA matrix. However, this only updates vertices in  $\mathcal{S}$ . In

order to seamlessly stitch the new vertex positions to the unmodified face region, we compute the Laplacian coordinates (discretized through cotangent weights and Voronoi areas (Botsch et al., 2010)) of the resulting mesh and then use surface reconstruction from differential coordinates (Sorkine, 2005). For the vertices of the face region and its 1-ring neighborhood, the Laplacian is computed based on the unmodified vertex positions  $\mathbf{x}$ , while for the rest of the vertices, the Laplacian is computed based on the modified vertex positions  $\tilde{\mathbf{x}}$ . Since the position of vertices of the face region is known and should not change, we treat the position of these vertices as *hard* instead of *soft* constraints as discussed by Botsch and Sorkine (2008). Setting  $\Delta \mathbf{d} = (\Delta w, 0, 0, 0, 0)^T$  then allows modifying the user's weight while keeping the other anthropometric measurements fixed. Keeping the face region fixed 1) preserves the identity of the user for high values of  $\Delta w$  and 2) avoids having to recalculate the position of auxiliary meshes of the avatar such as eyes and teeth (Figure 3). Results of the described body weight modification method are shown in Figure 4.

### 3.3.2 Interaction methods

To allow users to modify the avatars' body weight as quickly, easily, and precisely as possible, we compare in our evaluation three implemented interaction methods regarding their usability. Since interaction methods for human body weight modification have not yet been explored, we considered the guidelines for object modification presented by LaViola et al. (2017). Figure 5 gives a short overview of the body weight modification methods. After a pilot test of body weight modifications with multiple generated virtual humans, we restricted the body weight modification for all interaction methods to a range of  $\pm 35\%$  of the user's body weight to



FIGURE 4

The figure shows a generated female avatar (BMI = 19.8) with modified body weight corresponding to a BMI range of 16–32 in two-point increments.

avoid unrealistic, possibly unsettling shape deformation. To avoid providing any hidden cues, we have extended the possible modification range compared to the used modification range of our passive estimation task. The constants given in the formulas for calculating the velocity of body weight change were determined empirically in a further pilot test.

### 3.3.2.1 Body weight modification via controller movement gestures

To modify the avatar's body weight via gestures (see Figure 5, left), users have to press the trigger button on each controller simultaneously. Moving the controllers away from each other then increases the body weight, while moving them towards each other decreases it. The faster the controllers are moved, the faster the body weight changes. When active, the body weight changes by the velocity  $v$  in kg/s, determined by the relative distance change between the controllers  $r$  in m/s, and calculated as  $v = 3.5r^2 + 15r$ .

### 3.3.2.2 Body weight modification via joystick movement

To modify the avatar's body weight via joystick (see Figure 5, center), users have to tilt the joystick of either the left or the right controller. Selecting joystick for an initial modification leads to a deactivation of the other joystick for the remaining interaction. Tilting the joystick to the left decreases the body weight, tilting it to the right increases it. The stronger the joystick is tilted, the faster the body weight changes. When enabled, the body weight changes by the velocity  $v$  in kg/s, determined by the normalized tilt  $t$  of the joystick and calculated as  $v = 10t^2 + 5$ .

### 3.3.2.3 Body weight modification via controller movement towards objects

To modify the avatar's body weight via objects (see Figure 5, right), users have to touch either a virtual "plus" or a virtual "minus" object within the virtual environment. As long as an object is touched, the body weight increases or



FIGURE 5

The figure sketches the three body weight modification methods we used for our evaluation: Gestures (left), Joystick (center), and Objects (right).

decreases. The longer the object is touched, the faster the body weight changes. When active, the body weight modification velocity  $v$  in kg/s increases quadratically over a normalized contact duration  $d$  of 1.5 s in a normalized range  $r$  between 3 kg/s and 15 kg/s.

## 4 Evaluation

We tested our first system prototype in a structured UX evaluation based on multiple relevant qualitative questions and quantitative measures concerning the users' scan and VR exposure experience as well as their body image. The following sections contain a detailed explanation of the evaluation process.

### 4.1 Ethics

Since our technical system was developed with the aim of being tested on potential patients in a clinically relevant context as part of a later feasibility study, particular attention has already been paid to ethical aspects during the here reported development and evaluation of our system. As part of a conservative development and evaluation strategy, we decided to work with a relatively small sample of healthy participants in this initial formative evaluation. The system, as well as the evaluation, was designed in consultation with our clinical partners within the context of our interdisciplinary research project ViTraS (Döllinger et al., 2019). A detailed ethics proposal following the Declaration of Helsinki was submitted to the ethics committee of the Human-Computer-Media Institute of the University of Würzburg and found to be ethically unobjectionable. Free professional help services provided by the Anorexia Nervosa and Associated Disorders (ANAD)<sup>5</sup> organization were explicitly highlighted during the acquisition and evaluation process.

### 4.2 Participants

A total of 12 students (8 female, 4 male) of the University of Würzburg participated in our evaluation and received course credit in return. Before the evaluation, we defined four exclusion criteria queried by self-disclosure: Participants had to have 1) normal or corrected to normal vision and hearing, 2) at least 10 years of experience with the German language, 3) not suffered from any kind of mental or psychosomatic disease, or from body weight disorders, and 4) no known sensitivity to simulator sickness. None of the participants matched any exclusion criteria. The participants were aged between 20 and 25 ( $M =$

22.0,  $SD = 1.48$ ), had a BMI between 17.85 and 32.85 ( $M = 22.72$ ,  $SD = 3.98$ ), and had mostly very little VR experience. Nine out of the twelve participants claimed to know their current body weight. The mean deviation of the indication of their body weight compared to that measured by the experimenter was 0.29 kg ( $SD = 1.57$ ).

### 4.3 Design

The evaluation of our system included qualitative and quantitative measures regarding 1) the body scan experience, 2) the UX of the VR exposure and the different modification methods used, and 3) their impact on the body image-related measures body awareness and body weight estimation. To compare our three modification methods (see Figure 5), participants performed for each modification method a set of active modification tasks (AMTs) in a counterbalanced order using a  $1 \times 3$  within-subjects design. For comparing the novel AMT with the more traditional passive estimation task (PET), the participants performed a PET each before and after the AMTs (see Figure 6, right). All tasks and the timing of the measures will be further explained in Section 4.5.

### 4.4 Measures

#### 4.4.1 Body scan experience

We conducted a semi-structured interview to evaluate the body scan experience. It included questions concerning the participants' expectations, their physical and psychological comfort and/or discomfort during the body scan and the assessment of their body measures, and about the clarity and transparency of the process. A full version of the questions can be found in the supplementary material of this work.

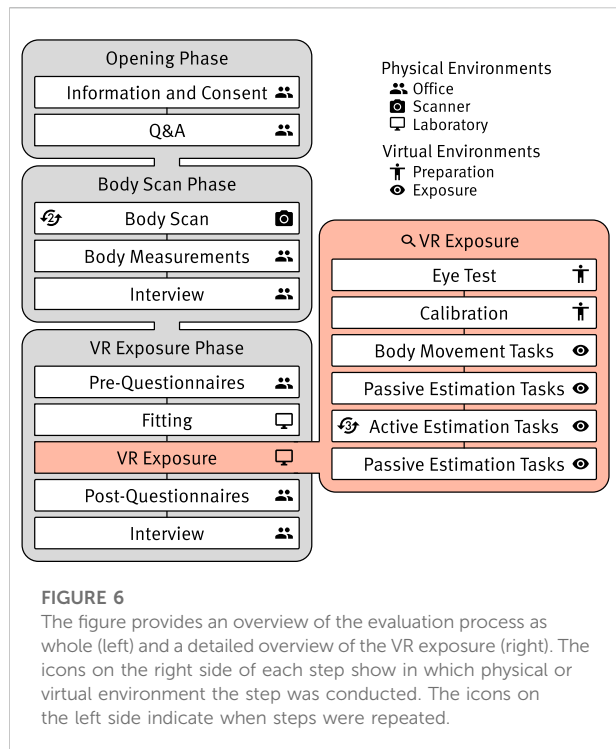
#### 4.4.2 VR experience

Regarding the VR experience, we included a variety of VR-specific and task-specific UX measures to get a holistic view of the system's overall UX. We used a combination of qualitative and quantitative measures, in virtual reality ratings, and pre- and post-questionnaires for the VR UX evaluation.

##### 4.4.2.1 Interview

We conducted another instead of a semi-structured interview with focus on the VR experience. It included questions concerning the participants' expectations and feelings towards the avatar, their favored body weight modification method and the perceived difficulty of the body weight estimation in general, their intensity of body awareness, and their affect towards their body. A full version of the questions can be found in the supplementary material of this work.

<sup>5</sup> <https://www.anad.de/>.



#### 4.4.2.2 Presence

We measured the participants' feeling of presence using the Igroup Presence Questionnaire (IPQ) (Schubert et al., 2001). It captures presence through 14 questions, each rated on a scale from 0 to 6 (6 = highest presence). The items are divided into four different dimensions: general presence, spatial presence, involvement, and realism. The questionnaire was provided directly after the VR exposure to capture presence as accurately as possible.

#### 4.4.2.3 Embodiment

As suggested by prior work, we divided the measurements for the feeling of embodiment into VBO and agency (Kiltner et al., 2012). Following Waltemate et al. (2018) and Kalckert and Ehrsson (2012), we presented one embodiment question for each dimension on a scale from 0 to 10 (10 = highest). Both questions based on items of the Virtual Embodiment Questionnaire (VEQ) of Roth and Latoschik (2020). To investigate possible differences in the feeling of embodiment caused by our interaction methods, the questions were provided multiple times during exposure.

#### 4.4.2.4 Simulator sickness

To test our system prototype regarding simulator sickness caused by latency jitter or other sources (Stauffer et al., 2018; Stauffer et al., 2020), we included the Simulator Sickness Questionnaire (SSQ) (Kennedy et al., 1993; Bimberg et al.,

2020) before and after the VR exposure. It captures the appearance and intensity of 16 different simulator sickness associated symptoms on 4-point Likert scales. The total score of the questionnaire ranges from 0 to 235.62 (235.62 = strongest). An increase in the score by 20 between a pre- and post-measurement indicates the occurrence of simulator sickness (Stanney et al., 1997).

#### 4.4.2.5 Avatar perception

For measuring the affect towards the avatar, we used the revised version of the Uncanny Valley Index (UVI) (Ho and MacDorman, 2017), including its four sub-dimensions: humanness, eeriness, spine-tingling, and attractiveness. Each dimension is captured by four to five items ranging from 1 to 7 (7 = highest).

#### 4.4.2.6 Workload

We measured workload to 1) determine the perceived effort during the calibration of the system and to 2) determine the perceived difficulty when modifying the avatar's body weight with our modification methods. To capture workload fast and efficiently during VR, we used a single item scale ranging from 0 to 220 (220 = highest) called SEA scale (Eilers et al., 1986), a German version of the Rating Scale Mental Effort (Zijlstra, 1993; Arnold, 1999).

#### 4.4.2.7 Preference rankings

Participants were asked to order the three body weight modification methods concerning their workload, perceived body weight estimation difficulty, vividness, contentment, and overall preference. Ranking scores were then calculated using weighted scores with reversed weights. A weighting of 4 was used for the highest rank, a weighting of 3 for the second highest, and so on. The overall rankings were summed up and averaged over the number of ratings. A high scores states high workload, difficulty, vividness, contentment, and overall preference.

#### 4.4.2.8 Calibration and modification time

To measure the efficiency of the avatar calibration and the interactions methods, we captured the average time needed from the beginning of calibration or modification until the end. Calibration time included potential amendments of the avatar skeleton and re-calibrations. A lower time states a higher efficiency.

### 4.4.3 Body image

#### 4.4.3.1 Body awareness

Similar to VBO, agency, and workload, we included (virtual) body awareness (VBA) as a one-item scale from 0 to 10 (10 = highest VBA) assessed at multiple times during exposure. The item was derived from the State Mindfulness Scale (SMS) (Tanay and Bernstein, 2013).



#### 4.4.3.2 Passive body weight estimation (PET)

The PET was adapted from prior work (Wolf et al., 2020, 2021, 2022a) and used to capture the participants' ability to numerically estimate the avatars' body weight based on a provided body shape. We repeatedly modified the body weight of the embodied avatar within a range of  $\pm 20\%$  incremented in 5% intervals in a counterbalanced manner resulting in  $n = 9$  modifications. To not provide any hints on the modification direction, the HMD was blacked-out during the modification. For each modification, the participants had to estimate the avatar's body weight in kg, which we used to calculate the misestimation  $M$ . It is based on the estimated body weight  $e$  and the presented body weight of the avatar  $p$  as  $M = \frac{e-p}{p}$ . A negative value states an underestimation, a positive value an overestimation. Additionally, we calculated 1) the average misestimation  $\bar{M} = \frac{1}{n} \sum_{k=1}^n M_k$  and 2) the absolute average percentage of misestimation as  $\bar{A} = \frac{1}{n} \sum_{k=1}^n |M_k|$ .

#### 4.4.3.3 Active body weight estimation (AMT)

The AMT was inspired by related work (Piryankova et al., 2014a; Thaler et al., 2018a,b) and used to capture the participants' ability to modify the avatar's body weight to match a requested numeric value. We also used it to analyze whether a certain interaction method for body weight modification influenced the participant's ability to judge the avatars' body weight. Participants had to modify the avatar's body weight by using one of our modification interaction methods until they thought it matched a presented numeric target weight in kg. The task was repeated for a target weight range of  $\pm 20\%$  of the actual avatar's body weight incremented in 5% intervals in a counterbalanced manner resulting in  $n = 9$  modifications. For each modification, we calculated the misestimation  $M$  based on the modified body weight  $m$  and the target body weight  $t$  as  $M = \frac{t-m}{t}$ . A negative value states an underestimation, a positive value an overestimation. Additionally, we calculated  $\bar{M}$  and  $\bar{A}$  as for the PET.

## 4.5 Procedure

The entire evaluation took place in three adjacent rooms (office, body scanner, laboratory) of the University of Würzburg and averaged 117 min. The procedure is illustrated in Figure 6.

### 4.5.1 Opening phase

First, participants were informed about the local COVID-19 regulations, received information about the experiment and the body scans, gave their consent, and generated two personal pseudonymization codes used to store the experimental data and the generated avatars separately. Then, the main experimenter answered potential questions and measured the participant's body height without shoes as required for the body scan.

### 4.5.2 Body scan phase

An auxiliary experimenter performed the body scan in normal clothes without any accessories. Afterwards, the main experimenter measured the interpupillary distance (IPD), body weight, and the participants' waist and hip circumference, and conducted the interview about the scan process. The duration of the interview averaged 4 min. All interviews were recorded by a Tascam DR-05 voice recorder.

### 4.5.3 VR exposure phase

Prior to the VR exposure, participants answered demographic questions and the SSQ as pre-questionnaires on a dedicated questionnaire computer. Then, an auxiliary experimenter demonstrated the participants how to fit the equipment, adjusted the HMD's IPD, and controlled that all equipment was correctly attached. After finishing the fitting, a pre-programmed experimental procedure was started, and participants were transferred to the preparation environment. For all virtual transitions during the VR exposure, the display was blacked-out for a short moment. All instructions were displayed on an instruction panel and additionally played as pre-recorded voice instructions. As the first preparation step, the participants had to undergo a short reading test to ensure the view was sufficient. Then, they performed the embodiment calibration in T-pose and judged its workload. During the whole VR exposure, participants had to answer questions and measurements verbally. Although interaction between the experimenter and the user may cause small breaks in presence (Putze et al., 2020), we considered this approach as part of the evaluation, since interaction between patient and the rapist would also likely occur in clinical settings and advanced in virtuo interaction to answer questionnaires might be difficult for novice users.

After the preparation finished, participants were transferred to the exposition environment. There, they performed five movement tasks in front of a virtual mirror while being instructed to alternately look at the mirror and directly on their body to induce the feeling of embodiment. Movement tasks were adapted from related work (Wolf et al., 2020) and had to be performed for 20 s. The first PET followed. Participants estimated the modified body weight of their avatar nine times. Between the estimations, the display was blacked-out briefly to cover the weight changes. In the next phase, participants conducted AMTs nine times for each body weight modification method in a counterbalanced manner. For all body weight estimation tasks, no feedback regarding the estimation error was provided to the participants. The second PET concluded the phase. After each AMT (see Figure 6), participants were asked to judge workload, agency, VBO, and



VBA in virtuo. The whole VR exposure took 36 min on average. After the VR exposure, participants answered IPQ, SSQ, UVI, and UX questions again on the dedicated questionnaire computer.

#### 4.5.4 Closing and debriefing phase

The questionnaires were followed by the second interview about the VR exposure that lasted on average 11 min. At the end of the session, the main experimenter thanked the participants and granted them credits for their participation. As part of the debriefing process after the session, the interviews were first transcribed and then two researchers summarized and clustered the answers into categories.

## 5 Results

In this section, we report the results of our evaluation separated into 1) the body scan experience, 2) the UX of the VR exposure including the different modification methods, and 3) their impact on body image-related measures. The statistical analysis of our results was partially performed using the software R for statistical computing (R Core Team, 2020) and partially using SPSS version 26.0.0.0 (IBM, 2020). All tests were performed against an  $\alpha$  of .05.

### 5.1 Body scan experience

#### 5.1.1 Feedback on the body scanning process

When asked whether the body scan procedure matched their idea of a body scan, four participants expected a different amount or arrangement of cameras, three participants expected a different scan process (e.g., one camera moving around the body, a laser measuring the body shape, or cameras only in the front), and one participant claimed to have no previous expectations on the body scan process. The other participants stated they already knew the body scan procedure from former experiments and did not remember expectations.

Most of the participants perceived the scan process as simple and clear. Only one participant stated not knowing what was happening between two scans. The experience during the scan process differed from “straightforward” and “easy” ( $n = 4$ ) over “interesting” or “cool” ( $n = 4$ ) to being “something to getting used to” or making one “feel observed” ( $n = 4$ ).

All participants stated positively they would do a body scan again. While most of them did not have suggestions for improvement ( $n = 8$ ). One suggested that the experimenter should be visible during the whole scan process to increase a feeling of safety. Others pointed out that a reduced number of cameras would ease the feeling of being watched and that the

stiff posture during the scan felt kind of uneasy after some time.

#### 5.1.2 Feedback on the body measurements

When evaluating the assessment of body measures, most participants claimed to perceive it as neutral or similar to being measured during a doctor’s appointment ( $n = 8$ ). Some others pointed out they would not expect it in a “normal” lab study but did not perceive it as awkward ( $n = 3$ ). One participant stated to perceive the measuring of their weight as very private and therefore uncomfortable.

### 5.2 VR experience

Since there was no comparison condition to the overall quantitative scores of the VR experience, we report the data, which were mainly collected on validated and comparable scales, descriptively. For measures captured multiple times during the experience, we calculated the mean value of all data points. The descriptive results of the VR exposure experience are summarized in Table 1.

To evaluate the possible occurrence of simulator sickness, we compared SSQ pre- and post-measurements. The observed increase in SSQ scores of 16.21 was below the indication threshold for simulator sickness of 20 points (Stanney et al., 1997), implying a safe use of the application with respect to potential simulator sickness-related impacts. Further, a two-tailed Wilcoxon signed-rank test revealed that the median ranks did not differ significantly between measurements,  $Z = 1.14$ ,  $p = .254$ .

#### 5.2.1 Feedback on embodiment and avatar perception

When asked about their feelings towards their personalized avatar, two participants used “neutral” or “okay” to describe their experience, and another four participants described it as “cool”, “interesting”, or “pleasant”. The remaining six participants described the experience as less positive, using words like “strange” and “irritating”. While one of the former emphasized the quality of the embodiment compared to other studies, three of the latter criticized the embodiment, especially concerning the lack of facial expression, eye movements and hand gestures. One pointed out that their “hands hold these controllers but the avatar does not”. The participants who found the experience rather irritating emphasized a lack of similarity in their avatar’s appearance.

The question of whether the avatar’s appearance met the participants’ expectations also received mixed responses. While one participant found it overall disproportional, six participants stated that the look of their avatar rather met their expectations. The remaining participants indicated that although the avatar’s

**TABLE 1** The table shows the descriptive values for our captured measurements concerning the VR experience. Detailed information regarding the measurements can be found in [Section 4.4](#).

| Measure              | Variable             | [Range]    | <i>M</i> ( <i>SD</i> ) |
|----------------------|----------------------|------------|------------------------|
| Presence             | IPQ General presence | [0–6]      | 4.58 (0.90)            |
|                      | IPQ Spatial presence | [0–6]      | 4.38 (0.95)            |
|                      | IPQ Involvement      | [0–6]      | 3.75 (0.89)            |
|                      | IPQ Realism          | [0–6]      | 3.22 (1.2)             |
| Embodiment           | VBO score            | [0–10]     | 5.49 (2.33)            |
|                      | Agency score         | [0–10]     | 7.22 (1.94)            |
| Simulator sickness   | Pre-SSQ              | [0–235.62] | 26.8 (23.7)            |
|                      | Post-SSQ             | [0–235.62] | 43.01 (39.21)          |
| Avatar perception    | UVI Humanness        | [1–7]      | 4.03 (1.10)            |
|                      | UVI Eeriness         | [1–7]      | 4.06 (0.95)            |
|                      | UVI Spine-tingling   | [1–7]      | 3.88 (0.88)            |
|                      | UVI Attractiveness   | [1–7]      | 4.25 (0.87)            |
| Calibration workload | SEA score            | [0–220]    | 20.83 (16.35)          |
| Calibration time     | Time in s            |            | 96.79 (50.29)          |

body looked as expected, they did not associate its face with themselves.

### 5.2.2 Comparison of the body weight modification methods

For comparing the three AMT conditions (gesture, joystick, and objects), we calculated a one-way repeated-measures ANOVA for each listed measurement (see [Table 2](#)) except modification times, where we calculated a Friedman test, and preference rankings, which are presented descriptively only. Test results showed significant differences between conditions only for workload,  $F(2, 22) = 13.95$ ,  $p < .001$ . Two-tailed paired-sample post-hoc *t*-tests revealed significant differences in the SEA score between body weight modifications with gesture and joystick,  $t(11) = 2.74$ ,  $p = .019$ , gesture and objects,  $t(11) = 2.8$ ,  $p = .017$ , and joystick and objects,  $t(11) = 4.86$ ,  $p = .001$ . Thus, the workload was considered to be highest when modifying body weight via objects and lowest when using the joystick.

### 5.2.3 Feedback on the body weight modification methods

When asked to explain their preference for an interaction method, most of the participants who preferred joystick ( $n = 8$ ) stated that it felt most controllable and least complicated. One participant additionally preferred the continuity of joystick-based interaction compared to the necessity of repetition in the gesture-based interaction. The participants who had preferred the gesture-based interaction ( $n = 4$ ) stated they found it most intuitive, flexible, and direct. They reasoned that controlling the speed of modification by extent and speed

of arm movements increased usability. None of the participants preferred modification via the objects.

## 5.3 Body image

In the following, we present the impact of our VR exposure on the body image-related measures of body awareness and body weight estimation as well as the related qualitative feedback.

### 5.3.1 Comparison of body awareness between body weight modification methods

We calculated a one-way repeated-measures ANOVA to compare the body awareness (VBA) during the three AMT conditions (gesture, joystick, and objects). As shown in [Table 2](#), VBA ratings differed tendentially between the three AMT conditions, with higher joystick ratings than the other conditions,  $F(2, 22) = 3.37$ ,  $p = .053$ .

### 5.3.2 Feedback on the intensity of body awareness

Seven participants stated they felt in contact with their physical body during the experience, while the other five stated they had lost contact to their body at least once. The latter stated, for example, they focused mainly on the task and the avatar. Others felt that their bodily awareness “got a bit lost” or that the situation and virtual surroundings made them forget reality, including their real body. On the other hand, three participants who stated being aware of their body during the experiment reasoned the embodiment as a main cause. One of them stated that “once before re-calibration, my avatar’s foot was kind of crooked, that’s when I paid attention to my real body. I made sure my knee was straight”. The other one focused on the avatar weight and claimed that “I was still aware of my body, but it was very strange because I was looking at a different mirror image, and sometimes, I felt much heavier when the weight of the avatar was lower than my actual weight”. Another reason why participants were aware of their bodily sensations was the physical contact with the floor or the proprioception during movements, which reminded them of their presence in the physical room ( $n = 2$ ).

### 5.3.3 Feedback on the affect towards the body

Eight of the participants stated that their feelings towards their bodies had changed during the experience. These changes concerned either their general awareness ( $n = 3$ ), their experienced body size ( $n = 2$ ), or their satisfaction with their body ( $n = 3$ ). The two participants stating a change in their experienced body size had either felt thicker or thinner in contrast to their avatar during the experience or felt thinner after the experience. Two of the participants whose bodily satisfaction changed stated an increased body

**TABLE 2** The table shows all descriptive values of the measures related to the comparison between our proposed body weight modification methods including *p*-values when calculated. Asterisks indicate significant *p*-values. Post-hoc tests for significant differences can be found in the corresponding text.

| Measure            | Variable  | [Range] | Gestures               | Joystick               | Objects                | <i>p</i> |
|--------------------|-----------|---------|------------------------|------------------------|------------------------|----------|
|                    |           |         | <i>M</i> ( <i>SD</i> ) | <i>M</i> ( <i>SD</i> ) | <i>M</i> ( <i>SD</i> ) |          |
| Embodiment         | VBO       | [0–10]  | 5.75 (2.63)            | 5.08 (2.68)            | 5.38 (2.5)             | .300     |
|                    | Agency    | [0–10]  | 7.25 (2.61)            | 7.16 (2.29)            | 7.33 (2.02)            | .915     |
| Modification time  | Time in s |         | 23.19 (2.94)           | 24.53 (10.37)          | 27.82 (7.72)           | .197     |
| Workload           | SEA       | [0–220] | 41.25 (27.97)          | 20.75 (13.37)          | 65.33 (33.06)          | < .001*  |
|                    | Ranking   | [1–4]   | 1.91                   | 2.73                   | 3.45                   | –        |
| Task Difficulty    | Ranking   | [1–4]   | 3                      | 1.81                   | 3.36                   | –        |
| Vividness          | Ranking   | [1–4]   | 3.09                   | 3.09                   | 2.27                   | –        |
| Contentment        | Ranking   | [1–4]   | 3.36                   | 3.45                   | 1.91                   | –        |
| Overall preference | Ranking   | [1–4]   | 3.27                   | 3.45                   | 2.09                   | –        |
| Body awareness     | VBA       | [0–10]  | 6.58 (1.98)            | 7.08 (1.93)            | 6.67 (2.06)            | .053     |

satisfaction or increased motivation to care for their bodily interests. In contrast, one participant reported increased dissatisfaction towards their physical body after the experience.

### 5.3.4 Feedback on the perception of body weight changes

Concerning the changes in the avatar's body weight, the participants equally rated them as “interesting” ( $n = 6$ ) or “weird” ( $n = 6$ ). Two participants especially pointed out that it was interesting to compare the avatar's body shape to their own former body, as they had lost or gained weight in the past. One stated “when I started my studies 5 years ago, I was 20 kg lighter than now, and it was kind of interesting to compare the avatar's look to the memories of my old body shape. It gave me a little perspective on how I want to look”. Four of the other participants liked the idea of seeing how they could look if they changed their eating/exercise behavior. Especially the modification towards a lower weight was perceived as threatening by some of the participants ( $n = 3$ ), as they thought it looked a bit unhealthy. To enhance the modification, two participants suggested more individual and fine-grained possibilities to manipulate only body parts instead of the body as a whole, for example, by including “two fixed points on the virtual body, one in the middle of the body and one at the shoulder area, to adjust the weight in these areas more exactly”.

### 5.3.5 Comparison of body weight estimations between body weight modification methods

For comparing the performance in body weight estimations between the AMT, we calculated a one-way repeated-measures ANOVA for  $\bar{M}$ -values, the percentage

body weight misestimation, and a Friedman test for  $\bar{A}$ -values, the absolute percentage body weight misestimation. The tests revealed that the three interaction methods did not differ significantly, neither in  $\bar{M}$ ,  $F(2, 22) = 0.66$ ,  $p = .529$ , nor in  $\bar{A}$ ,  $\chi^2(2) = 0.50$ ,  $p = .779$ , as summarized in Table 3.

### 5.3.6 Comparison of body weight estimations between estimation methods

We compared AMT and PET using two-tailed paired-samples *t*-tests for  $\bar{M}$ -values and two-tailed Wilcoxon signed-rank tests for  $\bar{A}$ -values. For  $\bar{M}$ , we showed that participants misestimated the body weight significantly less using the PET ( $M = 1.46$ ,  $SD = 8.4$ ) than when using the AMT ( $M = 3.1$ ,  $SD = 8.4$ ),  $t(11) = 2.47$ ,  $p = .031$ . For  $\bar{A}$ , the median ranks for PET,  $Mdn = 6.28$ , were tendentially lower than the median ranks for AMT,  $Mdn = 7.85$ ,  $Z = 1.88$ ,  $p = .060$ .

We further analyzed the results of AMT and PET concerning the modification levels ( $\pm 20\%$  in 5% steps) using linear regression. Our data violated pre-requirements for linear regression in terms of homoskedasticity and normality. Therefore, we calculated each linear regression using parameter estimations with robust standard errors (HC4) as recommended by Hayes and Cai (2007). Figure 7 shows the body weight misestimations  $M$  (left) and the absolute body weight misestimations  $A$  (right) for PET and AMT in relation to the modification levels.

For  $M$ , the results showed a significant regression equation for PET,  $F(1, 106) = 7.88$ ,  $p = .006$ , with a  $R^2$  of 0.069. The prediction followed equation  $M = -0.194 \times \text{Body Weight Modification in \%}$  instead of a simple point. The modification level did significantly impact on body weight misestimations  $M$ ,  $t(106) = 5.11$ ,  $p = .013$ . For AMT, we found no significant prediction of the modification level on the body weight

**TABLE 3** The table summarizes the body weight estimation performance (average misestimation  $\bar{M}$  and absolute average of misestimation  $\bar{A}$ ) of the comparison between our proposed modification methods.

|                | Gestures               | Joystick               | Objects                | <i>p</i> |
|----------------|------------------------|------------------------|------------------------|----------|
|                | <i>M</i> ( <i>SD</i> ) | <i>M</i> ( <i>SD</i> ) | <i>M</i> ( <i>SD</i> ) |          |
| $\bar{M}$ in % | 3.44 (9)               | 3.44 (8.9)             | 2.41 (8.05)            | .529     |
| $\bar{A}$ in % | 8.92 (4.58)            | 8.46 (5.10)            | 8.36 (3.66)            | .780     |

misestimations  $M$ ,  $F(1, 106) = 3.05$ ,  $p = .084$ , having a  $R^2$  of 0.028. The found prediction followed equation  $M = -0.120 \times \text{Body Weight Modification in \%} + 3.099$ . In consequence, the modification level did not significantly impact on body weight misestimations  $M$ ,  $t(106) = -3.46$ ,  $p = .094$ .

For  $A$ , the results showed a significant regression equation for PET,  $F(1, 106) = 5.27$ ,  $p = .024$ , with a  $R^2$  of 0.047. The prediction followed equation  $A = -0.101 \times \text{Body Weight Modification in \%} + 7.743$ . The modification level did significantly impact on body weight misestimations  $A$ ,  $t(106) = -2.09$ ,  $p = .039$ . For AMT, we found a significant prediction of the modification level on the body weight misestimations  $A$ ,  $F(1, 106) = 15.7$ ,  $p < .001$ , with a  $R^2$  of 0.129. The found prediction followed equation  $A = -0.147 \times \text{Body Weight Modification in \%} + 8.585$ . The modification level did significantly impact on body weight misestimations  $A$ ,  $t(106) = -17.9$ ,  $p < 0.001$ .

In addition to the linear regressions, we averaged body weight estimations for negative and positive modifications for both measurements to analyze differences between the modification directions. Again, we compared AMT and PET using two-tailed paired-samples  $t$ -tests for  $M$ -values and two-tailed Wilcoxon signed-rank tests for  $A$ -values. Test results for  $M$ -values showed that body weight misestimations in PET differed significantly between negative ( $M = 3.96$ ,  $SD = 11.13$ ) and positive ( $M = -1.09$ ,  $SD = 7.44$ )

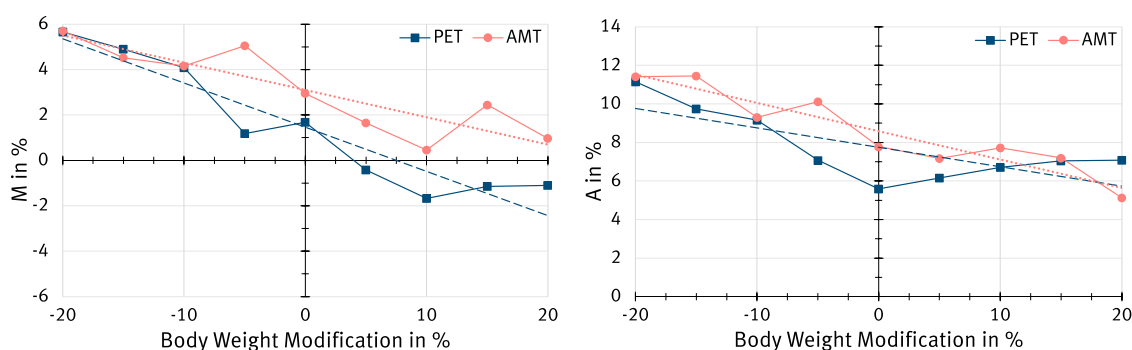
modifications,  $t(11) = 2.27$ ,  $p = .044$ , but misestimations in AMT did not differ between negative ( $M = 4.86$ ,  $SD = 10.57$ ) and positive ( $M = 1.38$ ,  $SD = 7.45$ ) modifications,  $t(11) = 1.63$ ,  $p = .131$ . For  $A$ -values, we found no significant differences between the median negative ranks,  $Mdn = 7.23$ , and the median positive ranks,  $Mdn = 5.80$ , modifications for PET,  $Z = 1.26$ ,  $p = .209$ , but found significant differences between the median negative ranks,  $Mdn = 9.51$ , and the median positive ranks,  $Mdn = 5.39$ , modifications for AMT,  $Z = 2.59$ ,  $p = .010$ .

### 5.3.7 Feedback on the body weight estimation difficulty

Regardless of the estimation method, estimating the body weight of the avatar was found to be difficult ( $n = 8$ ). Only three participants stated they found it relatively easy or only medium-difficult to estimate the body weight. The main reason why participants rated the task as difficult was the high number of repetitions ( $n = 2$ ) or a reduced perceptibility of their physical body, both leading to a “loss of perspective”. Additionally, one participant stated that the task difficulty depended on the distance of the avatar’s weight to their own.

## 6 Discussion

In the present paper, we introduced a prototype of an interactive VR-based system that aims to support body image interventions based on embodied, modulatable, and personalized avatars in future clinically relevant settings. We evaluated the system regarding 1) the body scan experience, 2) the general UX of the VR exposure including body weight modification interaction methods, and 3) the body-image specific UX of the exposure, namely the impact on body awareness and body weight perception. In the following, we summarize and discuss the results of our evaluation to ultimately derive guidelines supporting the design of systems for body image interventions. The guidelines are based on conclusions of the



**FIGURE 7**

The figure shows the body weight misestimations  $M$  (left) and absolute body weight misestimations  $A$  (right) in relation to the performed body weight modifications for PET and AMT.

qualitative and quantitative results accomplished by the researchers' observations and participants' comments during the evaluation. While these may overlap with existing best practices or established VR guidelines, we believe it is elementary to summarize them for the given context and to highlight their importance.

## 6.1 Body scan experience

Overall, the scan process was mainly rated as simple and interesting, although it took place in a separate room with great technical effort. Participants stated a high acceptance and willingness to be scanned again. In addition, the scan and the associated body measurements were seen as something that one would do in a clinical setting, and that does not trigger unpleasant reservations. This assessment strengthens the idea of using body scans in a clinical context.

Nevertheless, two main criticisms of the scanning process were the feelings of being watched and being left alone. The large number of visible cameras mainly caused the first while both can be attributed to the arrangement of the cameras surrounding the person in all directions. Curtains around the scanner also supported the feeling of being left alone during the scan process. In particular for our target group and the intended clinical usage, amendments seem necessary. Options to reduce the negative feelings could be a change in the arrangement of cameras, e.g., opening the space by placing them only on one side or reducing the number of cameras to a minimum as proposed by Wenninger et al. (2020) and supported by the results of Bartl et al. (2021). In addition, we suggest a constant dialogue about and during the process to counteract the feeling of being alone.

### Guidelines for Body Scanning

- Users should receive thorough information and instruction in advance about the body scan procedure to provide clarity and transparency.
- Body scans should be performed unobtrusively to protect privacy and avoid the feeling of being watched.
- The number and arrangement of cameras should be planned carefully to avoid the feeling of being watched.
- The number of people involved in the body scan should be minimized to increase privacy, and personal contact should be maximized to increase safety.
- Body-related measurements should be performed professionally while maintaining privacy.

## 6.2 User experience of VR experience

The feedback regarding preparation and calibration was consistently positive, confirming the decision for our

approach. This is empirically supported by the low measured calibration times requiring only a short time holding T-pose, and the low workload ratings during the calibration process. Nevertheless, there are further possibilities to reduce the effort for calibration and invasiveness, for example, by using completely markerless body tracking solutions (Wolf et al., 2022a).

Regarding VR-specific measures, participants rated their perceived feeling of presence on an acceptable level (Buttussi and Chittaro, 2018; Wolf et al., 2020), with lower ratings on involvement and realism. A reason for the lower observed involvement score could be the constant interaction with the experimenter during the tasks (e.g., confirming body weight estimations, rating experiences). Possible implausible content (e.g., body weight modification by interaction) could have impacted negatively on realism. Continuous communication between therapist and patient during weight modifications might be a crucial element in clinical settings. Therefore, further research on the role of presence (and its sub-dimensions) in VR body image interventions seems required, as the latest reviews did not address this topic (Riva et al., 2019; Horne et al., 2020; Turbyne et al., 2021).

Surprisingly, although participants rated their feeling of virtual body ownership descriptively higher compared to non-personalized avatars (Waltemate et al., 2018; Wolf et al., 2020), their ratings were lower than in prior work using personalized and photorealistic avatars (Waltemate et al., 2018). A reason for the noticed differences could be the particularly body-related nature of our task. Avatars created via body scans have a very high resemblance to the individual but still do not provide a perfect visual replica. In a task highly focusing on body perception, even minor inaccuracies may become noticeable, and participants might focus on these, experiencing a diminished feeling of virtual body ownership. Another factor could be the performed body weight modification leading to a reduced congruence between real and virtual bodies and, consequently, might decrease the feeling of virtual body ownership.

The ratings and especially the qualitative statements on avatar perception reveal similar results, as some of the participants stated their avatar to be uncanny or not fully recognizable as themselves. This raises doubts about the degree of personalization of avatars and whether the creation of highly photorealistic textures is currently necessary (and feasible). Tools such as Virtual Caliper (Pujades et al., 2019) can create in shape personalized avatars using only VR equipment. In conjunction with generic avatar generators, such as Meta Human (Epic Games, 2021), highly realistic avatars with personalized body shapes could be created with less effort. They would not resemble the person perfectly, but this lack of resemblance could make them less uncanny while remaining a still better quality in general. Additionally, a personalization in body shape would be sufficient for



simulating body weight changes. However, one counter-argument is provided by [Thaler et al. \(2018a\)](#), who clearly state that the body weight perception of avatars having personalized textures differs from generic ones. To address the question of whether personalization of avatars in our context should be achieved through photorealism or customization, further research seems necessary.

### Guidelines for VR Design

- The physical and mental effort for system calibration should be kept as low as possible.
- The animations of embodied avatars should be as authentic as possible and include facial expressions, eye movements, and hand gestures to increase realism and reduce eeriness.
- When using physical controllers, virtual controller representations should be displayed in VR and controlled by the avatar.
- When using personalized avatars, body shape and texture should aim for the highest possible conformity with the user to reduce uncanniness.

#### 6.2.1 User experience of body weight modification

When comparing the subjective rankings of the three modification methods, it becomes apparent that the interaction via virtual objects was the least preferred. It was rated as more demanding and difficult, and less vivid, resulting in lower contentment and overall preference than the other two modification methods. Modification via joystick and gestures were rated rather similarly with a slight preference towards the joystick interaction. The in vitro ratings of workload match these rankings. While joystick was rated quantitatively most positively, the qualitative analysis shows arguments in favor of gesture interaction, especially in terms of vividness and intuitivity. No impact has been noticed on the feeling of embodiment or performance in body weight estimation, which is particularly important in our context.

Regardless of the interaction method, the lack of body weight modification in relation to different body parts (e.g., abdomen, hips, thighs) and in relation to the composition of the body tissue (e.g., fat or muscle mass) was mentioned. The use of advanced body modification methods, such as those presented by [Maalin et al. \(2020\)](#) or [Pujades et al. \(2019\)](#) could allow for body weight modifications that go beyond using only BMI as a single parameter modifying the whole body's weight. However, having more complex body weight modification methods would also increase the complexity of user interaction.

### Guidelines for Body Weight Modifications

- Body weight modifications severely differing from the user's BMI or reaching unrealistic or considered unhealthy ranges should be avoided to reduce alienation.

- Body weight modifications should allow changing the body weight independently on different body parts considering different body tissue compositions.
- Body weight modifications performed directly via a hardware input device or body gestures should be preferred over virtual objects or buttons.

## 6.3 Body image-related outcomes

The comparison of body awareness between the three modification methods indicated a higher body awareness in joystick interaction compared to gestures and objects. However, the reported effects of the VR exposure on body awareness and affect towards the body were very individual, with participants reporting either a loss or an increase of body awareness during the experience. Future work with an increased sample size is necessary to further investigate the difference between the conditions and whether the individual differences are related to people's overall body awareness, as proposed by [Filippetti and Tsakiris \(2017\)](#). These insights will be crucial to determine what effects can be expected for a target group with low body awareness or negative body image.

In contrast to body awareness, body weight estimations did not differ between body weight modification methods. However, when comparing the accuracy of the type of estimations task, PET provided more accurate estimates than AMT. While estimating a person's weight based on their appearance is not an everyday task, it is surely more common than actively modifying a (virtual) body to a certain body weight. Thus, the difference might originate in the relative novelty of active modification compared to passive estimation. Another reason could be the different phrasing of the task instructions, which has been shown to have the ability to influence body weight estimation ([Piryankova et al., 2014b](#)). For both PET and AMT, the accuracy of the body weight estimation depended on the target weight, or in other words, on the deviation between the own real weight and the virtually presented body weight. This effect has been observed priorly for VR body weight estimation tasks ([Thaler et al., 2018a](#); [Wolf et al., 2020](#)) and is in line with the so-called *Contraction bias* as described by [Cornelissen et al. \(2016, 2015\)](#). It states that body weight estimates are most accurate around an estimator-dependent reference template (of a body) and decrease with increasing BMI difference from this reference. Thereby, bodies heavier than the reference tend to be underestimated, while lighter ones tend to be overestimated. Results on absolute body weight estimations show that although the average misestimations were comparatively low, they are subject to high deviations and uncertainties, which also has been observed priorly ([Thaler et al., 2018a](#)). The reasons for this probably lie in the nature of the task, since estimating body weight seems generally challenging, and body image disturbances are ubiquitous even in the healthy population ([Longo, 2017](#)). Qualitative feedback confirms the task difficulty. When further analyzing the absolute body weight estimations, it is particularly noticeable that they seem to be easier and more

accurate for increased than for reduced body weight. This is rather unexpected since *Weber's Law* suggests that differences in body weight become harder to detect when body weight increases (Cornelissen et al., 2016). A possible reason for the high uncertainties in the absolute body weight estimations and the contradiction to Weber's law could be the perspective on the avatar offered by the virtual mirror, which mainly shows the front side of the body (Cornelissen et al., 2018). More research on this topic seems required.

### Guidelines for Body Weight Estimations

- Body weight estimations capturing the current perception of the real body in VR should be performed at the beginning of an intervention, as the perceptibility of the real body might decrease over time.
- When performing body weight estimations, care should be taken to present the respective body equally from multiple perspectives.
- When analyzing body weight misestimations based on avatars, determining the average accuracy of the misestimations with healthy individuals helps avoid strong influences of the system properties.

## 6.4 Future research directions

The results of our work raise new research questions for future work. First, the high necessity of communication between therapist and user, potentially leading to breaks in presence, raises the question of the general impact of presence in body image interventions. This is also interesting when it comes to augmented reality, as already recognized by Wolf et al. (2022a).

Second, the observed ratings in body ownership despite using photorealistic, personalized avatars and the feedback on avatar perception leads to the question of how photorealism and personalization should be applied to body image interventions. Future work should explore whether avatars that are less personalized in texture are sufficient for our purpose as they might raise less uncanniness.

Third, the severe individual differences in the report of body awareness and affect towards the body raise the question, of which individual characteristics might predict the effects of a VR-based intervention on both variables.

Fourth, future work should further address the difference between active body weight modification and passive body weight estimation we found in this study. It remains unclear which underlying processes lead to differences between the two tasks and whether they impact differently on body image. Similar counts for the observed differences in body weight misestimations for avatars with decreased or increased body weight.

Finally, although our current work is situated in the context of body image disturbances, it aimed to test the usability and user

experience of our application regardless of the target population in a non-clinical setting. For subsequent work, we suggest directly incorporating our gained knowledge by considering the participants' feedback and the derived guidelines and testing the system with the intended target population in a feasibility study. To further improve the system in direction of an appropriate clinical setting, technical advancements, like low-cost avatar reconstruction techniques (Wenninger et al., 2020; Bartl et al., 2021), should be incorporated and domain expert opinions, like recently summarized by Halbig et al. (2022), further considered.

## 6.5 Limitations

Our system implementation and evaluation still have limitations. As stated earlier, some of our participants described mixed feelings toward their personalized avatar and a lack of similarity between their avatar's face and their own. Including animations of facial expressions and eye movements could help increase the association with one's avatar. However, previous work on facial animations has shown only little effect on the perceived embodiment (Gonzalez-Franco et al., 2020; Döllinger et al., 2022). Improving the scan quality in the facial area, i.e., by using more cameras in the facial area, could improve this problem.

While modifying the body weight of the personalized avatars, we keep parts of the face region fixed (see Figure 3). This does not completely accurately model weight gain/loss in this region, as the soft tissue in this area of the face changes with varying body weight (De Greef et al., 2006). Other methods (Piryankova et al., 2014a; Zhao et al., 2018; Tang et al., 2021) deform the whole face region or regularize the deformation of a region similar to ours (Xiao et al., 2020). These methods, however, produce other undesirable effects such as changing eye socket shape or pupillary distance due to the fact that the underlying statistical model produces one direction of change that is applied to all avatars. As the data measured by De Greef et al. (2006) shows, the soft tissue thickness in our fixed region does positively correlate with BMI. However, we note that the correlation for landmarks in our fixed region is smaller than for those outside the fixed region and as such we decided to keep the face region around the eyes, nose, and mouth fixed. As seen in Figure 3, this still produces plausible results while avoiding undesirable changes in face identity. For future work, weight modification models should incorporate information about the underlying bone and muscle structure (Achenbach et al., 2018; Komaritzan et al., 2021) in order to more accurately model changes in soft tissue thickness.

Although our sample included slightly overweight participants, the current design and development phase was limited to students without a diagnosed body image disturbance and predominantly with a BMI in a healthy range. The clinical applicability to our target group, which is

already in preparation as part of our ViTraS research project (Döllinger et al., 2019), is one next step after the here presented design and UX optimization phase. Further, given the small sample size of 12 participants and the comparatively narrow range of age, the results cannot be generalized to a wider population. However, the study provides valuable insights into such a system's user experience and facilitates further research.

Overall, the design and development phase would benefit from a larger test sample tailored to the final target group. However, this is not an easy endeavor since it blurs the separation between the usability and user experience tests in the development phases and the clinical application. Hence, it requires closer integration and supervision by therapeutically trained professionals and experts in obesity treatment. Ultimately, this integration would be necessary throughout all steps of technical developments to safeguard against unwanted effects for all participants during the design and development and UX optimization steps. Notably, two participants of our overall healthy sample already showed some emotional reactions when confronted with their modified virtual self. Given the uneasiness some participants felt when their avatar's body weight was modified, further research is needed on how to restrict body weight modifications levels for different populations.

## 7 Conclusion

In this work, we have presented and evaluated the prototype of an advanced VR therapy support system that allows users to embody a rapidly generated, personalized, photorealistic avatar and modulate its body weight in real-time. Our system already offers numerous positive features and qualities, especially regarding the execution of body scans and an overall enjoyable VR experience. The guidelines for designing VR body image therapy support systems that we derived from our results helps to facilitate future developments in this field.

However, more research is needed for a therapeutic application. Possible areas of investigation include the implementation of photorealism, which may need to be revisited when working on body image. More research is also required on the differences between active body weight modification and passive body weight estimation. Finally, investigations with more focus on the target group and the individual characteristics of future users will be necessary, especially concerning body image distortion, body dissatisfaction, and body awareness.

## Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

## Ethics statement

The studies involving human participants were reviewed and approved by the Ethics Committee of the Institute Human-Computer-Media (MCM) of the University of Würzburg. The participants provided their written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article.

## Author contributions

ND and EW conceptualized large parts of the experimental design, collected the data, performed data analysis, and took the lead in writing the manuscript. EW and DM developed the Unity application including the experimental environment and avatar animation system. MB and SW provided the avatar reconstruction and body weight modification framework. CW and ML conceived the original project idea, discussed the study design, and supervised the project. All authors continuously provided constructive feedback and helped to shape study and the corresponding manuscript.

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## Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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## Supplementary Material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/frvir.2022.935449/full#supplementary-material>

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# Eye Tracking in Virtual Reality: a Broad Review of Applications and Challenges

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## Abstract

Eye tracking is becoming increasingly available in head-mounted virtual reality displays with various headsets with integrated eye trackers already commercially available. The applications of eye tracking in virtual reality are highly diversified and span multiple disciplines. As a result, the number of peer-reviewed publications that study eye tracking applications has surged in recent years. We performed a broad review to comprehensively search academic literature databases with the aim of assessing the extent of published research dealing with applications of eye tracking in virtual reality, and highlighting challenges, limitations and areas for future research.

**Keywords** Eye tracking · Virtual reality

## 1 Introduction

Catering the stimulus to the user's actions, such as head movement, eye movement, and hand movement, is the core principle of virtual reality (VR). Head-mounted display (HMD)-based VR depends on the ability to track head movements and render visual scene motion contingent on head movement. This advance was made possible by improvements in head tracking technology. Moving forward, similar improvements in HMD-based eye tracking technology will allow for fundamental advances in VR applications based on eye movement.

Consumer VR HMDs have significantly advanced in recent years in terms of tracking, latency, refresh rate, resolution and optics (Koulieris et al. 2019) with major consumer platforms (HTC Vive, Oculus Rift, Sony VR) already having presented second or third generation HMDs. Eye tracking technology has been commercially available for decades on desktop displays, but in recent years, a number

of commercially available solutions have been developed that facilitate eye tracking on consumer VR HMDs. As a result, research and development surrounding eye tracking in HMDs has accelerated and expanded in recent years.

A number of literature reviews have provided an overview of general eye tracking applications (Duchowski 2002) and gaze-based interaction (Duchowski 2018) without focusing on VR. Additionally, there are several reviews, that we highlight in this review, that focus on specific applications of eye tracking in VR, including reviews by Rappa et al. (2019), Souchet et al. (2021), Lutz et al. (2017), Harris et al. (2019), and Souchet et al. (2021). However, to the best of our knowledge, no reviews exist that exclusively aim to provide a broad overview of eye tracking applications for VR HMDs. We believe that interest in this area is growing rapidly, given the large number of studies that utilize eye tracking in a VR setting. It is therefore timely to provide a broad overview of the applications, challenges and limitations of eye tracking in VR.

This paper provides a broad review of current and seminal literature of applications of eye tracking in VR and identifies limitations, challenges and areas for future research. The rest of this paper is organized as follows: Sect. 2 discusses background concepts about eye movements and eye movement tracking methods in VR that will help the reader to understand the concepts discussed in later sections. Section 3 presents the applications of eye tracking in VR by organizing them into seven broad application areas. Finally,

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Sect. 4 discusses the current challenges and limitations faced by eye tracking in VR.

## 2 Background concepts

### 2.1 Types of eye movements

There are several distinct types of eye movements that serve different functions and are controlled by distinct neural circuits. Optimization of eye tracking for VR applications often requires identifying and distinguishing these different eye movement types and exploiting knowledge of visual and cognitive processing during eye movements to achieve the desired outcome (Holmqvist et al. 2011).

The need for eye movements arises because human visual acuity is not uniform across the visual field. Visual acuity is highest at the fovea, a central region of the retina (Cowie and Rolls 1974). The foveal region has the highest density of photoreceptors that allows parts of the visual field that fall on it to be seen with the highest detail (Curcio et al. 1990). Outside the foveal region, visual acuity drops gradually towards the edges of the retina. Below we will describe several eye movement types and their relevance including: saccades that just move the eyes from target to target, smooth pursuit that allow tracking moving objects, reflexive eye movements that stabilize vision, and vergence eye movements that coordinate the two eyes to allow binocular fixation at different depths. In the rest of this section, we describe these eye movements and their functional role in greater detail.

The subjective experience of being able to see the entire visual field in high acuity is achieved by scanning the visual field with successive eye movements known as saccades and integrating the high acuity snapshots of the scene that are obtained, a process known as trans-saccadic integration (Melcher and Colby 2008). Saccadic velocities may reach as high as 400 to 600 deg/sec, with durations lasting from 30 to 120 msec, and amplitudes in the range 1 to 45 degrees; there is a systematic relationship between amplitude, duration, and velocity known as the saccadic main sequence (Leigh and Zee 2015). Under natural circumstances, gaze shifts of greater than about 30 degrees are typically achieved by a combination of saccade and head movement. Vision is typically blurred during a saccade which allows for manipulation of the visual scene to go largely unnoticed during the saccadic interval.

Between saccades, the eyes typically remain relatively still for an average duration of 200–300 ms to fixate on objects of interest in the scene. Analysis of fixation locations provides information about the user's attentional state and the scene content that the user is interested in. However, the eyes are never completely still during

fixations, because there are miniature fixational eye movements that occur during this time of relative stillness. Fixational drifts are small and slow movements away from a fixation point (Otero-Millan et al. 2014). Drifts are often followed by small faster movements known as microsaccades that bring the eyes back to the fixation point; these small movements are typically less than 1 degree (Otero-Millan et al. 2014).

Pursuit eye movements occur when the eyes track a slowly moving visual target in order to stabilize the image of that target on the retina. Targets with velocities of greater than 30 deg/sec cannot be effectively tracked by pursuit eye movements, but catch-up saccades are triggered to compensate for this lag. Most people are unable to initiate pursuit without a moving visual signal. Nevertheless, pursuit is considered to be a voluntary eye movement (Robinson 1965).

Another type of eye movement that serves to stabilize the image on the retina is known as the vestibulo-ocular reflex (VOR). These movements act to rotate the eyes to null foveal retinal image motion that would result from head movement relative to the stationary environment. The VOR is driven predominantly by vestibular signals that transduce inertial cues to head motion, both linear and angular (Ramaioli et al. 2019). Vestibular-driven movements are supplemented by visually driven movements that respond to uncompensated image motion at the fovea known as retinal slip. The resulting stabilizing eye movement, referred to collectively as the visually enhanced VOR, allows us to see the world as clear and stable despite head movement (Leigh and Zee 2015).

Vergence eye movements are the simultaneous movements of both eyes in opposite directions that allow the two eyes to change binocular fixation distance. The nearer the target, the greater the convergence that is required. Thus, the vergence angle of the eyes along with disparity in the images presented to the left and right eyes determines how users perceive stereoscopic depth. Because vergence is associated with target depth, there is a linkage between neural signals that drive vergence and those that drive accommodation, or focal state, of the lens of the eye, which also needs to vary with target depth to ensure that the retinal image is in focus (Toates 1974).

Several of these eye movement types can occur simultaneously, such that the total movement of the eye is the result of superposition of contributing movements, making it difficult to identify and quantify distinct types of eye movements. In addition, eye movements measured in head coordinates can be ambiguous and difficult to characterize without information about scene content and head movement. Inaccurate and/or imprecise classification and quantification in turn compromises the use of eye movement data for VR applications. Advanced identification



**Fig. 1** An HTC Vive Pro Eye HR HMD with integrated VOG-based eye trackers (left). A VR HMD with scleral search coil-based eye tracking system (right). Image courtesy of Whitmire et al. (2016)

and classification methods that make use of all available information are therefore an active area of research (Kothari et al. 2020).

## 2.2 Eye tracking methods in VR HMDs

Three methods have been used to track eye movements in HMDs: (1) electro-oculography (EOG), (2) scleral search coils, and the most common, (3) video oculography (VOG).

EOG measures the orientation of the eye by placing electrodes on the skin around the eye which measure the resting potential of the eye. The electrodes can be easily incorporated into the HMD where it contacts the face (Bulling et al. 2009; Shimizu and Chernyshov 2016; Xiao et al. 2019; Kumar and Sharma 2016). The method works because the eye is a dipole that is positively charged toward the cornea and negatively charged toward the retina. The difference in voltage across electrodes placed on opposite sides of the eye (e.g., left and right) map well to eye orientation (e.g., horizontal position of the eye) (Mowrer et al. 1935). One drawback is that EOG provides a fairly imprecise measure of eye position, but EOG is the only method that allows tracking while the eyes are closed.

The scleral search coil method works by tracking the orientation of a wire loop that is embedded in a contact lens worn by the user (Robinson 1963). The user's head is positioned between Helmholtz coils which generate a uniform magnetic field. When the eyes move in the uniform and known magnetic field, electric current is induced in the scleral coils indicating the horizontal, vertical and torsional orientation of the eye. This method is highly accurate and precise with reported spatial resolution of less than  $0.1^\circ$  and temporal resolution that is greater than 1KHz (Whitmire et al. 2016). However, this method is

challenging to implement in HMDs not only because of the need for wired contact lenses but also because the Helmholtz coils must be head-mounted and consequently the resulting magnetic field is much smaller in volume. Nevertheless, an HMD with scleral coil system (see Figure 1) has been developed Whitmire et al. (2016) for use in specialized cases where high-precision tracking in an HMD is required, for example to validate alternative tracking systems.

By far, the most common eye tracking method used in HMDs is VOG (Duchowski 2007). Currently commercially available HMD-based eye trackers including Tobii, Pupil Labs, Varjo, and Fove employ VOG-based eye tracking. Images of the eyes are captured by cameras mounted in the HMD and analysis of the video frames indicates eye orientation. Most often, analysis methods rely on identification of the pupil and perhaps other landmarks to recover eye orientation. There are many analysis methods for recovering eye position from eye images, and methods implemented by commercially available systems are often proprietary (Holmqvist et al. 2012). This is an active area of research, but in-depth discussion of eye image analysis methods is beyond the scope of the present paper. Regardless of the method, better quality eye images translate into better quality tracking. Camera position is an important factor limiting image quality. Systems that film the eye front-on through the HMD optics using a hot mirror typically perform better than systems that mount the cameras obliquely thereby bypassing the HMD optics (Richter et al. 2019). Spatial and temporal resolution of the eye cameras also limit tracking performance. The current standard in commercially available systems varies. For example, the Pupil Labs VR/AR plugin system has  $192 \times 192$  spatial resolution at 200 Hz, while the Varjo system has  $1280 \times 800$  spatial resolution at 100Hz.



## 2.3 Technical aspects of eye movement tracking

### 2.3.1 Eye tracking data quality

High quality eye tracking data are an important prerequisite for any application that uses eye movement data. Eye tracking data quality refers to the performance of an eye tracker as measured by various metrics including spatial precision, spatial accuracy, sampling rate, and latency.

*Spatial Precision* is related to the ability of an eye tracker to reproduce measurements accurately over time. Spatial precision of an eye tracker is commonly measured by calculating the sample-to-sample root-mean-square angular displacement (RMS) and the standard deviation of samples collected within a given time window. The amount of precision needed for an eye tracking-based application depends on the type of the eye movement that needs to be recorded and the type of eye tracking task. Applications that use small fixational eye movements, like tremors, drifts and microsaccades require high-precision eye trackers. Andersson et al. (2010) point out that for such tasks the eye tracker should have an RMS precision value that is lower than  $0.03^\circ$ .

*Spatial Accuracy* is the average angular deviation between the true gaze position and the gaze position estimated by an eye tracker (Holmqvist et al. 2012). Spatial accuracy of an eye tracker is measured as the distance, in degrees, between the positions of a target point and the average position of a set data samples collected from the eye tracker within a given time window. Lack of accuracy may not pose any problems when the areas of interest in an application are large and are distant from each other. However, in applications with closely spaced stimuli, small inaccuracies could be critical, and could lead to the information obtained from the user or the action executed by the user being different from what the user intended.

*Latency* refers to the delay between eye movement events and the time these events are received by our system. There are various factors that contribute to data latency in the eye movement tracking pipeline including the sampling rate, the time to compute the gaze position from the eye images, the time to filter and post-process the data and the time to transmit the data to our VR display. Latency could cause elements of the stimulus that are supposed to respond to eye movement to lag behind the eye movements. When this lag is big enough that it is perceptible by the user, it could severely affect the virtual experience. This could be particularly problematic in interactive applications or gaze-contingent displays which change some parts of the stimulus in real time in response to eye movements.

*Sampling Rate* or sampling frequency of an eye tracker refers to how many times per second the eye is recorded by the eye tracker. Sampling rate determines the type of

events that can be measured with your eye trackers and the suitability of the eye tracking data for a particular application. For example, the precise measurement of small amplitude eye movements such as fixational eye movements and low amplitude saccades requires eye trackers with high sampling rates. Andersson et al. (2010) argue that based on the Whittaker–Nyquist–Shannon theorem Shannon (1949), the sampling rate should at least be at least twice the speed of the eye movement to be recorded.

### 2.3.2 Calibration

Calibration is the process by which the eye tracking system finds a mapping function that maps coordinates reported by the eye tracker to the coordinates of a gaze point in the visual environment (Harezlak et al. 2014). Most HMD-based eye trackers use a standard point-based calibration procedure. The procedure successively shows a set (usually 5 to 16) of small point targets to the user and the user is asked to fixate on each target for a few seconds. The calibration procedure requires willful engagement and cooperation from the user to be successful, because the mapping function requires the user to fixate on each target for the full duration that the target is shown on the environment. Due to differences in eye attributes and the geometry of the eye tracking setup, most eye tracking use-cases require users to calibrate the eye tracking system before they can use it. However, some use-cases, such as smooth pursuit-based interaction, just employ the user's relative eye motion in reaction to scene objects; therefore, they do not need to calibrate the eye tracking system Zeng et al. (2020).

## 3 Eye tracking applications in VR

To improve readability, the presentation of the review is organized into seven broad eye tracking application areas: display and rendering; user interaction; collaborative virtual environments; education, training and usability; security and privacy; marketing and consumer experience research; and clinical applications.

### 3.1 Display and rendering

Knowing where the user is looking can provide advantages in both functionality and efficiency of rendering in VR. Building truly immersive VR experiences would require the development of HMDs that are capable of rendering content with visual quality that is close to that of the human visual system. To enable such display systems, it is estimated that we would need to deliver more than 100 Gb/sec data to the HMD (Bastani et al. 2017). Achieving this data rate on VR

systems is challenging. However, due to the physiological limitations of the eyes, the human visual system can only consume a small percentage of this data stream. The human eye has very high acuity in the central  $5.2^\circ$  region—the fovea—of the retina. Outside this region, visual acuity falls off as we move toward the periphery of the retina. The fovea covers around 4% of pixels on consumer VR systems (Patney et al. 2016) with about 96% of the rendered pixels in HMDs falling in regions of the retina that have low visual acuity. Gaze-contingent (foveated) display systems (GCDs) exploit this phenomena to increase performance and reduce computing costs in HMDs by rendering content that falls outside the fovea with low resolution or level of detail (Guenter et al. 2012). When coupled with high quality eye tracking, GCDs could help future HMDs to have a wide field-of-view with higher resolutions and refresh rates. On top of reducing the computational requirements and improving speed in VR systems, GCDs also have other highly relevant advantages for VR that include reducing streaming content bandwidth requirements, reducing visual discomfort in VR and alleviating VR sickness.

### 3.1.1 Improving rendering efficiency

Foveated Rendering renders the areas of the display that lie at the user's center of eye gaze with the highest resolution and degrades the resolution with increasing eccentricity (Patney et al. 2016; Guenter et al. 2012). This leads to improvements in rendering performance and rendering quality, and has been shown to achieve up to 50–70% in performance savings (Weier et al. 2017). Popular implementations of foveated rendering include those of Guenter et al. (2012) and Patney et al. (2016).

Compression of the peripheral parts of the scene, however, could introduce various perceptible artifacts such as tunnel vision, aliasing and flicker that could distract users and reduce immersion (Patney et al. 2016). To address these artifacts, Guenter et al. (2012) used three gaze centered concentric circles with resolution degrading progressively towards the periphery. Whereas Turner et al. (2018) propose a technique to reduce motion induced flicker in the periphery using phase-alignment—aligning the rendered pixel grid to the virtual scene content during rasterization and upsampling.

### 3.1.2 Reducing transmission load

Streaming immersive omnidirectional video (ODV), also known as  $360^\circ$  video, to VR devices is a growing trend. When ODV is viewed in VR, it allows the user to look around a scene from a central point of view and provides a more immersive visual experience than traditional 2D video playback. However, streaming ODV across content

delivery networks and displaying it in a VR device are challenging in part due to the large resolution requirement of the video. Foveated rendering could be used to reduce computational rendering cost once the streaming content is available in the user device. However, streaming the content from where it is stored to the end user's device is in itself a big challenge. As discussed above, viewers can only watch a small part of the streaming content due to the physiological constraints of the human eye. As a result, gaze-contingent (foveated) transmission techniques have been proposed to minimize the amount of data transmitted to the user's device (Lungaro et al. 2018; Romero-Rondón et al. 2018; Ozcinar et al. 2019). These techniques generally aim to reduce the amount of data transferred to the user's device using gaze-adaptive streaming. An exemplar gaze-adaptive streaming technique is that of Lungaro et al. (2018) where user's eye gaze is continuously tracked and gaze positions are transmitted to a *Foveal Cloud Server*, which, in return, transmits the content with high visual quality around the users' fixations points while lowering the bandwidth required to encode the content everywhere else. Evaluation of the technique showed that it could lower the bandwidth requirement for streaming VR content by up to 83%.

### 3.1.3 Reducing discomfort due to depth conflicts

Conventional stereoscopic 3D near-eye displays, like those used in VR HMDs, create 3D sensation by showing each eye a distinct 2D image where each image is rendered with slight differences to create binocular disparity. Binocular disparity is a critical stimulus to vergence, which is a critical depth cue. However, the distance between the user's eyes and the image is fixed by the location of the display screen. As a result, although the 3D imagery is displayed at various depths, the eyes are always focused at a single depth. Thus, the display does not depict correct retinal blur which leads to the inability of the eyes to focus or accommodate correctly, causing loss of accommodation—another critical depth cue. This mismatch is called vergence-accommodation conflict (VAC) and is a source of visual discomfort including eye strain, blurred vision and headaches (Kramida 2016; Shibata et al. 2011; Hoffman et al. 2008).

Matsuda et al. (2017) provide a review of several “accommodation-supporting” displays that have been proposed to address VAC, including varifocal displays, monovision displays, accommodation-invariant (EDOF) displays, multifocal displays, retinal scanning displays, light field displays and holographic displays. Out of these approaches, varifocal displays utilize eye tracking to mitigate VAC. Varifocal displays use eye tracking to

actively track the vergence of the eyes and use a focusing element with a variable focus to match the eye's vergence.

Ocular parallax is another important depth cue that current stereoscopic displays fail to render accurately (Krajancich et al. 2020). Ocular parallax is a depth cue caused due to the centers of projection and rotation not being the same in the human eye, thus, leading to the formation of small amounts of depth-dependent image shifts on our retina when we move our gaze (Kudo and Ohnishi 1998). Although ocular parallax is an important depth cue that could significantly improve the impression of realistic depth when viewing a 3D scene, conventional stereoscopic rendering techniques, that are widely used on VR systems, assume that the centers of projection and rotation are the same and do not render ocular parallax. Konrad et al. (2020) introduced a gaze-contingent ocular parallax rendering technique that tracks the user's eye gaze to render ocular parallax. The authors report that the technique improved perceptual realism and depth perception.

### 3.1.4 Reducing VR sickness

The potential of gaze-contingent displays to reduce the incidence of VR sickness has also been explored. Adhanom et al. (2020b) explored the utility of foveated field-of-view (FOV) restriction to reduce VR sickness. FOV restriction (tunneling) is a popular technique to reduce visually induced motion sickness (Fernandes and Feiner 2016) that involves blocking the peripheral view of users by a restrictor to minimize optical flow in the peripheral parts of the retina which are sensitive to it. Most current implementations of FOV restriction do not respond to eye gaze and could reduce immersion and the sense of presence. Foveated FOV restriction (Adhanom et al. 2020b), however, implements a restrictor that moves with the user's eye gaze that would allow the users to see a bigger part of the visual scene while still blocking their peripheral vision. This allows greater visual exploration of the environment when compared to fixed FOV restrictors.

### 3.1.5 Summary

Current gaze-contingent display techniques have shown great potential to make rendering more efficient and improve presence and comfort in VR. However, it is still unclear how degradation of the peripheral image impacts attentional behavior and task performance in VR. Previous studies have shown that non-immersive gaze-contingent displays affect task performance (e.g., reading speed) negatively (Albert et al. 2019); therefore, further research is needed to understand the effect of GCDs on task performance in VR. Moreover, most of the eye tracking devices integrated

in VR HMDs have high latency, lack precision and do not have simple calibration procedures. Latency could have a negative effect on gaze-contingent rendering by introducing perceptible delays to the rendering pipeline which could reduce immersion and cause discomfort in VR. Various approaches have been proposed to address latency including saccade end-point prediction (Arabadzhiyska et al. 2017), and machine learning-based gaze position prediction (Hu et al. 2019) techniques. However, there is still room for improvement and further research is needed to reduce latency in GCDs.

## 3.2 User interaction

Eye movement-based interaction interfaces employ real-time eye movements as a user interaction modality. Eye movement-based interaction could be particularly useful in situations where other interaction modalities are not preferred or available, for instance when the user has severe motor disabilities or when the user's hands are occupied with other tasks. Although, using eye movements for interaction may not be as accurate as using hand-based controllers in VR, eye gaze can be much faster than conventional input devices (Špakov et al. 2014; Sibert and Jacob 2000). By eliminating or reducing the number of hand-based gestures in VR, eye gaze-based interaction has the potential to reduce the so-called gorilla arm syndrome—arm fatigue due to prolonged hand-based midair gestures (Boring et al. 2009; Cockburn et al. 2011)—which has been shown to limit the amount of time users spend in VR (Jang et al. 2017). Similar to LaViola Jr et al. (2017), we classify the interactive applications into three categories: selection and manipulation, virtual locomotion and system control.

### 3.2.1 Selection and manipulation

Manipulation, one of the fundamental tasks in both physical and virtual environments, refers to interaction tasks that involve selecting and manipulating virtual objects in a virtual environment (VE). These could be distilled into basic tasks that include pointing at, selecting, positioning, rotating and scaling virtual objects (LaViola Jr et al. 2017).

*Selection* According to Bowman et al. (2001), a selection technique has to provide means to indicate an object, a mechanism to confirm its selection (confirmation of selection) and some form of feedback to guide the user during the selection task. Indication of an object can be accomplished through object touching, pointing, occlusion/framing or indirect selection. Eye gaze-based object indication is accomplished with pointing, whereas eye gaze-based confirmation of selection is accomplished through dwell and other bi-modal mechanisms (Bowman et al. 2001).

Pointing is a fundamental interaction task that allows the user to point at objects or interaction elements they intend to interact with. Eye gaze-based pointing allows the user to select and interact with objects that are at a distance. The most common way of implementing eye gaze-based pointing in VR is to use the 3D gaze direction vector provided by the eye tracker, and to observe which objects in the scene intersect with the direction vector (Mohan et al. 2018; Sidenmark and Gellersen 2019; Cournia et al. 2003). Usually, a ray is cast based on the direction vector, and the first intractable object that the ray intersects with is considered to be the item that is being pointed at.

Various studies show that gaze-based pointing is faster than hand-based pointing as we are able to move our gaze faster to a target than our hands (Sidenmark and Gellersen 2019; Tanriverdi and Jacob 2000). However, due to inherent physiological characteristics of eye movements and technological limitations of eye tracking, eye gaze-based pointing is inaccurate compared to other common pointing interfaces such as hand- or head-based pointing (Špakov et al. 2014; Hansen et al. 2018; Qian and Teather 2017; Luro and Sundstedt 2019). The two main forms of inaccuracies in eye gaze-based pointing interfaces are caused by natural noise in eye tracking data and low eye tracking data quality. These issues are discussed in detail in Sect. 4.

Confirmation of Selection allows the user to confirm the selection of an object after they indicate it with a pointing interface. Selection with eye gaze alone is a relatively challenging task, necessitating the implementation of further mechanisms to enable selection when using eye-based interaction in VR. An added benefit of implementing other selection confirmation techniques is remedying the Midas touch problem (Jacob 1990)—an inherent problem for eye gaze only interaction techniques. The Midas touch problem arises from the difficulty of distinguishing between an intentional gaze interaction from natural eye movements—“everywhere you look, something is activated; you cannot look anywhere without issuing a command” (Jacob and Stellmach 2016).

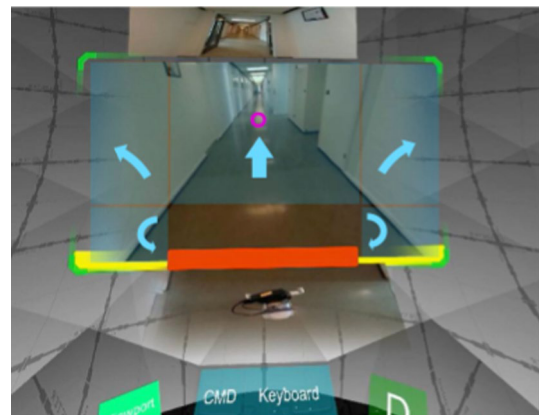
Various techniques have been used to implement selection confirmation for gaze-based interaction in VR. Hansen et al. (2018) implemented a technique that uses eye gaze-based dwell for selection confirmation. Sidenmark and Gellersen (2019) implemented two head assisted techniques: Eye & Head Dwell, a confirmation technique where a dwell timer is only triggered by head-supported gaze shift but can be paused and resumed with eyes-only gaze; and Eye & Head Convergence, an alternative technique to dwell for fast target confirmation that allows users to confirm selection by aligning both the eye pointer and the head pointer over a target. Kumar and Sharma (2016) implemented a technique that uses blink or wink gestures for selection confirmation. Pfeuffer et al. (2017) allows users to point at objects with

their eye gaze and select them with a pinch gesture with their hands. Pai et al. (2019) introduced a technique where users pointed at targets with their gaze and triggered actions using arm muscle contractions that were detected using electromyography. Qian and Teather (2017) used a keyboard button press for selection confirmation and eye gaze for pointing. Sidenmark et al. (2020) proposed a technique—Outline Pursuits—that utilizes smooth pursuits to allow users to select occluded objects in VEs.

**Feedback** A selection technique should provide feedback to the user to give them a clear indication of the system’s status: Is the pointer following the eye gaze accurately? Has the system accurately recognized the intended target? and has the system selected the correct object (Majaranta and Bulling 2014)? As the eyes are sensitive to visual changes in the field of view, they instinctively try to shift attention to these visual changes. Thus, care should be taken when providing feedback to the user, as visually prominent feedback mechanisms could have unintended consequences of shifting the users gaze unless the feedback is provided on the selected object itself. Using non-visual feedback, such as auditory feedback, Boyer et al. (2017) would be an alternative approach. Examples of visual feedback for eye gaze-based interaction include: highlighting the selected object (Blattgerste et al. 2018); displaying an outline around the selected object (Sidenmark et al. 2020), and showing confirmation flags around the selected object (Mohan et al. 2018).

### 3.2.2 Virtual locomotion (travel)

Virtual locomotion is the act of navigating VEs in VR. Designing efficient and universally accessible locomotion techniques presents considerable challenges (Al Zayer et al. 2020). Eye movements have been used to develop virtual locomotion interfaces that map eye movements to the control



**Fig. 2** A gaze-interactive user interface overlaid over a virtual environment. Image courtesy of Zhang et al. (2019)



of the virtual viewpoint's translation and orientation or to augment other navigation techniques Fig. 2.

Several implementations of eye movement-based virtual navigation use eye gaze-based steering interfaces, where 2D gaze interfaces are superimposed on the VE to allow users to issue steering commands with their gaze (Stellmach and Dachselt 2012). Zhang and Hansen (2019) proposed a steering-based virtual navigation technique that is used to train users who are disabled how to control wheeled tele-robots with their eye gaze. A similar implementation is that of Araujo et al. (2020), except that the proposed eye gaze-based virtual navigation technique is used to train people who are severely disabled on how to control wheelchairs in a safe simulated virtual environment. This review also included two other control interfaces in addition to overlaid steering user interfaces: continuous-control interface and semi-autonomous waypoint interface. In the continuous control interface, steering is implemented by measuring the depth and horizontal values of the gaze point intersection with the ground plane, and these measurements are used to calculate motor torque to drive the wheelchair. The semi-autonomous waypoint navigation allows the user to select waypoints (targets) in the ground plane using a dwell action; then, the application directs the motion of the user through the fastest route to the selected waypoint. The study found that the semi-autonomous waypoint-based method had superior performance compared to the two other techniques. Other implementations of eye gaze-based virtual navigation use point and fly techniques (Qian and Teather 2018; Zeleznik et al. 2005), where participants fly to a point in the VE by gazing at it. Eye tracking has also been used to implement orbital navigation techniques (Pai et al. 2017; Outram et al. 2018). Orbital navigation techniques—which are similar to fly-by camera shots in film and sports coverage—allow the user to move in an orbital path around points of interest. They maintain the point of interest in sight at all times and are particularly suited to observational tasks (Outram et al. 2018).

Eye tracking is also used to augment other popular navigation techniques like redirected walking. While real walking-based locomotion is the most natural way to travel in VEs, it is constrained by the size of the available physical tracking space. One interesting approach to overcome the limit of the physical space is using redirected walking techniques, whose primary goal is to allow the user to navigate VEs far greater than the physical tracking space. This is accomplished by manipulating the transformations of the user's movements within the VE to give them the illusion of walking in straight paths in the VE while in reality they are walking in a curved path (Al Zayer et al. 2020). Ideally, the manipulations should be subtle enough to be imperceptible to the user. Techniques that detect the

user's eye movements to apply subtle translation of the user's viewpoint during blinks (Nguyen and Kunz 2018) or saccades (Sun et al. 2018) are among the various approaches used to make translation changes imperceptible in redirected walking. Joshi and Poullis (2020) have proposed a technique that relies on inattention blindness and foveated rendering to apply spatially varying transformations to different parts of the scene based on their importance in the field of view. Other eye movement-based approaches use the user's eye gaze and probabilistic methods to predict the user's future locomotion targets to reduce the number and magnitude of translation changes (Zank and Kunz 2016).

### 3.2.3 System control

LaViola Jr et al. (2017) defines system control as an interaction task in which commands are issued to: (1) request the system to perform a particular function; (2) change the mode of interaction, or (3) change the system state. System control allows a user to control the flow of tasks in the system.

Perhaps the simplest form of eye gaze-based control mechanism is using blink as a binary input modality similar to a switch. Two example applications of blink-based control are the study by Xiao et al. (2019) that presented a technique where users can issue commands to the system, to control a VR-based music-on-demand system, by blinking in synchrony with a target button from several flashing buttons; and the study by Kumar and Sharma (2016) that proposed a technique where users could control a game using various blink commands including blink, double blink and wink eye movements. Requiring users to alter their natural blink rate, however, can cause eye strain, dry eyes and eye fatigue in users (Hirzle et al. 2020). Subjective results from the study by Kumar and Sharma (2016) also indicate that frequent blinking and winking leads to eye fatigue in users. Blink-based interfaces tend to be inaccurate because voluntary (intentional) blinks are hard to distinguish from natural blinks and thus require users to perform extended blinks. Extended blinks, however, have obvious disadvantages like slowing down the flow of interaction and blocking the user's sight for the duration of the extended blink. Consequently, eye gaze-based system control applications mostly rely on the point-and-select paradigm discussed in Sect. 3.2.1.

Symbolic input—the input of characters and numbers—is an important and fundamental system control task. Symbolic input in VR remains to be challenging due to the fact that users' eyes are obstructed from the physical world making the use of conventional text input devices, like physical keyboards, challenging, if not impossible.



As a result, eye gaze has been explored as a potential symbolic input modality for VR. Gaze-based text entry in VR could be considered as a special form of target selection, where user's use their gaze to select the keys of a virtual keyboard displayed in the VE. Rajanna and Hansen (2018) developed an eye gaze-based typing interface in VR and investigated how the keyboard design, selection method, and motion in the field of view may impact typing performance and user experience. The results of the study showed that gaze typing in VR is viable but constrained. Users' perform best when the entire keyboard is within view compared to the larger than view keyboard (10.15 WPM vs 9.15 WPM), in addition the larger than view keyboard induces physical strain due to increased head movements. Ma et al. (2018) developed a hybrid gaze-based text entry system for VR by combining eye tracking and brain-computer interface (BCI) based on steady-state visual evoked potentials (SSVEP). The authors designed a 40-target virtual keyboard to elicit SSVEPs and to track gaze at the same time. The study compared the performance of the hybrid system to single-modality eye tracking-based and BCI-based techniques. The results indicate that the hybrid method has better performance than both the single-modality systems achieving a typing speed of around 10 WPM. It should be noted that gaze typing is very slow compared to conventional typing interfaces such as a conventional keyboard.

### 3.2.4 Summary

Eye tracking has the potential to enable accessible and handsfree ways of input that require very little exertion (Majaranta and Bulling 2014). However, one technical challenge common to eye movement-based interaction applications is the rapid, accurate and precise identification of these eye movements that allows them to be used in real time as an input signal. Consequently, the exploration of new techniques that improve the quality of eye tracking data is an area that needs further research. With more sensors being embedded in VR HMDs, sensor fusion (i.e., combining eye tracking with other sensors) is another area for future research that holds promise to further increase the accuracy and robustness of using eye tracking for input. Eye tracking could be explored to enable individuals with severe motor impairments (i.e., quadriplegics or persons who are locked in) to navigate their avatar in VR. Though there might be significant challenges with using eye tracking for precise navigation, some sort of semi-autonomous approach could be used with a user providing broad navigation instructions. Of interest would be how this approach would affect VR sickness. Finally, our review reveals that there is scant literature on the use of eye-based input for manipulation tasks in VR.



**Fig. 3** Photorealistic virtual avatar created through reconstruction of gaze and eye contact. Image courtesy of Schwartz et al. (2019)

## 3.3 Collaborative virtual environments

Collaborative virtual environments (CVEs) allow multiple co-located or remote users to interact with each other and to collaborate on tasks. CVEs differ from other traditional group interaction or teleconferencing applications, such as video conferencing, instant messaging, and email, in that, instead of connecting users from different environments (locations), CVEs create an immersive virtual environment common to all the participants and immerse the users and task related information into that environment. This shared virtual environment provides users with a common spatial and social context for interaction and collaboration (Roberts et al. 2003). Eye tracking in collaborative virtual environments has been used to improve user representation, to aid multiparty communication and as a modality for user interaction Fig. 3.

### 3.3.1 Representation

Users in CVEs are generally represented as virtual avatars. Natural looking and personalized avatars have been shown to improve immersion and presence in VR and to aid better communication between users Waltemate et al. (2018). Rendering the human face accurately, however, is still a particularly challenging problem. This stems from the extreme sensitivity of humans to rendering artifacts in photo-realistic facial renderings which leads to the Uncanny Valley problem (Mori et al. 2012). Previous studies show that avatars with realistic eye gazes look more natural and realistic (Garau et al. 2003); therefore, virtual avatars that have human face models should be modeled with realistic eye movements or eye movements that are consistent with

human ocular behavior. To this end, eye tracking has been used to monitor users' eye movements to allow for better reconstruction of avatars' eyes and faces.

Different approaches have been used to create realistic artificial eyes for avatars in CVEs. The first approaches modeled the eyes as spheres or half spheres and used high resolution pictures of the human eyes as texture maps (Ruhland et al. 2014) and were not anatomically realistic. Later approaches, however, have considered the eye's physiological features and tried to recreate them in artificial eyes. An interesting example in this area of research involves modeling changes in pupil diameter that occur as reactions to changes in light or as a function of emotional arousal (Ruhland et al. 2014). However, in order for the artificial eyes to appear realistic, they should also try to replicate the characteristic movements of the human eye. To this end, various studies have tried to animate realistic eye movements on avatars. Most of these techniques rely on eye trackers to accurately track the user's eye movements and various models to reconstruct human eye movements based on the physiological characteristics of the eye. Ruhland et al. (2014) and Duchowski (2018) have provided an excellent review of the most widely used techniques for avatar eye movement animation.

Recent developments in deep learning have led to more efficient techniques for human face reconstruction with multimodal approaches that use eye tracking in combination with other modalities. An example application in this line of research is the recent work by Richard et al. (2020) which proposes a technique to generate photo-realistic face models using audio and eye gaze signals. In this work, the authors use codec avatars (Lombardi et al. 2018)—a class of deep models for generating photo-realistic face models that accurately represent the geometry and texture of a person in 3D that is almost indistinguishable from video. The authors develop a real-time solution for the animation of the avatars full-face geometry and texture using deep multimodal models trained on audio and gaze data. While simple face animation models may use simple reconstruction approaches, for example: eye gaze only affects eye shape and audio only affects mouth shape, multimodal approaches use multiple data streams, such as gaze and audio, to dynamically determine how these signals collectively affect each facial feature.

### 3.3.2 Communication

According to Burgoon et al. (1994), social human communication encompasses verbal and non-verbal communication, references to objects and references to the environment. The eyes play an important role in non-verbal communication (Steptoe et al. 2010) and in providing references to objects and the environment. Deictic gestures

are one of the most fundamental forms of communication that allow users to indicate objects they are referencing (Mayer et al. 2020). Although the typical deictic gesture in humans is performed through pointing by extending the arm and the index finger, previous work has shown that pointing in VR has limited accuracy (Mayer et al. 2018). As a result, eye gaze has been used as a natural way of providing deictic references (i.e., indicating where your partner is looking at) in CVEs (D'Angelo and Gergle 2016). Among others, the study by Pejsa et al. (2017) is a good example, where the authors developed a gaze model that would allow virtual agents to signal their conversational footing, i.e., signaling who in the group are the speakers, addressees, by-standers, and over-hearers.

### 3.3.3 Interaction

In CVEs, interaction techniques are aimed at allowing the cooperative manipulation of objects or interaction elements in the VE. Cooperative manipulation (Co-manipulation) refers to the situation where two or more users interact cooperatively on the same object at the same time. Eye gaze-based interaction, among other methods, has been used in unimodal or multimodal fashion to allow CVE users to select or manipulate objects cooperatively. The recent work by Piumsomboon et al. (2017) is a sample application of eye tracking for cooperative manipulation. The authors present a collaborative system combining both VR and AR that supports bi-modal eye gaze-based interaction. In this system, collaborating users could use their eye gaze to select objects and cooperatively manipulate them with their hands. The technique also allows collaborating parties to gaze at the same target object to trigger an action.

### 3.3.4 Summary

The eyes are an important part of the face, as a result any future advancement in avatar representation in CVEs needs to be complemented with techniques that allow for precise representation and rendering of the eyes and their natural movements. Although great advancements have been made on representing users with high fidelity virtual avatars, there is still room for improvement in this area.

## 3.4 Education, training and usability

Learning is an important application of VR, and eye tracking in VR shows great promise for the assessment of learning outcomes and improving the learning process. VR has the potential to provide an immersive simulation-based training environment, where learners can hone their skills in an environment that is safe to fail and allows correction and repetition with minimal costs, while also allowing the

trainer to control the learning environment down to the smallest details. As a result, VR and eye tracking has been used for education and training in several domains including transportation Lang et al. (2018), military, medicine Xie et al. (2021) and sports Bird (2020); Pastel et al. (2022).

The most common traditional approaches of evaluating learning performance in simulated learning environments include post-experiment interviews, questionnaires and think-aloud protocols (Xie et al. 2021). However, since the first two methods collect data after the experiment, the participants' memory might fade and events might be reconstructed, potentially making the results subjective and irrelevant (Renshaw et al. 2009). Eye tracking allows the researcher to quantitatively evaluate the learner's visual, cognitive and attentional performance during the experiment without interfering with the participant's learning experience.

In their review of the use of eye tracking in learning, Lai et al. (2013) summarized the eye tracking indices that are most commonly used in training environments into two dimensions: types of movements (such as fixation, saccade, mixed) and scale of measurement (temporal, spatial, count). Temporal measurement indicates that eye tracking is measured in a time dimension, e.g., the time spent gazing at a particular area of interest or time to first fixation. Temporal measures aim to answer the when and how long questions in relation to cognitive processes. Spatial scale indicates that eye measurement is measured in a spatial dimension, e.g., fixation position, fixation sequence, saccade length and scanpath patterns. Spatial measures answer the where and how questions of the cognitive processes. The count scale indicates that eye movements are measured on a count or frequency basis, e.g., fixation count and probability of fixation count. Following this, Rappa et al. (2019) have identified several themes on how eye tracking has been used to aid learning and evaluate learning performance in VR, and how the scales of eye measurement summarized by Lai et al. (2013) relate to each theme. In this section, we explore applications of eye tracking in VR along these themes.

### 3.4.1 Measuring cognitive skills

Traditionally, the interview procedure-based think-aloud protocol has been the most frequently used technique to understand cognitive processes during learning (Lai et al. 2013; Xie et al. 2021). However, as discussed above, this method could be subjective and suffers from validity issues. Eye movement tracking allows researchers to identify and measure the aspects of the learning environment that influence cognitive skill acquisition. Previous studies show that temporal characteristics of fixation, such as fixation duration, are associated with cognitive effort (Renshaw et al. 2009). According to Kiili et al. (2014), longer fixation

duration might indicate increased cognitive effort suggesting that participants are engaging in analysis and problem-solving. Whereas shorter fixation duration might suggest that participants might be glossing over content because of their difficulty interpreting and comprehending it. Moreover, a review by Souchet et al. (2021) identifies that cognitive load can be measured with pupil diameter in learning tasks.

### 3.4.2 Measuring affective skills

VR simulations have been successfully used to teach interpersonal, communicative and other affective skills. Affect refers to feelings, emotions and moods. The most common current methods of assessing affect and affective skill acquisition in simulated learning environments are self-reported measures that usually interrupt the learning experience to collect verbal or questionnaire-based responses from participants, or methods that collect data after the experiment, which may produce unreliable or biased results as discussed above (Tichon et al. 2014). Tichon et al. (2014) underscore the importance of uninterrupted and continuous measures of affect, and introduce a technique that measures the anxiety of pilot trainees in a flight simulation using eye movements and pupillometry. The results of the study indicate that fixation duration and saccade rate corresponded reliably to pilot self-reports of anxiety which suggests eye tracking-based measures could serve as reliable measures of affective skill acquisition during simulated training.

### 3.4.3 Measuring visual attention

Visual attention refers to the ability of the human visual system to selectively process only relevant areas of interest from the visual scenes (Borji et al. 2019). Modeling users' visual attention in VEs allows us to understand how people explore VEs and what elements of the VE attract their attention. Previous studies indicate that experts and novices show different visual attention patterns in learning tasks (Duchowski 2018). However, previous research findings have been inconsistent when looking at the average fixation duration exhibited by experts and novices. Some studies show that experts tend to exhibit shorter average fixation duration; they make better use of visual elements outside their foveal region and make use of a larger visual span area (Duchowski 2018). A meta-analytic review by Mann et al. (1998) also showed that experts used fewer fixations of longer duration compared to no experts. The review points out that experts' eye movements are moderated by several factors including the task type and the environment. However, a study conducted outside VR by Eivazi et al. (2017) has found the opposite results showing that experts in a neurosurgery task showed longer fixation duration compared to novices. Moreover, a study by Harris et al.

(2020b) has found that there is no significant differences in fixation duration between experts and novices. Although visual attention in simulated learning environments has been used to measure the expertise level of trainees and to understand visual learning strategies of experts, the inconsistency of findings shows that there is still need for more research in this area.

#### 3.4.4 Assessing learning outcomes

Eye tracking could be used to analyze learning outcomes and learning performance in VR-based training scenarios (Tichon et al. 2014). Previous studies (Rappa et al. 2019) have found that average fixation duration and number of revisits can predict the learning outcomes or the effectiveness of VR-based training. Identifying and predicting learning outcomes could allow participants to identify their areas of strengths and weaknesses in the given learning task. Additionally, a visual behavior known as the quiet eye has been shown to be a characteristic of high levels of expertise, particularly in tasks that require motor skills (Vickers 2000). Quiet eye is the final gaze fixation prior to the execution of a movement towards a target. However, in a study of quiet eye in virtual reality, Harris et al. (2021b) found that the quiet eye had little to no impact in skill execution.

#### 3.4.5 Measuring immersion

Immersion refers to the objective level of sensory fidelity provided by a VR system, whereas presence is defined as the subjective sensation of being there in a scene depicted by a medium, usually virtual in nature (Bowman and McMahan 2007; Barfield et al. 1995). In simulated learning environments, it is widely believed that better immersion leads to better learning outcomes (Jensen and Konradsen 2018). Previous studies show that eye movements correlate with the immersiveness of virtual environments and could be used to measure immersion in virtual environments (Jensen and Konradsen 2018). Results from a study by Jennett et al. (2008) indicate that the participants' number of fixations per second significantly decreased overtime in an immersive condition and significantly increased overtime in a non-immersive condition. These results indicate that participants were more concentrated in the immersive condition, while they were more distracted in the non-immersive condition.

#### 3.4.6 Measuring usability

Usability is often associated with ease of use and is an important part of user experience. Usability deals with measuring the aspects of the virtual environment that make it easy to use, understandable and engaging (Kiili et al. 2014). The design aspects of the environment and the

elements within it could affect the cognitive performance of the participant (Kiili et al. 2014). Thus, simulated learning environments should be designed with optimal task performance in mind. Eye tracking could help in identifying factors that affect usability and be an objective measure of usability in VEs (Kiili et al. 2014). In particular, average fixation duration and gaze points within areas of interest indicate which elements of the scene attract the user's attention and are correlated with virtual environment design features (Renshaw et al. 2009).

#### 3.4.7 Summary

Eye tracking in VR has been used to assess learning performance in simulated learning environments. Care should be taken, however, in translating the eye movement-based performance measurements. Kiili et al. (2014) remind that eye tracking data reflect what the user perceives, but do not tell whether or not the user comprehends the information they were presented in the experiment. Developing accurate metrics that could measure knowledge comprehension based on eye tracking data is an open area for future research. On the other hand, the degree to which skills gained in simulated environments are transferable to the real world needs further research (Harris et al. 2021a). Moreover, there are concerns that differences between the actual task and the simulated training task, and the limited interaction and realism provided by VR could be detrimental to VR-based training programs (Ruthenbeck and Reynolds 2015; Harris et al. 2020a). More research is needed in this area to develop better eye tracking data interpretation methods to measure learning outcomes; to ensure that VR-based training tasks are effective; and to ensure that the skills learned in simulated environments can be transferred to the real world.

### 3.5 Security and privacy

As the adoption of VR expands, the number and variety of VR applications has been growing steadily. Many of these applications require users to authenticate themselves or to enter their personal details—tasks that require a high level of security. With the current surge in the integration of eye trackers in HMDs, eye movement data could be used to implement robust and non-intrusive authentication methods. Personal authentication and recognition techniques based on eye movements (not to be confused with iris recognition) have been used successfully for several years (Katsini et al. 2020). Due to differences in gaze behavior and oculomotor physiology, there are certain eye movement characteristics that are unique for each individual. This difference can be exploited to implement a biometric identification system.

A user's patterns of eye movements depend on the user's distinct characteristics and are very different among users.



Visual stimuli produce varied eye movement behaviors and patterns among users. Pupil size, saccade velocity and acceleration, fixation behavior and pupil center coordinate and its displacement between consecutive samples have been identified as dynamic eye movement features that are specific to individual users and suitable to be utilized for authentication (Kinnunen et al. 2010; Eberz et al. 2015; Liang et al. 2012). These eye movement metrics can be measured by most VR-based eye trackers and can be fed into a decision or classification algorithm to develop an eye movement-based authentication system.

Additionally, due to the fact that users' eyes are totally obstructed from outsiders when using HMDs and the idiosyncratic characteristics of eye movements, using eye movement-based authentication could be a spoof proof authentication method. Eye gaze data can be employed to enable explicit or implicit authentication of VR users. When eye gaze-based authentication is used in an explicit fashion the user has to first define a password that involves consciously performing certain eye movements, then the user is authenticated by recalling these movements and providing them as input. Mathis et al. (2020) presented an authentication method where VR users can be authenticated by selecting digits from a 3D virtual cube using their eye gaze, head pose or hand controller. The authors report that while the eye gaze-based method was slightly slower than when using the two other input techniques, it was highly resilient to observation-based attacks with 100% of observation-based attacks being unsuccessful.

Implicit eye gaze-based authentication refers to authentication methods that aim to verify the identity of users implicitly without requiring the user to remember a secret. Implicit authentication is based on the user's unconscious eye movements and can be performed continuously in the background. Various studies have explored implicit eye movement-based biometrics in VR (Zhang et al. 2018; Iskander et al. 2019; Liebers and Schneegass 2020; Pfeuffer et al. 2019; Lohr et al. 2018). The main advantage of implicit authentication methods over explicit authentication methods is the fact that they can be performed in the background without interrupting the user and can be performed continuously throughout a session to ensure intruders do not access already authenticated sessions. Studies show that implicit eye gaze-based authentication methods have comparable or better accuracy than explicit authentication methods, with some studies reporting that 86.7% accuracy can be achieved with only 90 seconds of data (Zhang et al. 2018).

### 3.5.1 Summary

Eye movement data holds great potential to be used for explicit or implicit identification of VR users without

interrupting their virtual experience. On top of developing better algorithms for implicit eye movement-based authentication, future work in this area could include the development of intrusion-proof transmission and storage techniques or protocols for eye tracking data that take into consideration the highly sensitive nature of the data. Moreover, the development of eye tracking data privacy practices and policies that outline and restrict the use of the data only for necessary purposes is an important area for future research.

## 3.6 Marketing and consumer experience (CX) research

Eye tracking can help us to better understand consumer's visual attention and provides rich information about the cognitive and emotional state of consumers (Lohse 1997). Most of the current research on consumer behavior that involves collection of eye tracking data is conducted in laboratory settings, with a majority of studies using 2-D pictures and desktop eye trackers, however the behavior observed in the laboratory may be different from what is observed in the field (Kahn 2017). Therefore, various studies suggest that consumer research should be done in the real world or at the point of sale (Meißner et al. 2019). However, conducting research in the real world—in the field—is cumbersome and could introduce uncontrolled factors to the experiment which in turn could make the experiment hard to replicate. Several reviews (Bigné et al. 2016; Wedel et al. 2020; Alcañiz et al. 2019; Loureiro et al. 2019) provide an overview of the various advantages of using VR for customer research. VR allows the experimenter to control every detail of the experiment similar to experiments conducted in laboratory settings, while providing an immersive virtual environment that feels like reality and provides the customer with freedom of movement similar to a real-world shopping experience. As a result, eye tracking in VR is a powerful tool for CX research. Lemon and Verhoef (2016) conceptualized CX as a customer's journey over time during the purchase cycle, and divided the purchase journey into three stages: the pre-purchase stage, the purchase stage and the post-purchase stage.

*The pre-purchase stage* encompasses the CX from when the customer recognizes the need to purchase a product up to the time the need is satisfied with a purchase. Eye tracking is widely used in the pre-purchase stage. This stage includes various marketing practices such as advertising, design of product attributes such as package design and testing new products.

Eye tracking plays an important role in understanding the effectiveness of advertisements. Advertisement has been a key application area for VR and augmented reality (AR) in marketing. In 2017, \$167 million was



spent on VR and AR advertising alone, and this figure is expected to grow to 2.6 billion in 2022. This growth is in part driven by recent findings that show that VR and AR ads have higher engagement and click through rate than traditional advertising methods (Grudzewski et al. 2018; Van Kerrebroeck et al. 2017). For example, Hackl and Wolfe (2017) mention that marketing campaigns that use AR have an average dwell time of 75 s, compared to an average dwell time of just 2.5 s for traditional radio and TV ads, and that 71% of shoppers would shop at a retailer more often if they used AR. Wang et al. (2019) have proposed a system for the analysis of dynamic user behavior when watching VR advertisements. The experiment used eye tracking data to measure the participants' visual behavior around the virtual ads.

Product attributes, such as its package's design, play an important role in marketing and influencing the consumer's purchase choice. Creusen and Schoormans (2005) identified that product appearance plays an important role, which includes drawing the consumer's attention among other roles. Various studies have shown that consumers choose the products that attract their visual attention (Bigné et al. 2016). Visual attention (see section 3.4) can effectively be measured using eye tracking. Most studies on product appearance are conducted in laboratory settings which fail to capture the visual context in which the product will be situated. Product appearance studies in the real world, on the other hand, fail to control all experiment variables which could greatly affect experiment results. VR may help researchers to close this gap by allowing them to rapidly develop new product prototypes and place them in immersive environments while giving the experimenter the ability to control all aspects of the experiment environment. A particularly good example is a recent study conducted by Meißner et al. (2019) that combines VR with eye tracking to provide a naturalistic environment for research on package design. Rojas et al. (2015) conducted a study in which the consumer perception of a virtual product with photographic representation of the

same product. Eye tracking data were used to analyze the participants' gaze behavior during the experiment Fig. 4.

*The purchase stage* includes all customer interactions with the brand and its environment during the purchase event itself. At this stage, eye tracking is mainly used to examine the user's behavior to understand how they process the information at the point of sale so we can get insights into their product choice, customer experience, and shopping behavior. This insight could then be used to drive the user towards new purchases (e.g., recommender systems) and to customize the shopping experience. Example applications of eye tracking and VR at this stage include a study by Bigné et al. (2016) where the authors developed a system that uses data from eye movements, and a VR-based virtual shopping experience, to track consumer behavior and report consumer paths, seeking behavior, purchase behavior, and the time a person spends on each task. Pfeiffer et al. (2020) demonstrates that eye tracking can be used at the point of sale to tailor recommender systems to individual customers' shopping motives. They conducted two experiments where one was conducted in VR and the other was conducted in a physical store. The results of the experiment indicate that the information search behavior of users in VR might be similar to the one used in the physical store.

*The post-purchase stage* encompasses customer interactions with the brand and its environment following the actual purchase. Although there is an increasing number of VR applications in this stage, we could not find applications that use both VR and eye tracking.

### 3.6.1 Summary

Eye tracking coupled with VR provides a powerful tool that can help advance marketing and CX research in various areas including new product development, advertising, and assessment of consumer behavior and engagement. The applications of VR with eye tracking are under-utilized at the post-purchase stage, where there are potential promising applications that could help companies to analyze consumer behavior and interactions with the purchased product or service.

## 3.7 Clinical applications

Clinical applications of eye tracking in VR mainly include diagnostics and assessment, therapeutic uses and interactive uses in a clinical context.

### 3.7.1 Diagnostic applications

The potential of using eye movements to identify markers of psychiatric, neurological and ophthalmic disorders is well researched (Trillenberget al. 2004; Clark et al. 2019).



**Fig. 4** A combined image showing which items in a virtual store attract the user's visual attention. Image courtesy of Tobii AB

Because different eye movement types are controlled by different brain regions and different neural circuits, examination of the kinematics of various types of eye movements can provide clues to disorders in the underlying neural structures (Trillenberget al. 2004; Tao et al. 2020). Targeted VR applications that mimic diagnostic tasks used by clinicians can be used to elicit eye movement abnormalities associated with psychiatric and neurological disorders. Eye tracked VR, thus, can be used to develop complete diagnostic tools for psychiatric, neurological and ophthalmic diseases.

Eye tracking in VR has shown great potential in the diagnosis of neurodegenerative conditions such as Parkinson's disease (PD) and Alzheimer's disease (AD). The diagnosis of neurodegenerative disease is usually made by physicians based on visible signs and symptoms. However, these signs and symptoms may take years to develop. Previous studies show that early stages of PD and AD can be detected through the observation of eye movement abnormalities. For example, Orlosky et al. (2017) developed a VR and eye tracking-based system to diagnose neurodegenerative disease and evaluated the system by conducting experiments on patients with Parkinson's disease. The main focus of the system is to evoke eye movement abnormalities using virtual tasks in VR associated with neurodegenerative disease so that a correct diagnosis can be made by observing these abnormalities.

Furthermore, VR with eye tracking has been employed to develop diagnostic tools for various ophthalmic diseases and disorders. Tatiyosyan et al. (2020) used a VR headset with an eye tracker to develop an optokinetic nystagmus (OKN)-based tool to test contrast sensitivity. Measuring contrast sensitivity in low vision patients is used to determine the stage of visual impairment. Miao et al. (2020) used a VR headset with integrated eye tracking to develop an automated test for ocular deviation for the diagnosis of strabismus in patients. Current diagnosis of strabismus uses cover tests that rely on the doctor's experience, which could be susceptible to human error.

### 3.7.2 Therapeutic applications

In the clinical context, eye tracking in VR has been used for neuropsychological, physical and ophthalmic therapeutic interventions (Lutz et al. 2017). In therapeutic applications, eye tracking is commonly used as an objective measure of the patient's symptoms. The objective metric could then be used to assess the progress of treatment or to personalize the treatment experience to the patient.

Eye tracking has been used as an objective metric in the VR-based treatment of Generalized anxiety disorder (GAD) and various phobic disorders. GAD is a mental health condition marked by excessive, exaggerated and consistent

anxiety and worry about everyday life events. Phobias are similar to anxiety disorders, but the anxiety is specific to an object or situation. The treatments for those disorders commonly require patients to confront the situations they fear through a process known as exposure therapy. Although exposure therapy has been proven to be highly effective, recreating the feared situations in real life is challenging and could put the patient in danger, for example, exposing a person who has fear of heights in an actual elevated space. VR, on the other hand, can provide a safe simulated environment for exposure therapy. Various studies have demonstrated that persons with psychological disorders show attentional biases and different eye movement patterns when they are exposed to the situation they fear. Consequently, eye tracking has been an integral part of virtual reality exposure therapy (VRET). For example, in a VR-based therapy for social phobia, Grillon et al. (2006) used eye tracking movement tracking to objectively assess eye gaze avoidance, a persistent symptom in social phobia.

VR-based applications have a demonstrated potential for treatment of various psychiatric and neurological disorders. However, the treatment tasks should be managed effectively to keep the user engaged and to ensure that the task is within the user's capability. Bian et al. (2019) discusses that a task that is too difficult for the patient could be overwhelming and cause anxiety while a task that does not fully utilize the patient's capability might cause boredom. Eye tracking in VR can be used to provide real-time data to assess the performance and engagement of patients and create a feedback loop to dynamically update the treatment experience in response to the user's engagement and performance.

### 3.7.3 Interactive applications in a clinical context

Various clinical applications of VR require the user to interact with the environment. Many patients, however, might not have the physical ability (e.g., patients with motor disabilities) to use a hand-held VR control device. Previous studies have shown the suitability of eye tracking-based interaction interfaces for clinical VR applications. Eye tracking-based interaction is mainly used to increase the immersiveness of the VR experience to illicit a stronger sense of virtual presence in patients.

Al-Ghamdi et al. (2020) evaluated the effectiveness of VR-based therapy at reducing the pain experienced by patients with severe burn wounds during non-surgical wound debridement procedures. However, the injuries prevent most patients from using conventional VR controllers to interact with the virtual environment. Thus, they investigated whether eye tracking-based interaction can enhance the analgesic effectiveness of the VR-based distraction for pain management. The results of the study indicate that

interactive eye tracking improved the immersiveness of the virtual environment and as a result increased how effectively VR reduced worst pain during a brief thermal pain stimulus. Another study by Pulay (2015) proposed the use of interactive eye tracking to motivate children with severe physical disabilities (like Tetraparesis spastica) to take an active role in their VR-based rehabilitation programs.

### 3.7.4 Summary

Eye tracking in VR in the clinical context is used for diagnostic, therapeutic, and interactive purposes. Neuro-ophthalmic diagnosis is traditionally conducted in a very rudimentary manner at the patient's bedside. Development of uniform HMD-based diagnostic tools with precise stimulus control to elicit specific and relevant eye movements, e.g., pursuit, saccades, nystagmus, etc., along with automated analysis of the resulting eye movements holds great promise.

Clinical applications could also benefit from the development of more usable applications that are easy to comprehend and use for the patient as well as the clinician practitioner. This has the potential to allow patients to self-diagnose or self-treat neurological disease and provide clinicians with easy-to-use tools.

Most current clinical applications of VR and eye tracking use consumer hardware that may not be appropriate for clinical use. For example, Lutz et al. (2017) mention that most HMDs have to be modified by removing, enclosing or replacing their textile foam and Velcro components in order to comply with clinical hygiene regulations. Most HMDs and their eye tracking components also cannot withstand clinical disinfection procedures. Thus, there is still work to be done to produce clinical grade HMDs.

In conclusion, as the data quality of eye trackers, and the VR hardware and software continue to improve, we expect the clinical applications of eye tracked VR to continue to grow.

## 4 Challenges and limitations

In this section, we discuss the inherent challenges and limitations of eye tracking and how these challenges affect eye tracking in VR.

### 4.1 Technological limitations

#### 4.1.1 Eye tracking data quality

Issues with eye tracker data quality are arguably the biggest technological challenges for eye tracking in VR. Table 1 shows the manufacturer reported data quality specifications for the currently most used HMD-based eye trackers.

*Spatial Precision:* Applications that use small fixational eye movements, like tremors, drifts and microsaccades require high-precision eye trackers. Andersson et al. (2010) point out that for such tasks the eye tracker should have an RMS precision value that is lower than  $0.03^\circ$ . Table 1 shows that not all manufacturers of VR-based eye trackers report the precision value of the eye trackers, and those that have reported precision values seem to have precision values that are worse than the recommended range needed to detect small fixational movements.

*Spatial Accuracy:* Table 1 shows that the most popular HMD-based eye trackers report accuracy values between  $0.5^\circ$  and  $1.1^\circ$ . However, manufacturer reported specifications could be misleading, as these metrics are often measured under ideal conditions and do not reflect the accuracy under realistic usage scenarios (Adhanom et al. 2020a).

Additionally, most current HMD-based eye trackers have the highest accuracy and precision in a small central region of the FOV. Outside this region, accuracy and precision drop substantially. For example, the HTC Vive Pro Eye reports accuracy values of  $0.5^\circ$  -  $1.1^\circ$  within the central  $20^\circ$  of the FOV. Outside this region, the accuracy is not guaranteed. This limits researchers or developers from using the whole FOV for eye tracking-based experiences.

**Table 1** Current widely used HMD-based eye trackers and their manufacturer specifications. Note that the manufacturer specifications are calculated under ideal conditions and may be hard to reproduce in realistic settings

| Device             | Accuracy                  | Precision    | Sampling rate              | Latency                                                       |
|--------------------|---------------------------|--------------|----------------------------|---------------------------------------------------------------|
| Fove-0             | $1.15^\circ$              | Not reported | 120 Hz                     | Not reported                                                  |
| HTC Vive Pro Eye   | $0.5^\circ$ - $1.1^\circ$ | Not reported | 120 Hz                     | Not reported                                                  |
| Pico Neo 3 Pro Eye | Sub-degree gaze accuracy  | Not reported | 60/90 Hz                   | Not reported                                                  |
| Pupil Labs         | $1^\circ$                 | $0.08^\circ$ | 200 Hz                     | 8.5 ms Camera latency and 3–4 ms processing latency on i5 CPU |
| Varjo VR-1         | $1^\circ$                 | $0.2^\circ$  | 100 Hz (default) or 200 Hz | 20–30ms                                                       |
| Varjo VR-3         | Sub-degree accuracy       | Not reported | 200 Hz                     | Not reported                                                  |

Regardless of the application, performance across all categories would benefit from improvements in the precision and accuracy of eye tracking and also from improvements in identification and labeling of eye movement types. Advances in both of these areas can be achieved by leveraging additional sources of information beyond the eye images that underlie VOG. Simultaneous use of eye data and scene information can help constrain eye position estimates based, for example, on the assumption that users are much more likely to look at an object than the empty space next to an object (Tafaj et al. 2012). Similarly, simultaneous analysis of both head and eye movement data allows reconstructing gaze in world coordinates, which allows identifying eye movement types that would be difficult to identify based on eye movement alone (Hausmann et al. 2020; Kothari et al. 2020). As with most current information processing challenges, application of machine learning techniques has shown promise in advancing eye tracking precision, accuracy, as well as identification of eye movement types (Kothari et al. 2020; Yiu et al. 2019).

**Latency:** High latency could cause elements of the stimulus that are supposed to respond to eye movement to lag behind the eye movements. When this lag is big enough that it is perceptible by the user, it could severely affect the virtual experience. This could be particularly problematic in interactive applications or GCDs which change some part of the stimulus in real time in response to eye movements. Albert et al. (2017) points out that foveated rendering is tolerant to latencies of about 50–70 ms. For latencies beyond this range, foveated rendering could be perceptible to the user and lose its effectiveness. Although latencies have greatly decreased in current HMD-based eye trackers, a recent study analyzed the latency for most of the currently available HMD-based eye trackers and reported that the latencies for the mostly used HMD-based eye trackers ranged from 45 ms – 81 ms (Stein et al. 2021). Although these latencies do not cause significant issues for non-real-time applications, they could hamper real-time use of eye tracking data in VR.

**Sampling Rate:** As discussed above, the precise measurement of small amplitude eye movements such as fixational eye movements and low amplitude saccades requires eye trackers with high sampling rates. In Sect. 2.3.1, we have discussed that the sampling rate should be at least twice the speed of the eye movement to be recorded. Table 1 shows that current HMD-based eye trackers have sampling rates that fall in the range 100–200 Hz. These relatively low sampling rates show that most of the commonly used HMD-based eye trackers are not well suited to accurately record low amplitude eye movements.

#### 4.1.2 Calibration

The time consuming and repetitive nature of the calibration procedure could be an obstacle for the wide adoption of eye tracking and could make eye tracking unattractive for applications that require instant use.

Moreover, some users, such as children and users with attentional deficits, have difficulty completing the calibration procedure as they lose interest in the procedure after a few targets have been shown resulting in an unsuccessful calibration procedure (Blignaut 2017). Alternative eye tracking procedures have been explored to address the issues with the calibration procedure, the most common of which use smooth pursuit eye movements to dynamically calibrate the eye tracker without explicitly asking the user to look at point targets (Blignaut 2017; Drewes et al. 2019). Although these methods require less time and can be performed without the user being aware of the calibration procedure, they generally produce lower quality eye tracking data. As a result, these calibration methods are not in use in any HMD-based eye tracker we are aware of (Drewes et al. 2019).

After the calibration procedure, changes in lighting, eye geometry and the relative position of the eye tracking camera with respect to the user's eyes could cause calibration errors. The latter is the main cause of calibration error in HMD-based eye trackers, as small movements of the VR headset due to the user's movements could cause calibration errors. All these factors together cause the calibration to decay—the calibration error and spatial accuracy of the eye tracker worsens over time. We call this decay *drift*. Drift is a common cause of low quality eye tracking with some eye trackers showing calibration drift of about 30% in the first 4 minutes and 30 seconds after calibration (Ehinger et al. 2019). Calibration errors are hard to deal with due to their dynamic nature. However, the severity of calibration errors could be reduced by making sure the HMD does not move relative to the head after calibration and repeating the calibration procedure multiple times during long sessions. Further research is still needed to develop calibration procedures that are easy, comfortable, and robust to drift.

#### 4.2 Data privacy and security challenges

With eye tracking becoming ubiquitous in new HMDs, there has been growing concern about the privacy of eye tracking data collected on these devices. Kröger et al. (2020) and Steil et al. (2019) explained that eye tracking data contain rich information content that could be used to infer a vast amount of personal information about the user including: the user's interest in a scene; the user's cognitive load and cognitive state; various mental disorders including Alzheimer's, Parkinson's and schizophrenia; the user's



personality traits; and other sensitive data including the user's gender, age, ethnicity, body weight, drug consumption habits, emotional state, skills and abilities, fears, interests, and sexual preferences (Kröger et al. 2020). Adams et al. (2019) point out that a majority of VR developers do not follow proper privacy practices to guard user data collected in VR systems. They also mention data collected from users in VR systems could be collected or transferred to third parties without the user's knowledge or be leaked through known security vulnerabilities.

The near infra-red cameras used in HMD eye trackers collect tens or hundreds of high-quality images of the eye per second. The susceptibility of these images to attacks is a major concern. John et al. (2019,2020) point out that these images contain iris-patterns of the user and if an intruder gets access to even a single image from this data stream, they have effectively captured a gold standard biometric—iris authentication. John et al. (2020) introduces a hardware-based technique to degrade the images collected by the eye tracker so that the images cannot be used for iris authentication while still allowing the utility of gaze tracking. The results of the system indicate that the average Correct Recognition Rate—the rate at which users' can be identified from their eye images—was reduced from 79% to 7% when using the system.

Given the amount of sensitive information that can be gathered from eye tracking data and the lax privacy practices in the current VR ecosystem, Steil et al. (2019) explains how eye movements recorded using HMD-based eye trackers could be a potential threat to users' privacy. They point out that there is an urgent need to develop privacy-aware eye tracking systems—systems that provide a formal guarantee to protect the privacy of their users. They also propose a method that could be used to protect users' privacy in eye tracking based on differential privacy—a method that adds noise to eye tracking data to hide the privacy-sensitive information in the data while still allowing the data to be used for the desired task.

Despite these findings, there is a need for more research in privacy and security as it relates to eye tracking in VR. The rapid technological advancement in VR hardware and the increased integration of eye tracking in HMDs have resulted in a surge of customer-facing applications that use eye tracking data. Therefore, it is important for researchers and practitioners to develop tools and standards that preserve the user's privacy and enhance the security of eye tracking data.

### 4.3 Safety issues

Current customer-facing HMD-based eye trackers are VOG-based eye trackers that use infra-red (NIR) radiation to enhance the contrast between the pupil and the iris.

Most eye trackers use near IR (NIR) light sources with wavelengths around 880 nm which are invisible to the human eye. However, previous studies have shown that prolonged exposure to IR could have a damaging effect on the user's eyes (Kourkouvelis and Tzaphlidou 2011). The proximity of the IR source, the length of exposure and the number of IR sources have been identified as factors that could potentially increase damage to parts of the eye including the cornea, the lens and the retina (Kourkouvelis and Tzaphlidou 2011). Considering that eye trackers in VR HMDs use multiple IR sources in close proximity to the eye, and that VR applications are designed for prolonged use, the potential long-term hazards of IR radiation from HMD-based eye trackers should be thoroughly investigated.

## 5 Conclusion

It is clear based on the many applications mentioned above that eye tracking will soon become an integral part of many, perhaps most, HMD systems. We therefore expect research and development surrounding eye tracking in HMDs to accelerate and expand in the coming years.

Eye tracking in VR has shown great potential to improve rendering efficiency in VR systems and to help enable more comfortable and immersive VR experiences. The most prominent example in this area is gaze-contingent (foveated) rendering. However, more research is needed in this area to understand the perceptual effects of gaze-contingent rendering. More work also needs to be done to reduce the end-to-end latency of eye tracking systems in VR to make gaze-contingent changes in the environments imperceptible to end users.

Moreover, as the user base in VR grows and the need for developing accessible interaction techniques increases, eye tracking has been explored as powerful modality to develop accessible and hands-free interaction techniques for VR systems. Eye tracking could be explored to enable individuals with severe motor impairments to interact with VR systems. There are still challenges with using eye tracking for precise interaction, and more work could be done to enable multimodal or semi-autonomous interaction techniques that solve the challenges with eye tracking-based interaction. Moreover, with more and more sensors being embedded in VR HMDs, sensor fusion (i.e., combining eye tracking with other sensors) is another area for future research that holds promise to further increase the accuracy of eye tracking-based interaction.

VR is used for developing learning and skills training applications as it provides trainees an immersive environment that is safe to fail and allows repetition and correction with minimal cost. Eye tracking in VR has been used for the assessment of learning outcomes and improving



the learning process. More research is needed in this area to develop better eye tracking data interpretation methods to measure learning outcomes; to ensure that VR-based training tasks are effective; and to ensure that the skills learned in simulated environments can be transferred to the real task.

Eye tracking could supplement VR to provide powerful tools that can help advance marketing and CX research in various areas including new product development, advertising, and assessment of consumer behavior and engagement. Previous research has mostly focused at the pre-purchase and purchase stages of the customer journey. However, there is potential for interesting future research on the post-purchase stage that could help companies to analyze consumer behavior and interactions with the purchased product or service.

Eye tracking has been used in the clinical context for diagnostic, therapeutic and interactive uses. However, there is still need for further research to develop uniform and more usable clinical applications that are easy to comprehend and use for the patient as well as the clinician practitioner. This has the potential to allow patients to self-diagnose or self-treat neurological disease and provide clinicians with easy to use tools. There is also need for the development of VR and eye tracking hardware.

Additionally, eye tracking data can be used to easily identify and authenticate users in an implicit way without interrupting the user's main task. On the other hand, this makes eye tracking data vulnerable and sensitive to data privacy and security challenges. Therefore, more work needs to be done to develop tools and standards that preserve users' privacy and strengthen security of eye tracking systems.

In summary, eye tracking, i.e., the ability to precisely and accurately measure the user's eye gaze, has potential to become a standard feature on consumer VR headsets. The availability of high-precision, low-latency and low-cost eye trackers in VR HMDs has led to the emergence of a host of new applications spanning multiple disciplines. This paper provides a broad review of these applications and highlights some areas for future research.

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## Declarations

**Conflict of interest** The authors declare that they do not have any conflict of interest.

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# A Separate Reality: An Update on Place Illusion and Plausibility in Virtual Reality

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We review the concept of presence in virtual reality, normally thought of as the sense of “being there” in the virtual world. We argued in a 2009 paper that presence consists of two orthogonal illusions that we refer to as Place Illusion (PI, the illusion of being in the place depicted by the VR) and Plausibility (Psi, the illusion that the virtual situations and events are really happening). Both are with the proviso that the participant in the virtual reality knows for sure that these are illusions. Presence (PI and Psi) together with the illusion of ownership over the virtual body that self-represents the participant, are the three key illusions of virtual reality. Copresence, togetherness with others in the virtual world, can be a consequence in the context of interaction between remotely located participants in the same shared virtual environments, or between participants and virtual humans. We then review several different methods of measuring presence: questionnaires, physiological and behavioural measures, breaks in presence, and a psychophysics method based on transitions between different system configurations. Presence is not the only way to assess the responses of people to virtual reality experiences, and we present methods that rely solely on participant preferences, including the use of sentiment analysis that allows participants to express their experience in their own words rather than be required to adopt the terminology and concepts of researchers. We discuss several open questions and controversies that exist in this field, providing an update to the 2009 paper, in particular with respect to models of Plausibility. We argue that Plausibility is the most interesting and complex illusion to understand and is worthy of significant more research. Regarding measurement we conclude that the ideal method would be a combination of a psychophysical method and qualitative methods including sentiment analysis.

**Keywords:** virtual reality, presence, place illusion, plausibility, body ownership, questionnaires, measurement

## INTRODUCTION

Some physicists argue that there are an infinite number of parallel universes and even that these universes interact with one another at the quantum level (A very interesting discussion of the multiverse can be found in (Deutsch, 2011)). It is very hard for people not trained in mathematics or physics to have any idea what this might mean. However, with Virtual Reality (VR) we do have an example of how a parallel Universe (of sorts) can occupy the same physical space as our Universe in space and time. You are in your living room, and you don a head-mounted display. You perceive an



alternate computer-generated world through a wide field-of-view head-mounted display (HMD), in stereo. As you move your head the images that you see and the sounds you hear update in predictable ways, enabled through 6 degrees of freedom head tracking. If you see a close object, you can move your head to see what is behind it. If you bend down, you might see under it. In this case the world that you perceive is a concert that happened in 1983. You look around and see that you are in a theatre. There are thousands of other cheering people there. You see several people run onto a stage and when they are in position some guitar music starts to play. From a loudspeaker booming out across the theatre you hear “Good evening ladies and gentlemen, would you please welcome to the stage—Dire Straits!”

Where are you? Of course, you are in your living room. But your sensory system only perceives the reality of the theatre. The band starts to play “Sultans of Swing” and the audience around you dance in rhythm with the music. You find that you are dancing too. You look around at the people close to you. One of them appears to be staring at you. Another quickly turns away when you look at him. What am I doing here? a woman alone at this concert, with some quite creepy men around me. Maybe one of them will come over and start to try to engage with me. This is a nightmare.

This actually happened. Research that started from computer vision techniques to extract 3D geometry, texture and animation from video footage of a Dire Straits concert, and that used crowd simulation to create the audience transforming the whole into a VR scenario, was experienced as a “nightmare” by some participants in a pilot study (Beacco et al., 2021). Although unexpected and apparently negative, this was a highly informative result. In reality there was no theatre, no band playing and no audience. Nevertheless, the objective reality of lit pixels on two small screens and the sound from loudspeakers on the HMD were experienced as a live scene where many events were happening—including, for some women, frightening ones such as audience members “ogling” and “pretending not to be staring at” them, arousing fears of possible sexual harassment. In fact, none of these negative events at all were programmed into the scenario—there were no virtual audience members staring at participants (except a momentary glance by chance)—yet such illusory events were perceived by about half of the 20 participants in the pilot study.

How is it possible for people to have such experiences? We argue that this occurs through a number of illusions that are generated through VR experiences. People act out of these illusions as if they were real. In the rest of this paper we discuss those illusions, how they have been measured, and follow-on with some open questions and controversies.

## THE ILLUSIONS OF VR

VR and other immersive media such as augmented reality (AR, considered later) can generate at least three unique *illusions* that are not possible with other media. Here we mean illusions in the sense that people have perceptions that arise from digital sources that are totally different from what is actually being perceived.

The perceptions are real perceptions, and people may act on them as if they were real (Chalmers, 2017, 2022; Slater and Sanchez-Vives, 2022). For example, a participant in VR perceives a  $0.5\text{ m}^3$  cube stationary in space which reflects light that suggests that it is metallic, and which can be walked around and looked at from various locations, including ducking underneath it. There is an illusion of seeing a heavy cube somehow floating in the air. But what is actually happening? The participant is looking at two small screens modulated by an optical system, which displays illuminated pixels in a variety of colours. One shows a left eye view and the other a right eye view computed from the perspective projections of computer graphics. The brain integrates all this information into the perception of a cube. What is perceived is nothing whatsoever like the source of the perception. While standing in front of the cube it suddenly falls to the ground. Most participants would jump backwards—an automatic response caused by fear of the cube landing on their toes. This “falling” is nothing more than a sequence of colour changes of some of the illuminated pixels. Although one could make the same argument about perception in reality (we do not see “reality”—atoms, quarks etc. but an interpretation constructed by our sensory apparatus and brain processing) the source is quite different. For example, the entire physical apparatus responsible for perception in VR (computers, head mounted displays etc) can also be decomposed into atoms, quarks etc.—but the physical atoms cannot be decomposed into pixels on a display. So the VR apparatus and system is at a different level to physical reality, and the fact is that in spite of perceiving and reacting to a cube, there is no cube there.

The first two illusions we consider are part of the concept of “presence” originally considered as the sense of “being there” in the virtual world (Held and Durlach, 1992; Sheridan, 1992, 1996; Sanchez-Vives and Slater, 2005). Presence has been decomposed into two dimensions—“Place Illusion” (PI) and Plausibility (Psi). PI is the illusion of being in the place depicted in the VR (“being there”) in spite of the sure knowledge that this is not the case. “Plausibility” (Psi), is the illusion that events in the VR are actually occurring, that what is perceived is happening (a cube is falling towards your toes), again in spite of the knowledge that the events are digitally generated and nothing that is apparently happening is actually happening in reality (Slater, 2009). Events in VR correspond to changes in the illumination of pixels, changes in generated sound, and possibly other digitally generated sensory information such as haptics or olfactory.

The basis of PI is that perception in VR is through natural sensorimotor contingencies (O’Regan and Noë, 2001)—that is, participants perceive through using their body following much the same rules as in physical reality—turning the head, bending down, reaching out, looking around objects. A head turn, for example, results in updates to the displayed images so that participants would see visual changes and hear auditory changes corresponding to what would occur in reality. This is based on 6 degrees of freedom head-tracking, and possibly eye tracking. This integrates both interaction and display—since natural sensorimotor contingencies demand that what is displayed conforms with body movements. For example, if the participant looks closely at an object and sees pixels, then this breaks PI.

Psi relies on 1) the virtual environment responding to actions of the participant (for example, the participant looks towards a virtual human character that then responds by looking back); 2) events in the environment contingently referring to the participant (for example a virtual human character smiles towards the participant without any prior cause). 3) Where the virtual reality is a simulation of something that could occur in reality, depicting a situation in which the participant has expertise, then virtual events should meet expectations. This may be highly specific to the individual. In the virtual concert scenario discussed above, some participants noticed that the drummer was sometimes not moving in sync with the beat, or even that the light reflected from the guitar strings of the performers did not match the sounds that they were making. Also the concert was supposed to be in the 1980s but no one in the audience was smoking. This last example required that the participant concerned knew that smoking in theatres was still possible in London, United Kingdom, in the 1980s. Other participants who did not know this would not find the lack of smoking to be a failure of expectations. In a study of medical doctors who were in consultation with virtual patients, many complained that they could not read the screen of the computer on their (virtual) desk that they would typically be doing in reality (Pan et al., 2016). Skarbez et al. (2018a) introduced the term “coherence” to describe some of the factors that go into producing Psi, and we will return to this concept later.

PI and Psi are conceptually distinct and can be considered as orthogonal axes that make up an overall response that might be labelled as “presence.” Empirically they may be correlated, although they have been found to be different in, for example, (Slater et al., 2010a; Hofer et al., 2020) but with unclear results in (Brubach et al., 2022). Psi is beginning to be extensively studied e.g. (Bergström et al., 2017; Skarbez et al., 2017; GalvanDebarba et al., 2020). The important point of this is that when both PI and Psi operate people tend to respond realistically to situations and events in the VR, even though they know for sure that these are illusions and not reality.

Psi is the more complex and interesting aspect, since a seemingly unimportant feature of the environment that does not fit expectations can result in its loss. For example, in early versions of a study of bystander responses to violence, where the bystander was in a bar talking to one person who was attacked by another over an issue to do with soccer (Rovira et al., 2009; Slater et al., 2013), we were told by some participants that the scenario was not consistent with what would be expected in reality. Why? because a bar with that type of decoration would never be patronised by soccer fans. On the other hand in an environment that was used for cognitive behavioural therapy for people with fear of heights, participants looked into an atrium formed by several tall buildings (Freeman et al., 2019). A whale was “swimming” between the buildings, yet people simply accepted this without comment. In a 3D chess scenario, when participants touched a chess piece it would fly to its next location (Slater et al., 1996). No participant found this odd, and when asked about it, one said that: In this world that is the way things are.

The third illusion is “body ownership.” When participants wear a wide field-of-view head-tracked stereo head-mounted

display and they look down towards themselves, they will see a life-sized virtual body substituting their own, from their first person perspective (1PP) (if this has been programmed). Utilising real-time body tracking, as the person moves, their virtual body can be programmed to move synchronously and in correspondence with their own movements (referred to as visuomotor synchrony). If something is seen to touch their virtual body, then it can be arranged that a corresponding tactile stimulation is applied synchronously to their own real body (visuotactile synchrony). We refer to this as *embodiment*, which involves multisensory integration of the 1PP view of the body, and visuomotor or visuotactile synchrony, which will typically lead to the illusion that the virtual body is their own—even though they know for sure that it is not. Based on the original finding of the Rubber Hand Illusion (Botvinick and Cohen, 1998) ownership over a virtual body has been demonstrated multiple times—for example, (Petkova and Ehrsson, 2008; Slater et al., 2008; Slater et al., 2010b; Banakou and Slater, 2014). Moreover, changing the type of body can lead to physiological, behavioural, attitudinal, and cognitive changes in the participant. For example, multiple replications have shown that embodying White people in a Black virtual body will lead to sustained reduction in their implicit racial bias (Peck et al., 2013; Maister et al., 2015; Banakou et al., 2016; Hasler et al., 2017; Banakou et al., 2020). However, if the affective situation in which this occurs is a negative one, then implicit bias tends to increase rather than decrease (Banakou et al., 2020).

There is a fourth illusion that can be thought of as a corollary of these three, referred to as copresence (Nowak and Biocca, 2003). When in a virtual environment simultaneously with other remotely located people copresence refers to the extent to which the participant has the illusion of being there with the others, or virtual togetherness (Durlach and Slater, 2000). Participants should be represented in some form with virtual bodies, otherwise it would not be possible to know that they are there or where they are. Sensorimotor contingencies lead to the possibility of the participant having the illusion to be in the same space as the others. For example, each person has a representation that in principle the participant can walk around, look behind, hear the voice from different locations, reach out and touch, and so on. Exactly the same requirements for PI incorporate this aspect of copresence—i.e., the illusion of sharing the same space. Then Psi is needed for participants to be able to take the interaction events as really occurring. Hence the characters should respond when interacted with, for example, change gaze direction, and characters should be able to initiate interaction with the participants (Garau et al., 2005). The characters should move appropriately depending on context, so that the expectation “this is how a human should behave” is met. The requirement to satisfy expectations is the most difficult because it depends very much on context. For example, if the characters are represented as high quality human models, then they should behave according to this, but if they are represented, say, as clearly cartoon characters, then expectations about their behaviour might be different.

Copresence leads to results that would be expected under similar circumstances in reality. A good example of this is

proxemics, where people maintain different distances from one another depending on their relationship at the time: intimate, personal, social and public (Hall, 1973). Bailenson et al. (2003) found that proxemics predictions from reality also operate in VR. Llobera et al. (2010) found that skin conductance varied with distance of virtual human characters from the experimental participant in accordance with proxemics theory. Kastanis and Slater (2012) found that a Reinforcement Learning agent, represented in VR as a human character, learned that if it moved close to experimental participants they would back away (since the character entered their personal or intimate space causing discomfort) and then it used that to manipulate participants to a target spot in the virtual environment. A recent overview and review can be found in (Williamson et al., 2021).

There is a question as to whether the issue of copresence only refers to the situation of remote participants meeting in the same environment, or whether it can also apply to sharing the space with virtual characters controlled by an algorithm. In fact, there is a range of possibilities between these two (full control of the avatar by an embodied person, or control of an avatar by a computer program). Suppose that the avatar is only partially controlled by a remote person—for example, the person controls the location and gaze direction, but not the bodily gestures because there is no motion capture. Or suppose that the remote person only controls the speech, which is then mapped to appropriate lip-sync on the virtual human character, but the facial expressions and body gestures are wholly determined by an algorithm. In any of these situations the issue of copresence (do participants feel together with such representations?) is the same. Therefore, it can be argued that copresence first, is not a separate phenomenon from PI, Psi, or body ownership, if these are operating correctly and applied to the representations and actions of virtual characters, then there will be copresence. Second, copresence can occur whether the other characters are wholly representative of remote people or wholly determined by a computer program, or any variant in between these two extremes.

Recent reviews on the concept of presence can be found in (Parola et al., 2016; Skarbez et al., 2018b; a; Felton and Jackson, 2022) with a meta study in (Cummings and Bailenson, 2016). Grassini and Laumann (2020) focus on the variety of measures that have been used, and Souza et al. (2021) present a meta-study of over 1,200 papers on methods of measurement of presence. It is noteworthy that 86% of these papers used subjective measures and 12% used both subjective and objective. In the next section we discuss different approaches to measurement.

## THE PROBLEM OF MEASUREMENT

### Questionnaires

Since presence refers to subjective illusions (PI, Psi) the obvious way to elicit the sense of presence has been through questionnaires. Schwind et al. (2019) identified 15 questionnaires in their review. A widely used one is by Witmer and Singer (1998), which focuses on asking opinions of participants about a number of factors that have been thought to promote presence, but it has no questions about the sensation

of “being there” itself (Slater, 1999). An early questionnaire that came to be known as SUS, though never formally validated, appeared, for example, in (Slater et al., 1994), and concentrated on the participant’s sense of “being there” in the virtual world, the times when the virtual world became the participant’s reality so that they forgot about the real world, and the extent to which participants evaluated the virtual world as “somewhere that you saw” or “somewhere that you visited.” A fully validated questionnaire based on factor analysis was developed by Lessiter et al. (2001), which was intended also as cross-media (not only for immersive virtual environments). Their factor analysis pointed to three major components. The first is the “sense of physical space” which is closely related to the general view of presence (in particular PI) as “being there.” The second factor “engagement” refers to how much participants are involved and interested in whatever is happening in the virtual world. The third factor is “ecological validity,” which is concerned with “believability and realism” and overall consistency between the different sensory streams. This is most closely related to the concept of Psi, but is not the same. The fourth factor is “negative effects” such as discomfort, simulator sickness, and so on. Another comprehensive factor analysis study carried out by (Schubert et al., 2001) led to a similar three component model: spatial presence (“being there”), involvement, and realness. This has come to be known as the Igroup presence questionnaire.

There are fewer examples of questionnaires specifically aimed at copresence. A series of three studies where groups of three remotely located people carried out tasks together in VR, developed one of the first questionnaires. These included questions such: “There was a sense of being with the other people,” “The computer interface seemed to vanish and there was direct working with the other people,” “Rate how closely your sense of being together with others in a real-world setting resembles your sense of being with them in the virtual room” (Tromp et al., 1998; Steed et al., 1999; Slater et al., 2000). These types of questions were derived simply from the idea of copresence. Drawing on the extensive collaborative virtual environments literature Garau et al. (2001) reported an experiment on the impact of different eye gaze models on quality of communication between two remote individuals represented by avatars via a video tunnel. The quality of communication was assessed by a questionnaire eliciting four aspects: the extent of *face-to-face* communication that was perceived (e.g., “I could readily tell when my partner was listening to me” and five other questions), the degree of *involvement* (“I found it easy to keep track of the conversation,” “I felt completely absorbed in the conversation”), *partner evaluation* (e.g., “My partner was friendly,” and four other questions), and *copresence* (“I had a real sense of personal contact with my conversation partner,” “I was very aware of my conversation partner”). These same questions were used in a later study where participants met via immersive VR (Garau et al., 2003). Bailenson et al. (2005) introduced a measure of copresence using three questions (‘Even when the “other” was present, I still felt alone in the virtual room,’ “I felt like there was someone else in the room with me,” “I felt like

the “other” was aware of my presence in the room.’ Bulu (2012) carried out a study using the (2D) Second Life as the shared virtual environment, and assessed copresence as the sense of being part of the group and each participant’s assessment of the copresence of the others. A psychometric scale for copresence was developed by Poeschl and Doering (2015) in the context of a fear of public speaking scenario and in the German language. As can be seen while there is a lot of variability in the questions that researchers have used for copresence, there is the common aspect of simply the sense of being with other people.

There are several problems with questionnaires if used alone. One is that they are typically administered after an experience rather than while the experience is taking place. Schwind et al. (2019) showed that questionnaires can be administered during the actual experience so potentially overcoming this disadvantage. However, contrary results were found by (Graf and Schwind, 2020). Moreover, a problem with this is that it forces participants to take a meta-view of their experience during the very time that it is required to just experience it. A second problem with the use of questionnaires alone is that the questionnaire itself may bring about the very feelings that it is supposed to measure. For example, Slater (2004) carried out a study about an entirely invented concept called “the colourfulness of your day.” Participants were asked to think about their previous day and answer various questions including how “colourful” it was. Correlations were found between this and various other factors in the questionnaire, just as correlations are found with the sensation of “being there.” The problem of course is that before being introduced to this idea of a day being “colourful” participants almost certainly did not ever think in those terms. Similarly, it is possible that the feeling of “being there” never occurs to participants in a VR experience, but that this is introduced to them solely through the questionnaire. In other words, the researcher’s conceptual framework is imposed on the participants.

## Behavioural and Physiological Surrogates for Presence

In order to obtain a more objective approach to the measurement of presence (PI), behavioural or physiological surrogates have been used. Going back to our opening example, some participants clearly had a strong response to being at the virtual Dire Straits concert, feeling that they were alone, that people were staring at them, and so on. This indicates a high level of both PI (they are at the place of the concert) and Psi (that this was really happening) and also copresence (with the virtual audience around them)—otherwise, becoming stressed about their situation would have made no sense. Meehan et al. (2002) formalised the idea of using physiological measures of stress as a surrogate for presence to measure these types of response. People were placed in different conditions standing by a precipice and their heart rates were measured, the argument being that increased heart rate would indicate arousal, which is what occurred. Spanlang et al. (2007) carried out an experiment where a fire broke out in a virtual bar, and at first other (virtual) people in the bar ignored it—which resulted in physiological responses such as heart rate, heart rate

variability and skin conductance responses indicating significant arousal. Martens et al. (2019) had one group of participants go up a building in an interior elevator and another group in an external elevator, and found differences in a range of physiological responses, with greater arousal amongst those in the external elevator. Ríos and Pelechano (2020) placed participants in a train station, where suddenly virtual travellers started running in a particular direction, as if there were an emergency. Participants tended to follow the escaping virtual characters. Putze et al. (2020) investigated the relationship between physiological responses and questionnaire administration during and after the VR experience, and the results suggested less disruption to presence for questionnaires administered during the experience, and the result of Graf and Schwind (2020) is also relevant to this.

As well as physiological measures of arousal, brain activation has also been measured with EEG, for example (Baumgartner et al., 2006; Kober et al., 2012; Kober and Neuper, 2012; Clemente et al., 2014; Petukhov et al., 2020). One study suggested that placing participants in a stressful situation in VR can lead to extreme stress as indicated by brain activation responses (Fadeev et al., 2020).

Ochs et al. (2022) introduced a new approach for presence and social presence. During the course of social interaction, involving a doctor delivering bad news to a patient, various multimodal measures were recorded, mainly concerned with speech (e.g., length of sentences). Place presence and copresence were measured using the questionnaires in (Schubert et al., 2001) and (Bailenson et al., 2005) respectively. Then machine learning methods were used to successfully predict the levels of reported presence in each case. Association between subjective reports and other non-subjective measures (physiological, behavioural) is one of the best ways to demonstrate that there is indeed an underlying phenomena, which can be measured consistently in independent ways.

A sure sign that participants in VR exhibit correlates of presence such as emotional, behavioural and physiological changes is that one of the main areas where VR has been used in the past 30 years is in clinical psychology. In these applications VR is typically used as an adjunct to cognitive behavioural therapy, where participants are gradually exposed to the anxiety causing situation. Using VR this exposure can take place in the office of the clinician rather than trying to arrange real situations for exposure or giving the client “homework” that they can report on during the next session. This can only work because presence (both with respect to PI and Psi) operate—so that participants, even though they know that the situation is entirely virtual—nevertheless experience similar anxiety as they would in physical reality. Without this level of anxiety, therapeutic techniques would not work. For reviews see (Rizzo and Kim, 2005; Freeman et al., 2017).

The problem with using behavioural and physiological surrogates and EEG for presence is that this technique can only be used in limited circumstances—in particular for environments that cause measurable arousal (whether with negative or positive affect). In circumstances where there are no predicted arousal effects then this method cannot be used, or only where there are specific triggers that are there solely in order



to elicit physiological or brain responses. So, although behavioural and physiological measures, and especially in combination with questionnaires go some way towards a reliable and objective measure, they cannot provide a general answer.

## Breaks in Presence

The third main method for assessing presence is based on the idea of “breaks in presence” (BIP). This has its basis in our approach to understanding what presence is and its relationship to sensorimotor contingencies. The argument is that the normal state is that participants will have the sensation of PI in a virtual environment which has at least 6 degrees of freedom head-tracking (whether a Cave or wide field-of-view stereo HMD). However, every so often there may be a glitch, such as a sudden change in frame rate, or an event that breaks consistency or expectation, or even something simple such as bumping into a (real) wall. If we think of PI as essentially “always on” but with occasional failures, then the number of failures and their occurrences through time would provide an interesting measure of presence. This was first put forward in (Slater and Steed, 2000). In this framework we can think of the responses to a post-experience questionnaire as the integral over time of these continual periods of “on” instead of “off” so that the number of BIPs should correlate inversely with post-experience presence questionnaire scores. The problem with BIPs, however, is to know when they occur. In the original paper a BIP was signified by the participant, the argument being that if a BIP had already occurred then reporting that fact could not in itself diminish presence, because it had already been diminished. Another method has been to try to find physiological correlates for BIPs. Slater et al. (2003) studied how breaks in presence were signified by patterns in physiological responses, specifically ECG and skin conductance. Rey et al. (2011) found a relationship between blood flow velocity responses and breaks in presence forced through deliberately caused glitches in the VR. Liebold et al. (2017) viewed BIPs as orienting responses towards the virtual or real world and differentiated between different types of BIPs on this theoretical basis. They found associations between various types of BIP and physiological responses.

Breaks in presence have the advantage over questionnaires that they are intrinsically based on what is experienced during, rather than evaluations after, the VR exposure. Whether a questionnaire is posed during or after an experience it inevitably forces the participant to assess the experience in terms of phrases that are imposed from the outside. A BIP can be just a sudden failure of whatever it is that the participant takes as their presence in their separate reality induced by the VR, and from the point of view of the participant can be considered as conceptually neutral: it broke, we do not really need to know what they mean by that, it is just an observed fact. The problem is in the observation. Do we allow participants to self-report BIPs? perhaps they might forget to report an occurrence in the excitement of the moment, or maybe they feel impelled to report *something* because this is the expectation imposed on them. If we do not allow participants to self-report then some other means needs to be introduced—for

example, physiological responses that indicate a BIP. In this case there would therefore be a double layer of assessment—first, whether the physiological measures really indicate a BIP with all the problems of false negatives or positives, and second, instead of directly assessing whether a BIP has occurred it is now an indirect measure. It is more cumbersome, with the high probability that an additional layer of technology involves an additional layer of error. Although relatively neutral compared to a questionnaire, and while the BIP idea is conceptually interesting, it still has potential problems (Skarbez et al., 2020).

## Configuration Transitions

Why are measures of presence (whether PI or Psi) useful at all? There are at least two reasons. First, VR uniquely delivers PI (in the sense meant in this paper), and if a system or application does not at least result in that, then there seems to be no point to using VR. Second, generally application designers need some criteria against which to understand trade-offs between different factors that could be included in their designs. Understanding how different trade-offs may influence PI and Psi therefore requires some way of measuring these. To reiterate our view: there is immersion, i.e., the objective capabilities of the system. Presence is a subjective response. The goal is to understand how varying these different immersive capabilities influence presence.

The situation has some parallels with colour perception. There is physics, light being emitted or reflected at different wavelengths into the eye. This is the objective aspect. Then there is the human perceptual response where the wavelength distributions are interpreted as colours. However, the situation is not so straightforward. There is not a 1-1 mapping between wavelength distributions and perception of colour. Different wavelength distributions can result in the same colour perception (metamers) and different people may interpret the same wavelength distribution quite differently. The human visual system builds colour perception from three types of cones in the retina (so-called “red,” “green,” and “blue”), yet the wavelength distributions are infinite. Colours are quantified, however, through colour matching experiments. For example, a target colour is projected onto a screen and an experimental subject has control over three projectors emitting red, green or blue light, combined onto a different patch. The subject can manipulate the intensities of the three projectors, so mixing red, green and blue in different proportions. The subject’s task is to find the intensities of the red, green and blue that result in a colour that perceptually matches the target. By repeating this over a number of people, each colour of interest can be associated with a particular combination of proportions of red, green and blue intensity. Hence there are no questionnaires to elicit a colour (e.g., “How red is this on a scale of 1–7?”) but only the colour matching process. This does not require any knowledge of how the person “really sees” the colour only that they match it to the target colour.

In (Slater et al., 2010a) we applied a method analogous to this for the assessment of PI or Psi. Corresponding to wavelength distributions there were factors that could be set at different levels. In this first experiment the factors were: 1) the type of illumination model used in the scenario (Gouraud shading, static view-independent global illumination, dynamic global



illumination including real-time shadows and reflections) 2) the field of view (small or large); 3) display type (a simulated power wall or a head-mounted display) 4) the self-representation of the participant (without a virtual body, with a static virtual body, with a virtual body that moved with the participant's own movements based on real-time motion capture). Participants first experienced the scenario with all these factors set at the highest level (dynamic global illumination, large field of view; head-mounted display, full body representation with real-time movements). They were asked to pay attention either to their sense of being there (PI) or that the situation was really occurring (Psi). Then they had a training session that introduced them to the four factors and how they could change levels within each factor (always in the direction of lowest to highest). After this training period they were placed back in the scenario but now with all the factors set at a low level. At specific times they could choose to increase the level of one of the factors. They were asked to continue to do this until they felt that they had achieved the same sensation of PI or Psi as in their first exposure. The changes that they could make were under cost constraints.

We refer to any set of factor levels as a *configuration*. In this experiment there were therefore  $3 \times 2 \times 2 \times 3 = 36$  configurations. Each time that a participant chooses to change the level of a factor there is a transition from one configuration to another. Taking the entire set of transitions, it is possible to estimate a transition probability matrix  $\mathbf{P}$  with entries  $p_{ij}$ , the probability of transitioning to configuration  $j$  given that the current configuration is  $i$ ,  $\sum_j p_{ij} = 1$ . Now there are several ways that  $\mathbf{P}$  can be used. From Markov Chain theory it is simple to calculate the long run equilibrium probabilities of the system from any starting point. These specify for each configuration the probability that in the long run this would be the one into which the system settles down (i.e., no more transitions would occur). Second, we can trace the order of transitions—i.e., which are those that are made first to move towards the matching state, which are made second, and so on, and hence we can obtain information about the factor levels that most contribute towards the desired state of PI or Psi. Third we can compute probabilities for the individual factors: for example, taking all the final configurations reached, to compute the probability that a particular factor level would be contained in the final configuration. There are many other possibilities too. In addition this method can result in equivalence classes—i.e., sets of configurations that result in similar probabilities of PI or Psi. Such equivalence classes were argued for long ago by Ellis (1991), and are useful for engineers since they can use these to trade between different configurations, for example, based on a cost function.

What is important to understand is that participants are never asked their opinions or asked to give rating scales. The method is premised on observable events only—the fact of the transitions. The method is entirely based on the participants' decisions that a particular configuration results in a match of their sensation of PI or Psi, i.e., the only important thing are the matches that people make, and not the meanings that might be attributed to these. In the experimental study of this method participants who had been asked to focus on matching their state of PI transitioned between

the configurations differently than those who had been asked to match Psi, and the final resulting configurations were different.

This method has been applied several times. Azevedo et al. (2014) combined the method with EEG measures of engagement. Bergström et al. (2017) used this method to evaluate Psi for four different factors influencing the responses to a virtual string quartet performance. Skarbez et al. (2017) evaluated how a number of factors determining the behaviour of virtual characters influenced Psi. Gao et al. (2018) assessed the believability of a VR rock climbing scenario. Fribourg et al. (2020) turned the method towards evaluation of body ownership rather than PI or Psi. GalvanDebarba et al. (2020) studied Psi in relation to body animation features. Clearly the same method could be used for the evaluation of different subjective responses, not only PI of Psi, which we turn to next.

## Beyond Presence

Murcia-López et al. (2020) used the configurations transitions method to evaluate the impact of several factors controlling an avatar giving a TED-style talk to participants. There were two main differences from the previous application of the method. First, participants could choose to change a factor level in any direction, so that there was no assumption about levels being ordered, that one factor level was somehow superior to another. The second main change was that the goal was not to improve presence but simply to maximise *preference*. In other words, participants could decide to make a transition from configuration A to B only because they preferred B to A, irrespective of the reason. A consequence of these changes is that the method did not require participants to first experience the optimal configuration, since there was no *a priori* assumption about which configuration would be the best. The method produced coherent results across participants. This fits the idea mentioned above of not imposing criteria of researchers, such as presence, on participants.

This approach has been taken still further by Llobera et al. (2021). Instead of participants themselves choosing how to change the configuration, i.e., choosing which factor to change to which level, an AI agent does this. In this setup a Reinforcement Learning (RL) agent occasionally proposes to the participant a change of level of one of the factors in the current configuration. The participant can reject the proposed change, leaving the current configuration as the one in force, or can accept the change. Over time the RL agent learns which changes are likely to be accepted or rejected, i.e., it forms a policy consisting of probabilities of acceptance of proposed configuration changes given the current configuration. The process stops when the participant has reached a stable state and continues to reject further changes. The only criterion for participants to accept or reject a change is what they prefer. So over time the RL agent finds an optimum (though possibly only a local optimum rather than global) with respect to which configuration is most preferred. In the experiment reported by Llobera et al. (2021) this method was used for four binary factors. The RL was applied to each individual separately rather than cumulatively across all individuals. The optimal solutions happened to conform to what we would expect from previous

research on presence, though to emphasise, the issue of presence (PI or Psi) was never mentioned to participants, only that they should choose according to their preferences.

## Sentiment Analysis

A problem with the configuration transitions method is that although it is based either on the fact of a “match” between a target state and an actual state with respect to presence, or simply based on preference, it does not obtain any deeper insight into the reasons for people’s choices. In the opening paragraph of this paper, we described the rock concert scenario. This was the first time that we had created such a scenario, one quite unusual for VR. We did not want to impose our notions of presence on participants in the experimental study, so we simply asked them to write a short essay about their responses to it (Beacco et al., 2021). We then applied sentiment analysis (Liu, 2012; Bakshi et al., 2016) to the resulting essays. Sentiment analysis is a machine learning technique that will score text for positive or negative sentiment based on a pre-trained word-to-vector data base of millions of pre-classified words (Mikolov et al., 2017). Hence each essay carried a sentiment score, and the set of scores fell into distinct clusters (from a very low sentiment cluster to a very high sentiment one). The contents of essays of each cluster were then analysed to determine common themes within that cluster. For example, a common theme was “realism,” where sentiment scores were higher the more that participants mentioned something about the environment that seemed to be very real. Another theme was “disturbing,” where people found the crowd around them to be disturbing in some way, for example, one woman mentioning that she thought two people (of course virtual characters) were acting as predators towards her, by staring at her and then looking away whenever she looked back. Another concerned “failure of expectations” for example, the drummer visually not in sync with sounds of the drumbeats, or sounds of clapping but no one around actually visually clapping. In this way we were able to obtain very deep insight into participant responses to the concert and discover reactions that we would never have discovered through a presence questionnaire, BIPs or configuration transitions used alone. We found that the concert was highly plausible, but for some participants, not at all in the way that we expected. For some (not all) of the participants the concert was a “nightmare.” Being a nightmare of course demonstrated a high level of plausibility—since if people had not had the illusion that the events were really happening, then there could be no reason to become disturbed. Similar results were found in (Slater et al., 2022). It should be noted that as with any method the quality of the results depends on the quality of the input. Participants might be reluctant to write even a short essay after experiencing a VR scenario, or what they write might be too short for analysis. Recently we have started recording what participants say in a post experiment interview, rather than requiring them to write.

## OPEN QUESTIONS

### PI and Sensorimotor Contingencies

PI is the illusion of being in a place. In physical reality we are always in a place. In VR we may have the illusion of being in a different place, with the corollary that we know this is not true

(and this knowing it is not true is itself part of the feeling). In reality we might be completely uninterested in what is going on in the physical space in which we are located, but this will not destroy the sense of being there. Our minds might wander, and we start thinking about other things, but this does not change our perception of where we are. Now suppose we are in a VR that renders this same boring place, and similarly we become uninterested and not engaged in what is happening. This actually is not incompatible with the illusion of “being there” because it leads to the same responses as if we were really there. Hence questionnaires that include categories such as “engagement” or “involvement” are not assessing PI, even though in some applications the degree of engagement may be an important, but separate issue. Similarly a relationship between presence and task performance has been posited, with the argument being that greater presence enhances task performance, a debate that goes back a long way (Welch, 1999). Now in physical reality we might be trying to draw money from a bank teller machine (ATM) but not be successful because of the machine’s poor user interface. Carrying out a similar operation in VR might also lead to failure for the same reason. But here the failure would not be incompatible with presence—our poor task performance in reality should map to a poor task performance in virtual reality. The user interface is the issue, not PI.

In our view PI is intimately bound up with sensorimotor contingencies for perception. If we carry out bodily actions for perception (turning the head, moving the eyes, turning around, bending down, stretching up, looking around, looking over, looking under, turning our head to hear a sound better, touching, pushing, smelling) and the multisensory displays deliver integrated sensory outputs that correspond to those that would occur with similar actions in reality, then the simplest hypothesis for the brain to adopt is that what we see, hear, feel, . . . signifies where we are.

In our whole lives whenever we turn our head the visual images that we see change in a predictable way. This produces a massive probability for our sense of place: the probability that I am in the place that I see, hear and can touch. This is why, for example, if we are looking at a virtual environment displayed on a screen in a desktop system in front of us, we cannot experience PI in the sense meant in this paper. We only have to turn our head away from the display and we see a different reality, that of the real location in which we actually are carrying out this activity. In (Slater, 2009) we argued that this leads to a partial order over systems. In an immersive virtual reality, for example, realised through a high quality HMD, it is straightforward to simulate the activity of looking at an external screen that portrays a virtual environment. We referred to this as the HMD system being more “immersive” than the desktop system, since the first can be used to simulate experiences of the second, but not vice versa. Here immersion refers to objective capabilities of the system and not the subjective responses. Therefore, it was argued that the qualia associated with PI in the more immersive system cannot possibly be the same as in the less immersive system. In the desktop system one might attain some sensation of “being there” through employing additional attentional resources and imagination,

but in the HMD system it is just a fact—a head turn does not shift your perception out to the real world but is confined within the virtual separate reality. The qualia associated with the HMD and desktop system are at different levels, they are not comparable since they do not refer to the same underlying situation.

In this view there is no “book problem” (Biocca, 2002; 2003) regarding presence. When reading a book we might imagine ourselves into the story line, imagine being in the place depicted in the book. But this is not PI, the perceptual illusion of being in a place, as evidenced by the fact that you can carry out different acts of perception and the sensory stream remains consistent with being in that place. In VR it is simple to program the appearance of a book and allow people to read it while in VR. While they are reading the book they are in a virtual place, at any moment they can glance away from the book and they are still in that virtual place. In the book they might read about a fire breaking out in a train station with all the passengers running away, but they will not run away from that. In VR they might run (Ríos and Pelechano, 2020). The qualia associated with reading a book compared with sensorily being located in a virtual space are quite different. Of course one may choose to call the qualia associated with reading a book “presence,” but this is not what we mean.

If PI is so bound up with sensorimotor contingencies (SCs) for perception, why separate these two concepts? This is because SCs can fail, or not be consistent with one another, or delivered with too great latency, and for many other reasons. They are not “perfect” SCs corresponding exactly to real perception. Moreover, people may behave quite differently to one another. If person A stands and simply looks around an environment but B actively explores it looking close up at objects, trying to touch things, moving rapidly, then B might experience quite a different level of PI than A. This also shows that any measurements of PI need to implicitly take account of possible individual differences. Another way to think about this is to consider the level of PI as a random variable even if, as could be argued, that its possible values are restricted to 0 (no PI) and 1 (PI). In this view the expected value of PI (between 0 and 1) would be a function of the sensorimotor contingencies. However, any actually observed value would vary around its expected value, and one of the contributors to departure from the mean would be individual differences. In this approach individual differences could also be modelled as contributing to the variation.

## Place Illusion and Plausibility

The study of presence has been a major part of VR research since the early 1990s. We have argued for the view that presence consists of two orthogonal dimensions, Place Illusion and Plausibility. These are logically separable concepts in that it is possible for PI to occur independently of Psi and vice versa. For example, you can have a strong sense of being in a place, but the virtual characters there may have no plausibility—they do not respond to your actions, nor initiate any actions towards you (Garau et al., 2004). On the other hand, you can be interacting with a virtual character on a non-immersive device (e.g., a normal screen) not have any illusion of being in the place, but totally accepting that this is really happening—the person seems to be real, responds to you and initiates interactions with you (and this

happens every day in teleconference video calls). It is really happening but you are not there in the space where the other person is depicted. Of course, despite this logical separability between PI and Psi we might find them to be empirically related, but largely because of the great difficulty (or even impossibility) of having measures that are so precise that they can perfectly distinguish between the two. This is especially the case with questionnaires.

The concept of Plausibility arose out of the fire in the bar experiment discussed above (Spanlang et al., 2007). This was carried out in a Cave, had the best quality rendering that we had ever used, and yet the questionnaire scores on presence were one of the lowest we had seen. It was realised that this was because of failure of one component—repetitive motions of an avatar. Participants only had one way to signify this failure—by giving low scores on a presence questionnaire. So here the measurement of PI was confounded by Psi.

For future research Psi is by far the more important and interesting issue. As mentioned above our view is that in a typical VR with at least head-tracked stereo wide field-of-view HMD that PI will be the default sensation, since most of these systems offer a level of sensorimotor contingencies that are good enough to fool the brain into the illusion of being in the place depicted by the VR to our senses. The major research question remains to be able to understand in greater detail exactly which sensorimotor contingencies are critical, and under which conditions. For example, suppose that the HMD does not have 6 degrees of freedom tracking but only 3 degrees of freedom, so that there are no parallax effects. If participants do not move their heads, or only rotate them without translation, then the 3 degrees of freedom may be good enough. However, for example, in a simulation of a game such as table tennis where players are moving from side to side and back and forth very quickly all the time, the loss of the 3 degrees of freedom would severely diminish the experience (and probably result in simulator sickness). The fundamental research question for PI is therefore the map from sensorimotor contingencies to Place Illusion, irrespective of how this might be measured. A second question is how to model the impact of individual differences.

Although PI might be the default state, low Plausibility might also be default. PI is just part of what VR is (even though it can fail). However, Psi requires deliberate design: how can a scenario be designed and implemented so that participants buy into it? This does not necessarily involve photorealism, nor realism in the sense that the scenario might not be anything that can happen in real life.

Although we have said that PI and Psi can be thought of as conceptual frameworks imposed on participants by researchers, there are some applications where they are nevertheless crucial. For example, consider a therapeutic application for an anxiety disorder such as fear of crowds, one that employs an exposure therapy-based approach, where participants are gradually exposed to the situation that causes anxiety. So first the participant might be in a VR with one other (virtual) person, then two, then four, then ten, and so on, until over time the participant learns to control their anxiety while being in a large crowd. PI is needed because these events have to take place

somewhere, and if the participant does not have the illusion to actually be there, then there is little chance of anxiety being provoked. Second, the crowd should behave like a crowd, for example, it should part as the participant tries to move through it, and virtual crowd members should occasionally look towards the participant or acknowledge them in some way such as a wave or a smile or say “Excuse me.” Hence Psi is also critical to the success of this application. This illustrates the point above—that although PI might occur by default, Psi has to be deliberately designed. A corollary of this is that if PI fails (a BIP) it is likely to re-form again, since the sensorimotor conditions that gave rise to it are still there, even if they had temporarily failed in some way. However, if Psi fails it does not re-form. For example, once participants realise that a virtual human character is unaware of their presence, they just lose interest and move on (Garau et al., 2004). Although all this follows from the concepts of PI and Psi there is little empirical evidence available to date. In our view the major part of the research interest should be a focus on Psi—under what conditions do people take events and situations in the VR as actually happening, and when does this fail?

## Coherence

Skarbez et al. (2020) introduced the notion of coherence, which is the extent to which a virtual environment “behaves in a reasonable or predictable way.” More fully: coherence is defined ... as the set of reasonable circumstances that can be demonstrated by the scenario without introducing unreasonable circumstances, where a reasonable circumstance is a state of affairs in a virtual scenario that is self-evident given prior knowledge” (Skarbez et al., 2018b). In this view coherence stands in the same relation to Psi as immersion (the actual objective affordances of the system) stand to PI. However, this is not straightforward. While “immersion” is something objective and describable independently of any effect that it might cause (e.g., the field-of-view is  $x$ , it supports stereo vision with adjustable interpupillary distance, the resolution is  $y$ , the colour resolution is  $z$ , the tracking capabilities are  $w$ , and so on) behaviour in a “reasonable or predictable way” is not something that is objectively describable—it includes evaluations by people. One person may feel that a scenario is reasonable and events predictable, but another may feel that this is not the case. What is self-evident to one may not be to another. We have proposed that the three factors that contribute to plausibility are: 1) the reactivity of the environment to participant actions, 2) contingent references by elements of the environment to the participant, and 3) credibility of expectation—i.e., the environment is constructed based on evidence of what is supposed to happen in real life where this is relevant, so that the application is supposed to be a simulation of events that occur in reality. These are a question of the extent to which these elements are *supported by the hardware and have been programmed*. The *satisfaction of expectations* or the *realisation of environment being coherent* in the sense meant in (Skarbez et al., 2018b; Skarbez et al., 2020) cannot be programmed, since these are the responses of the participant. What is possible though is to carry out research, for example,

using co-design methods in order to attempt to meet these requirements. Another way to think about this is that Psi may fail if the environment never responds to the participant, or if nothing in the environment ever addresses the participant personally, or if there is something that fails to meet expectations. All of these are a function of the hardware and programming and the design of the application. To the extent that coherence might also be regarded in the same way—i.e., there is an attempt to satisfy coherence but it fails—then coherence can be thought of as a way of referring to requirement (3).

This discussion emphasises our distinction between “immersion” and “presence.” The “immersion” is fully under the control of the implementation. Whether a virtual character smiles back at you when you smile towards it, will have to have been programmed. The extent to which a virtual environment can meet expectations depends on prior research amongst potential participants about the critical elements that a scenario must support, and then actually programming these. Whether this investigation has been carried out or not, and then the required elements programmed into the application is a matter of fact. Then “presence” (PI and Psi) refers to how people respond to the “immersion.” As mentioned above this is not deterministic, since people have different prior experience, personalities, knowledge, and so on. We can think PI and Psi as conditional probabilities, and consider: conditional on a particular immersive configuration what are the probabilities of PI or Psi occurring? Given different immersive configurations these probabilities may change. Thus immersion sets the ground for PI and Psi. Similarly we use the terms “embodiment” to refer to the multisensory factors that provide evidence about the body—for example, it is seen as life-sized from the perspective of the eyes of that body, it moves synchronously with the person’s real movements, when something touches it the person feels this on their real body, and so on. Embodiment configurations, which are completely determined by the hardware and programming, may give rise to the illusion of body ownership. Again, this is not deterministic but provides the basis. For example, ballet dancers, always acutely aware of the exact disposition of their bodies might have reduced body ownership in VR because they are more likely to notice small discrepancies between the position and movement of virtual limbs compared to the true positions and movement.

Latoschik and Wienrich (2021) introduced a new model for Plausibility. However, the term “Plausibility” in this model does not have the same meaning as considered here, and it is confusing that the same term was used to describe something quite different. Latoschik and Wienrich (2021) write that ... in contrast to the discussed presence models, we don’t assume an illusion of plausibility but define plausibility as a state or condition during an XR experience that subjectively results from the evaluation of any information processed by the sensory, perceptual, and cognitive layers” (p5-6). In their model each of *cognition* (“social-cognitive processes” and “higher order cues”), *perception* (“proximal perception experiences” and “proximal perception cues” and *sensation* (“genetics or life-long habitude perceptions” and “habitual sensory cues”) contribute to a level of coherence (as defined by Skarbez et al.). Then Plausibility is a weighted function of these



**TABLE 1 |** Summary of types of measurement and their properties.

| Method                        | Type of data                                                                                                                                                       | Applied                                                                                                                                                                                                      | Analysis                                                                                                                                          | Advantages/disadvantages                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Questionnaire                 | Subjective responses on an ordinal scale                                                                                                                           | Typically after the experience, though it is possible to include questionnaires as part of the experience, with the danger of forcing participants to a meta level (thinking about rather than experiencing) | Frequency tables, median, interquartile range, ordered logistic or probit regression                                                              | Very simple, can be applied to any concept. Universal—can be applied to any scenario.<br>Surface results only, lack of intersubjective equivalence in meaning of scores, imposes researcher concepts on participants.                                                                                                                                                                                                                                                                                                                          |
| Behavioural and Physiological | Observed behaviours, e.g., based on tracking, eye movements, heart rate, skin conductance, respiration, etc                                                        | During the experience                                                                                                                                                                                        | Means and standard deviations, ANOVA, regression, time series                                                                                     | The observed values are based on what participants do, or on their automatic responses and not on their opinions.<br>The measurement process is more difficult, especially with physiological measures. Problems such as drift in skin conductance responses and artefacts—e.g. in ECG data—have to be overcome. Interpretation of results is open to question. The VR scenario requires events or situations that deliberately spark the behaviour or physiological responses which may not be appropriate. It is therefore not universal.    |
| Breaks                        | Discrete times when breaks occurred, count of the number of breaks                                                                                                 | During the experience                                                                                                                                                                                        | Rate of breaks per unit time, times between breaks, survival analysis, mean, standard deviation, Poisson regression or equivalent                 | Based on observational data rather than subjective. A break occurring is an observable event. It is universal—can be applied to any scenario.<br>It is difficult to ascertain when breaks occur. The subjective element clearly applies when participants are asked to self-report breaks.                                                                                                                                                                                                                                                     |
| Configuration transitions     | Transitions between configurations and their timing, and when matches to a specific quale (e.g., Place Illusion) are reported (depending on the particular method) | During the experience                                                                                                                                                                                        | Transition probability matrix, Markov Chain analysis, conditional probabilities                                                                   | Based on observed data—the actual transitions, and matches (if used in the method). The method can be combined with Reinforcement Learning.<br>Applicable when the goal is to find the optimal configuration for a number of factors. Very large data sets may be required for transition probability matrix estimation when the number of factors is above 4 and the number of levels of factors are above 3.                                                                                                                                 |
| Sentiment analysis            | Freely written or spoken verbal reports, resulting in quantitative sentiment scores                                                                                | After the experience. Talking aloud during the experience is possible but with the danger of forcing participants to a meta level                                                                            | Mean, standard deviation, regression, ANOVA, cluster analysis for clusters of sentiments, keyword analysis to find commonalities amongst clusters | The analysis gets beneath the surface of questionnaire or behavioural and physiological responses to allow participants to freely reflect on their responses to the VR experience, yet nevertheless resulting in quantitative measures.<br>The method only applies to sentiment, although researchers are able to create their own dictionaries that concentrate on other aspects of an experience (e.g., presence).<br>Requires participants to write essays or speak about their experiences at enough length to make the analysis feasible. |



three aspects of coherence. Plausibility in turn contributes to several different qualia: presence, social presence, copresence, placeness, body ownership—Figure 2 in Latoschik and Wienrich (2021). The authors state that “In our opinion, the proposed model possesses predictive and explanatory power of modern XR experiences.”

Brubach et al. (2022) tested this theory with two experiments where participants interacted with some cubes. In experiment 1 ( $n = 40$ ) there were two binary factors: manipulation of the perceptual layer (within group): cubes followed expected physical laws or they floated. The second factor was a cognitive manipulation (between groups): participants were told that they would be weightless, or nothing was said about this. Hence the conditions (floating cubes, weightlessness) and (cubes follow physical laws, no mention of weightlessness) were compatible with coherence, and the other two possibly with incoherence. A questionnaire was used to assess plausibility, which included questions such as “I am used to objects behaving this way,” “I had a prior expectation of how the objects would behave,” “the behaviour of the objects made sense,” **Table 1** in (Brubach et al., 2022). An assessment of presence was based on the Igroup questionnaire (Schubert et al., 2001). The results showed that each individual item on their Plausibility scale had higher scores for the normal cube behaviour on the perceptual factor. However, there were no significant results for the cognitive factor (i.e., the explanation about weightlessness had no effect), and no interaction effects. Presence as measured by the Igroup questionnaire showed a significantly higher mean for “realism” for the cognitive factor “weightlessness” level, whereas “involvement” had a higher mean for the perceptual “floating cubes” level. Participants were asked to choose which perceptual level (the within group factor) was more plausible, and 39/40 chose the normal cube behaviour.

The second experiment ( $n = 71$ ) was on the same lines, but with a much richer story line for the cognitive factor. Similar results were found for the Plausibility questions, but no significant differences for the Igroup presence questionnaire (except in overall scores). The authors concluded that both experiments together showed that “The manipulation of object behavior on the perceptual layer leads to a break in plausibility.” However, the hypothesis “Breaks in plausibility lead to break in presence” could only be partially accepted (based on experiment 1), and no conclusion could be reached on whether “Perceived coherence between the cognitive and perceptual manipulation has an effect on presence.” Overall the authors argued that the two experiments provide empirical evidence in favour of their model described above.

There are several points to be made about this model and experiment. First there is the unfortunate confusion of terminology, as mentioned. A definition of “plausibility” is given that has little relationship with the original one proposed in (Slater, 2009). However, considering the operationalisation of this via the questionnaire about plausibility, we can see that in this model plausibility refers to evaluations of object behaviour—how real this seems and how much it conforms with expectations. This is not the original meaning of Psi, which is not an evaluation of conformity to

realism but refers to the illusion that *events are actually happening* (even though cognitively the participant knows that nothing like those events are really happening). Psi is like an exclamation “This is really happening!” The cube falls towards your toes and you jump out of the way, an avatar smiles at you and you smile back—these are automatic reactions that follow from the possibly implicit exclamation “This is really happening!” At some level the brain does not know about virtual reality. It responds and acts based on its sensory surroundings. The point is that *there might be a falling cube!*, so the safest action to take is to get out of the way. In our whole lives when we have seen something falling towards our toes, there really was something falling towards our toes, so the safe thing to do is to act on that.

The second problem is that if the idea of plausibility as exhibited in the questionnaire is followed, then applications that depict fantasy worlds, or events that cannot happen in reality, can never be plausible. People adopt different models of reality in VR though, they can fly and be a superhero (Rosenberg et al., 2013), they can see their own virtual body separate itself from their location and carry out some actions (Goris et al., 2021) and have an out-of-body experience (Bourdin et al., 2016), or even a simulated near-death experience (Barberia et al., 2018). If these applications were judged on the basis of the plausibility questionnaire adopted in (Brubach et al., 2022), then none of them would be plausible, since events occurred that clearly broke physical laws. Yet each of these examples were effective in leading to changes in participant attitudes and behaviours.

The third issue is the complexity of the model, with three layers, several elements in each layer, and five resulting qualia. Moreover, the idea that “Plausibility emerges from a function of weighted congruence relations (activations)” does not offer a way forward—how could this ever be tested, or rejected? The approach we have proposed is much simpler. On one side is everything relevant that goes into the making of an application that can be objectively described—the hardware (e.g., computational, display and tracking), the programming (which determines the events that occur, the affordances and the valid actions that participants can carry out). On the other side are the responses—PI, Psi and body ownership. This relationship can even be expressed as conditional probabilities:  $P(\text{Psi} | \text{an immersive configuration})$ , and this has been done, for example in (Slater et al., 2010a; Bergström et al., 2017; Skarbez et al., 2017). This model could also include psychological assessments of participants to account for individual differences.

However, there is a different question that concerns the mechanisms that might explain how a particular immersive configuration becomes translated into PI, Psi or body ownership. The approach of Latoschik and Wienrich (2021) could be regarded as a step on the road in that direction.

With respect to Psi, there remains a huge amount of research to do in order to investigate what the illusion that events are “really happening” means and can be assessed, how this might vary from application to application, what types of contingent reference to the participant are important, how faithfully should the environment conform to expectations when it is a simulation

of events that could happen in real life. The issue of how to maximise Plausibility is a massive research area, that probably would benefit from input from theatre and film studies.

## Measurement

As we have seen multiple methods of measurement have been proposed and used. These are summarised in **Table 1**. Questionnaires have the advantage that they are relatively simple, and may help answer specific questions, but they are not neutral. If we want to find out what happens during an experience from the point of view of the participants, questionnaires are not ideal—they impose a conceptual framework which might have nothing to do with the actual experiences of participants during the exposure. Physiological and behavioural measures including EEG as surrogates for presence are useful and are more objective measures, but they are not universal. They require environments where there are events that are specifically included to induce automatic responses, or where these responses are well-understood and predictable and would only occur if there were presence. BIPs provide another neutral way to measure presence, and although interesting theoretically, they suffer from the problem of knowing when they occur. The configuration transitions method provides a psychophysical approach to quantifying presence, but this method does not easily give in-depth information about the underlying reasons for transitions. Using sentiment analysis does provide quantitative and qualitative information about responses to a scenario. However, questionnaires are going to be continued to be used because of their simplicity and universality (applicable to any concept and any scenario). Good evidence that the questionnaire responses are meaningful is when their results correlate with a completely different behavioural or physiological measure. For example, the experiment with medical trainees and doctors carried out by Pan et al. (2016) examined the extent to which they would prescribe antibiotics to (virtual) patients, even when the indications suggested a virus, only because the patients vociferously demanded the antibiotics. It was found that whereas almost all the trainees inappropriately prescribed antibiotics, the doctors were less likely to prescribe the greater their reported levels of PI and Psi elicited from questionnaires. In other words the more that they experienced the situation as really happening, the more appropriate their behaviour. The introduction by Meehan et al. (2002) of physiological measures for a stressful virtual environment also included a questionnaire for presence, and the results supported each other. Recently, Archer et al. (2022) used a questionnaire and physiological measures to assess presence in response to odours in the virtual environment, and found correlated results from the two measures. For further discussion of the relationship between questionnaires and physiological measures see (Grassini and Laumann, 2020). We propose that ideal measurements should involve this type of triangulation between several completely different approaches, such as a combination of configuration transitions and qualitative approaches, including sentiment

analysis, and where questionnaires are used then further backup with behavioural or physiological measures where these make sense in the context.

## PI and Psi in Augmented Reality

How does this discussion relate to Augmented Reality (AR) where the participant sees the physical surrounding into which are projected virtual entities? PI becomes inverted. VR aims to place the participant into the virtual world. In AR the problem is to incorporate virtual objects into the real world. Although the problem is inverted the solution is the same: the extent to which the AR system supports sensorimotor contingencies for perception of virtual entities. To bring a virtual object into the physical world participants must be able to perceive it by using their bodies as they would objects that are really there. Hence, they must be able to look around it, see it from different orientations, ideally reach out and touch it. It is more complex than in VR since in VR everything is under program control. But in AR virtual objects should reflect light from the real world as well as virtual light, should influence illumination of the real world, should cast shadows on real objects as well as virtual objects, should visually obscure objects that are behind them from the perspective of the participant, and real objects should obscure virtual objects that are behind them, and so on. Although this is much more technically demanding than in VR, the fundamental issue is the same. Sensorimotor contingencies for perception will bring virtual objects into the real world just as in VR SCs place the participant inside the virtual world. Some of the complexities involved in this have recently been discussed by Regenbrecht and Schubert (2021) who have provided a questionnaire for presence in AR and discussed the reasoning behind it. With respect to Psi, however, there is no fundamental conceptual difference from VR. Plausibility in AR is based on the same principles and causes as in VR—virtual entities should respond to actions of participants, entities should be able to initiate personal interactions, and—most difficult of all—there should be consistency between real and virtual objects and coherence as discussed earlier.

## CONCLUSION

In this paper we have reviewed the concepts of Place Illusion and Plausibility and argued that if these are achieved, then we would have created a separate reality, one parallel with the real world in which the whole experience is taking place. We have also pointed out that PI and Psi are concepts essentially imposed on participants, and while critical in some applications, it is also useful to have methods that rely on the actual experiences of participants rather than what we think those experiences should be. We have reviewed a number of methods for measurement, and found problems with all of them, and argued that a combination of a psychophysical method such as configuration transitions with qualitative methods including sentiment analysis would be the way

forward, but in any case triangulation through multiple approaches would be ideal. Some of these issues have been studied now for about three decades. With the entry of VR (and to a lesser extent AR) into the consumer market, there is the opportunity for conducting very large studies with participants from all walks of life (not just the typical student subjects), with experiences in their own homes (Steed et al., 2016; Mottelson et al., 2021). It also becomes even more pressing to address the vast amount of confusion there is in the literature. Our philosophy is to keep it simple: be there, experience what is going on as really happening, and therefore respond realistically. Additionally, allow participants to express themselves leading to researchers discovering new ways of thinking about the effects of VR on people.

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## AUTHOR CONTRIBUTIONS

MS conceived of the article and wrote the first version. All authors commented and contributed to the writing of the submitted version.

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